

Wireless Classroom • Instructor Notes

What each simulator teaches, what every control is for and why it is there, and a demo that works for each one.

VERSION Toolbox 1.11.0

DATE September 28, 2026

COVERS 48 simulators on four shelves, and the 11 course handouts

A reason marked *(inferred)* is our reading, not a recorded design decision.

It is reasoned from what the control lets the class see. Every other reason comes from the build spec or research, the in-app help, or one of Keith's recorded rulings. On the course handouts, the same mark flags where to hand a card out.

Not for distribution to students. This is a separate document for instructors. It is not included in the app.

How to use these notes

Present mode

Every simulator has a **Present** button in its title bar. It appears whenever the window is at least 720 by 480 pixels, so you will see it on a laptop or tablet but not on a phone.

Present gives you one screen with no scrolling: the animation fills about two-thirds of the width, the controls sit in a panel on the right, and the most important number moves onto the stage in large type so the back of the room can read it. On a Mac it goes full screen.

Text and lines scale up with the window, because a projector makes any window the same size on the wall.

Set the scene first. Open a tool, press the 'Present' button on the top right to have it go full screen. Then set it up the way you want to start (a wall material, a channel plan, a number of stations). I've found it easier to set the various options while already in Presenter mode.

Predict, then reveal

Before you press a key, ask the room what will happen. "If I move this client twice as far from the AP, how much signal do we lose?" "If one client drops to 6 Mbps, what happens to everyone else?" Take the guesses, then press the key. The simulators are built so one control makes one visible change, which is exactly what you need for this. Students remember the moment they were wrong far longer than a slide that told them the answer.

From the Teacher's Guide in the app (Toolbox 1.11.0). Every demo in these notes follows this pattern.

What each entry holds

Part	What it holds
Teaches	The one idea the class should leave with.
Controls	Every control, by its on-screen label, what it does, and why it is there.
Present mode	The keys the tool adds or uses. F, ? and Esc work in every tool; the key index lists the rest.
A demo that works	About three minutes: set up, ask the room, reveal, then an optional follow-up.
What it leaves out on purpose	The simplifications, for when a student asks "is it really like that?"
Pairs with	Related simulators and lessons, and the lab handout where one exists.

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Present-mode key index

The keys below work in presenter mode. The first table is the same in every simulator. The second lists the keys only some tools use, by key, with the page of the tool's entry.

Table 1. Keys in every simulator (Teacher's Guide)

Key	What it does
Space	Play or pause
Right arrow	Step forward one step (a slot, a frame, a message, a segment)
R	Reset
Up and Down	Move the tool's main control (distance, SNR, channel width, and so on)
F	Switch full screen on or off
?	Show every key for this tool
Esc	Leave presenter mode

Tools with nothing to play or step ignore Space and the Right arrow, and some tools use Up and Down for something other than their main control. Each entry's Present mode note says what the keys do in that tool. On the RF and Propagation shelf, [and] repeat Down and Up.

Table 2. Keys only some tools use

Key	Tool	What it does	Page
1 to 4	Fourier and FFT	Switch modes, in every mode.	40
1, 2, 3	Why a Long Wi-Fi Password Matters More on WPA2	Choose WPA2, WPA3 or transition.	119
1, 2, 3	PoE: Why the New AP Runs at Half Strength	Pick 802.3af, 802.3at and 802.3bt.	121
A	Airtime Anatomy	Steps through the arrangements.	61
B	A Frame's Journey	Picks the next band.	50
B	Channel Utilization Meter	Adds a 1-second burst.	79
B	Predict, Then Measure	Walks both sides of every wall.	101

Key	Tool	What it does	Page
B	Where Am I? Signal Strength vs Round-Trip Timing	Blocks or clears AP 1's direct path.	103
C	A Frame's Journey	Corrupts a bit or sends the frame clean.	50
C	Airtime Anatomy	Corrupts one MSDU or repairs it.	61
C	What an Interferer Costs (and how a NIC hears the air)	The next channel.	68
C	Predict, Then Measure	Clears the walk.	101
D	Voice Priority, End to End	Starts or stops the download.	59
D	DFS and Radar	Fires radar now.	109
D	Down the Stack, Across the Air, Up the Other Side	Shows the next To DS / From DS case.	113
E	Heat Map Builder	Switches heat map, truth and error map.	99
M	Uplink vs Downlink	Turns the AP down to match, and back.	31
M	Voice Priority, End to End	Switches the AP mapping.	59
M	Predict, Then Measure	Steps through the maps.	101
M	Where Am I? Signal Strength vs Round-Trip Timing	Also switches the method.	103
M	Conference Wi-Fi Runs Out of Addresses	Turns rotation on or off.	111
N	What an Interferer Costs (and how a NIC hears the air)	Picks the next source.	68
N	Multi-Link Operation	Draws new random traffic.	84
N	Heat Map Builder	Is the next step of the three-dots lesson.	99
O	Predict, Then Measure	Walks one side only.	101

Key	Tool	What it does	Page
P	Decibels in Your Head: the Rules of 3 and 10	Shows or hides the answer.	8
P	How to Measure Wall Attenuation	Shows or hides the answer.	15
P	Uplink vs Downlink	Shows or hides the answer.	31
P	Body Loss	Shows or hides the answer.	34
P	Voice Priority, End to End	Asks, reveals and closes the question.	59
P	Heat Map Builder	Steps the power through 1, 2, 4 and 6.	99
P	Where Am I? Signal Strength vs Round-Trip Timing	Steps the -70 dBm question.	103
P	Conference Wi-Fi Runs Out of Addresses	Asks, reveals and closes the question.	111
P	Why a Long Wi-Fi Password Matters More on WPA2	Reveals the answer.	119
P	PoE: Why the New AP Runs at Half Strength	Reveals the answer.	121
Q	Why a Busy Line Lags	Switches SQM.	107
S	How to Measure Wall Attenuation	Switches sides.	15
U	Predict, Then Measure	Updates the model.	101
V	Airtime Anatomy	Switches Time and Structure.	61
V	Down the Stack, Across the Air, Up the Other Side	Switches between the journey and the address fields.	113
W	Fourier and FFT	Steps to the next window (FFT mode).	40
W	Channel Utilization Meter	Skips ahead one window.	79
W	Heat Map Builder	Shows or hides the hidden wall.	99

Key	Tool	What it does	Page
Left	Decibels in Your Head: the Rules of 3 and 10	Takes 3 dB away.	8
Left	How to Measure Wall Attenuation	Moves the laptop 10 cm (4 in) toward the source.	15
Left, Right	Body Loss	Turn the holder 15 degrees.	34
Left	The Number on the Box vs the Number in Your Hand	The step before.	46
Left	Association, Frame by Frame	Takes one back.	115
Left	802.1X and EAP Ladder	Takes one back.	117
Page Up, Page Down	Decibels in Your Head: the Rules of 3 and 10	Move 10 dB (what a presentation clicker sends).	8
Tab	Where Am I? Signal Strength vs Round-Trip Timing	Switches signal strength, timing and both while no control has focus (Shift+Tab moves into the controls).	103
] and [	Modulation Simulator	Move the SNR 1 dB per press, as Up and Down do.	37

RF and Propagation

The RF and Propagation shelf takes a signal from the decibel arithmetic, through distance, walls, reflections and antennas, to what a real device reports and what people and regulations cost it. Every tool has Present and Copy in its title bar (ten also have the m / ft switch, which sets lengths for every Classroom tool at once), and in presenter mode F, ? and Esc always work and [and] repeat Down and Up, so each entry below lists only the keys that tool adds. A reason marked (*inferred*) is our reading of what the control lets the class see, not a recorded design decision.

Decibels in Your Head: the Rules of 3 and 10

db-rules

Teaches: dB is a ratio, so +3 dB doubles the power and +10 dB multiplies it by ten, which makes -67 dBm about twice the power of -70 dBm.

Controls

Control	What it does	Why it's there (the teaching point)
Signal level (slider, -80 to -60 dBm in whole dB, default -70)	Moves the signal against a fixed -70 dBm reference. The linear milliwatt bar, the lines from the dB ruler and the readouts follow it.	The research brief asked for exactly one control with one visible change: a dB slider over a linear bar that doubles every 3 dB. The range was chosen so -80, -70, -67 and -60 dBm fill 1%, 10%, 20% and 100% of the bar; a wider range would have shrunk the -70 vs -67 comparison to a few pixels.
-10 dB, -3 dB, +3 dB, +10 dB (buttons)	Step the signal by one rule. A step that would leave -80 to -60 dBm is turned off.	Lets you act out the rules path the readout prints (+1 dB is +10 -3 -3 -3) one press at a time on the bar. (<i>inferred</i>)
Reset	Returns the signal to -70 dBm and hides the answer.	Puts the signal back on the reference (1x, the same power) so the next question starts from zero difference. (<i>inferred</i>)
Reveal the answer / Hide the answer	Shows the answer to "Is -67 dBm a little or a lot stronger than -70 dBm?"	The house predict-then-reveal pattern: the room commits to a guess before the screen answers.
Show -67 dBm against -70 dBm (appears after the reveal)	Sets the signal to -67 dBm.	Puts the prediction on the bar: 10% of the length becomes 20%, the step the dB ruler hides.
Open dBm / Watt Converter, Open dB Reference	Open the two calculator tools.	For any level outside -80 to -60 dBm or a conversion to watts; this tool is the picture, those are the calculators.

Present mode: Up and Down move 1 dB. Right adds 3 dB and Left takes 3 dB away. Page Up and Page Down move 10 dB (what a presentation clicker sends). Space shows -67 dBm against -70 dBm. P shows or hides the answer. R resets.

A demo that works (about 3 minutes)

- 1 Set up:** open the tool as it starts, signal at -70 dBm, answer hidden.
- 2 Ask the room:** "Is -67 dBm a little or a lot stronger than -70 dBm?" Many will say a little: it is only 3.
- 3 Reveal:** press P, then Space. The bar goes from 10% to 20% of its length: 100 picowatts to 199.5, about double.
- 4 Optional follow-up:** from -70, press Page Up. The bar goes to full length, ten times. Then ask what -69 dBm is and read the rules path: +10 -3 -3 -3, ten times then halved three times, 1.25x against the exact 1.259x.

What it leaves out on purpose: Nothing here is measured or illustrative; it is arithmetic. +3 dB is 1.995 times, not exactly 2, which the readouts show beside the rule. The range stops at -80 and -60 dBm so the bar stays drawable; outside that, use the dBm / Watt Converter. It says nothing about noise or SNR.

Pairs with: FSPL Simulator, next on the shelf, where every doubling of distance costs 6 dB. dBm / Watt Converter and dB Reference for the numbers. No lab handout yet.

FSPL Simulator

fspl-simulator

Teaches: every doubling of distance costs 6 dB in free space, and higher bands lose more because the receive antenna is smaller at a shorter wavelength, not because the air absorbs more.

Controls

Control	What it does	Why it's there (the teaching point)
2.4 GHz, 5 GHz, 6 GHz (band chips, all on)	Show or hide each band's curve.	Puts the bands on one chart so the gap between them is visible. Each band has its own line style and marker (solid circle, dashed square, dotted triangle) because the design system has no categorical colors, so the class can tell them apart without color.

Control	What it does	Why it's there (the teaching point)
Received / Path loss (toggle, default Received)	Plots received power in dBm, or free-space path loss in dB.	Path loss is what the formula gives; received power is the unit a design target is written in, and the target lines only appear in Received. (<i>inferred</i>)
100 m / 1 km (300 ft / 3,000 ft in imperial)	Sets the end of the distance axis.	Keeps the default view on indoor distances while a point measured past 100 m (330 ft), or an outdoor link, can still land on the chart. (<i>inferred</i>)
Log / Linear (toggle, default Log)	Switches the distance axis.	Keith, 2026-09-27: log shows the straight 6 dB-per-doubling lines; linear shows the curve people are used to. Same numbers either way.
Drag or tap the chart; Cursor (slider)	Move the distance cursor. The table under the chart reads path loss and received power per band, and the difference between bands.	The readout makes the second lesson checkable: the difference between two bands is the same at every distance.
2.4 GHz channel, 5 GHz channel, 6 GHz channel (pickers; defaults 6, 100, 37)	Set each band's center frequency.	The band gap is $20 \log_{10}(f_2/f_1)$, so it depends on the channel, not just the band: 6 GHz channel 37 and channel 117 differ by about 0.5 dB. (<i>inferred</i>)
Tx power (0 to 30 dBm, default 20)	Raises or lowers every received curve.	Received power = Tx power + gains - path loss - losses; moving it shifts every band equally and leaves the gap alone. (<i>inferred</i>)
Tx antenna gain, Rx antenna gain (-5 to 15 dBi, default 0)	Add antenna gain at either end.	0 dBi is the isotropic antenna the aperture term assumes. A higher-gain receive antenna has a larger effective area and wins some of the high-band loss back.
Other losses (0 to 30 dB)	Subtracts a fixed loss from every curve.	A stand-in for cable, body or wall loss without modeling any of them. (<i>inferred</i>)
Show log-distance curves (switch); Path-loss exponent n (2.0 to 4.0, default 3.0, off until the switch is on)	Overlays the indoor log-distance model.	Free space is the best case; this shows how much faster an indoor path loses. Labeled a model with a chosen exponent, not a measurement; $n = 2$ reproduces free space.

Control	What it does	Why it's there (the teaching point)
Measured point: RSSI, Distance, Measured on (band), Clear measured point	Plots a reading you type as a diamond, with its gap to the free-space curve in dB.	The gap between the model and a measurement is where the design work happens. Manual entry only; nothing is read from the Wi-Fi adapter.
Design targets: Level, Label, remove (x), Add line (up to 4), Reset to -67 voice / -70 HD video	Edit the dashed target lines in the Received view. Saved on the device for every lesson.	Keith asked on 2026-09-27 for the lines to be editable with a number and a label. The defaults are read from the Signal Thresholds tool so the two tools cannot drift apart.
Show controls / Hide controls (tune button, phone only)	Opens the controls sheet under the chart.	Phone first: the chart owns the screen and the controls collapse to one row of band chips.

Present mode: Up and Down double and halve the cursor distance, the lesson's own step (every press is 6 dB). There is no Space, Right or R in this tool.

A demo that works (about 3 minutes)

- 1 Set up:** all three bands on, Received, Log, cursor at 10 m (33 ft). On channel 6 the path loss is 60.2 dB and, with 20 dBm and 0 dBi antennas, received power is -40.2 dBm.
- 2 Ask the room:** "Move twice as far away, to 20 m (66 ft). How much signal do we lose?"
- 3 Reveal:** press Up. Every band drops 6.0 dB. Press it again at 40 m (131 ft): another 6 dB, the same as the first doubling. Then point at the Why panel: the spreading bar is identical for every band; only the aperture bar changes, and channel 100 sits 7.1 dB under channel 6 at every distance.
- 4 Optional follow-up:** turn on Show log-distance curves at $n = 3$ and press Up again. A doubling now costs about 9 dB. Or switch to Linear to show the steep drop near the AP is the same 6 dB per doubling, squeezed into the first few meters.

What it leaves out on purpose: Free space has no walls, floor bounce, people or fading, so real readings usually sit below the curve. The aperture term assumes a 0 dBi receive

antenna. The indoor curve is a model with an exponent you choose. Measured points are typed in by hand.

Pairs with: Decibels in Your Head before it. Wi-Fi Through a Wall and 6 GHz Power and PSD after it (the "Why 6 GHz behaves differently" sequence). Rate vs Range and Uplink vs Downlink for the link budget. Lab: the FSPL lab, still a draft, at [Deliverables/2026-09-27-teachers-guide-labs/fspl/FSPL-Lab-DRAFT.pdf](#); Keith ruled it ships as a student version and an instructor version.

Wi-Fi Through a Wall

wifi-through-a-wall

Teaches: one wave meets one wall, part reflects, the rest shrinks inside and continues behind at the same frequency, and a layered or thin wall can lose less at a higher band because the echoes from its faces cancel.

Controls

Control	What it does	Why it's there (the teaching point)
Band (2.4 / 5 / 6 GHz, default 5 GHz) and Channel (picker; defaults 6, 100, 117)	Set the frequency.	Channel-accurate wavelength. For thin and layered walls the echoes add or cancel depending on how the gap compares with the wavelength, so moving a few MHz can move the loss.
Tx power (0 to 30 dBm, default 20)	Sets the level at which the wave reaches the wall. The wave height is drawn in dB above a -95 dBm noise floor.	Lets the class see that even thick concrete leaves a wave behind the wall while the signal is above the noise floor. There is no distance in this tool, so free-space loss is not included.
Wall (picker, default Interior stud wall): Interior stud wall, Concrete elevator-shaft wall, Solid wood door, Double-pane window, Single-pane window, Brick wall, one brick thick, One material, any thickness	Loads a real wall (its layers are listed under the picker and in the strip under the plot) or switches to one material.	Wall loss is usually quoted as one number per material. Real walls computed with ITU-R P.2040 show where the number comes from, and two of them break the "higher band loses more" rule: the stud wall and the double-pane window. Keith kept the double-pane window on 2026-09-27.

Control	What it does	Why it's there (the teaching point)
Material (ITU-R P.2040 Table 3) (One material only)	Picks one of ten materials.	Compare materials one at a time on the same model the other wall tools use.
Thickness (field and slider, 1 cm to 1 m (0.4 to 39 in), One material only)	Sets the slab thickness. The slider is logarithmic.	Keith, 2026-09-27: thickness in cm, and the wall drawn small (not to scale) so a 1 m wall does not swamp the picture. The log scale gives thin glass and thick concrete the same slider travel. (<i>inferred</i>)
Angle of arrival (0 to 80 degrees, default 0)	Tilts the wave from head on toward skimming along the wall.	A glancing wave reflects more.
Polarization (TE / TM, default TE)	TE: electric field along the wall face. TM: tilted with the angle.	Shows that reflection off a slant depends on polarization; head on the two are the same.
Play / Pause	Animates the wave, one cycle every 2 seconds. It plays when the tool opens; with reduced motion on it starts frozen.	A pause is always offered for accessibility, and a frozen frame is what you point at.
Set the wall to (thickness) (button in the measured values card)	Switches to one material at the published specimen's thickness.	Puts the model and the published measurement side by side at the same thickness. The card names each source and never averages them.

Present mode: Space plays or pauses the wave. R returns to the opening wall (Interior stud wall). Up and Down step through the Wall list; on One material they change the thickness instead, so use the Wall picker or R to get back to the real walls. In presenter mode Tx power, angle and polarization fold into one panel.

A demo that works (about 3 minutes)

- 1 Set up:** the tool as it opens: Interior stud wall, 5 GHz channel 100, head on, TE, Tx power 20 dBm, wave playing.
- 2 Ask the room:** "This stud wall at 2.4, 5.5 and 6.5 GHz. Which band loses the most?" Many will say 6 GHz.
- 3 Reveal:** read the All three bands card: 2.7, 2.8 and 1.4 dB. The two plasterboard sheets absorb only 0.5 to 1.2 dB of it; the rest is how the echoes across the air gap add or cancel, and at 6.5 GHz they cancel.
- 4 Optional follow-up:** press Up to the Concrete elevator-shaft wall: 14.8, 26.9 and 30.4 dB, so concrete does follow the rule. Then the Double-pane window: 0.5 dB at 2.4 GHz and 8.9 dB at 6.5 GHz, nearly all reflection.

What it leaves out on purpose: Smooth, uniform layers only: no studs, insulation, wiring, rebar, mortar, frames or low-E coatings (a low-E coating can add far more loss than the glass, and P.2040 has no class for it). The real walls use common construction sizes, not a measured wall. Published measurements often disagree with the model (NIST measured 102 mm of fresh concrete at about twice the model's loss). The drawing is one line straight through the wall, slowed, with the wall not to scale and the ripple inside it not drawn.

Pairs with: FSPL Simulator (free-space loss is a separate effect), How to Measure Wall Attenuation (same materials), Room Propagation (same wall model on a plan), 6 GHz Power and PSD, and the RF Attenuation calculator. No lab handout yet.

How to Measure Wall Attenuation

measure-wall

Teaches: measure a wall by averaging a series of readings close to each side with the source 4 m (13 ft) or more away; put the source close and the extra free-space loss on the far side reads as wall.

Controls

Control	What it does	Why it's there (the teaching point)
Drag the person with the laptop, or tap a spot in the room	Places the laptop on either side of the wall. Where it stands is where that side's readings are taken.	Acts out Keith's field method on the stage: one series in front of the wall, one behind.
Drag the RF source	Changes the source's distance from the wall.	The free-space curve is steep near the source, so the class watches the error grow as the source comes close.
Scenes: Tight, Loose, Source too close (buttons)	Set the source and both reading distances in one press: Tight is source 4 m, readings 10 cm (4 in); Loose is readings 1 m (3.3 ft); Source too close is source 1 m. The wall and readings settings stay.	Tight is Keith's method. Keith ruled on 2026-09-27 that the tool opens on Loose, so the error is on screen from the start. Source too close shows the worst case.
Source to wall (0.5 to 15 m (1.6 to 49 ft), default 4 m)	Same as dragging the source.	Keyboard and exact placement for the lesson's main variable.
Laptop on the: Near side / Far side (toggle); Go to the far side / Go to the near side (button)	Switches sides and keeps each side's spot.	A measurement needs a series on both sides; switching keeps both spots so the difference is always live.
Near reading, in front of the wall; Far reading, behind the wall (sliders)	Set how far from the wall each series is taken. The near reading stays at least 10 cm (4 in) from the source.	The error term is $20 \log_{10}(\text{far distance} / \text{near distance})$; these two sliders are what make it large or small.
Band (2.4 / 5 / 6 GHz, default 5)	Locks the laptop to channel 6, 100 or 117 for both series.	Lock the measuring device to one channel so every reading is of the same signal. The free-space error itself does not depend on the channel.
Material (ITU-R P.2040 Table 3), Thickness (1 cm to 1 m, default 10.2 cm (4 in) of concrete)	Set the wall.	The true wall loss comes from the same slab model as Wi-Fi Through a Wall, so the readout can compare the measured number with the truth.

Control	What it does	Why it's there (the teaching point)
Readings per side (1 to 30, default 10); Fading spread, standard deviation (illustrative) (0 to 6 dB, default 2)	Set how many readings each series takes and how much they scatter.	Average a series rather than trust one reading. Set the spread to 0 to see the geometry error alone.
Take new readings, near side / far side (button)	Draws a fresh series on the side the laptop is on.	Shows that the fading residual changes every time while the geometry error stays put. (<i>inferred</i>)
Reveal the answer / Hide the answer; Show it	Answers "Source 2 m from the wall, readings 1 m either side of it. Is the measured wall loss too high, too low, or right?" and puts that scene on the stage.	Predict, then reveal. The scene was set by Larry's correction during the build: the far reading is 3 m from the source and the near one 1 m.

Present mode: Right and Left move the laptop 10 cm (4 in) away from and toward the source, crossing the wall at a face. Space takes new readings. S switches sides. Up and Down move the source 0.5 m. P shows or hides the answer. R resets.

A demo that works (about 3 minutes)

- 1 Set up:** the tool opens on Loose: 5 GHz, 10.2 cm (4 in) of concrete, source 4 m (13 ft), readings 1 m (3.3 ft) either side. True wall loss is 14.3 dB and the measurement reads 19.2 dB, 4.6 dB of it geometry.
- 2 Ask the room the question on screen:** source 2 m, readings 1 m either side. Too high, too low, or right?
- 3 Reveal:** press P, then Show it. Too high: $20 \log_{10}(3 / 1) = 9.5$ dB of free-space loss that is not the wall, about 9.8 dB with this wall's thickness added.
- 4 Optional follow-up:** press Tight. Source 4 m, readings 10 cm (4 in): the measurement is 15.2 dB and the geometry error is down to 0.6 dB. Set the fading spread to 0 to show what is left is geometry, then back to 2 and press Space a few times.

What it leaves out on purpose: The reflection off the near face, which makes readings close to it rise and fall over a short distance (averaging several readings over a small area handles it in the field). Paths around the wall through doors and other rooms. The laptop's

own body and antenna pattern. The fading spread is illustrative. The source is a fixed 20 dBm with 0 dBi antennas, and nothing below a -95 dBm noise floor can be read.

Pairs with: Wi-Fi Through a Wall (how the wall takes the loss) and Predict, Then Measure (a predicted wall against a measured one); the tool has buttons that open both. FSPL Simulator for the free-space term. No lab handout yet.

Multipath Simulator

multipath-simulator

Teaches: the direct signal and its reflections add like arrows, so moving a few centimeters walks the receiver from a peak into a dip, and the dips repeat every half wavelength.

Controls

Control	What it does	Why it's there (the teaching point)
Scene (picker): One wall (two-ray), Standing wave (walk to a wall), Many paths (Rayleigh fading); default One wall	Switches between the three models.	The three steps of the lesson: one reflection, the standing wave in front of a wall, then random fading from many paths. It is a picker, not segments, because the labels truncated at phone width.
Band (2.4 / 5.5 / 6.5 GHz, default 2.4)	Changes the wavelength.	The dips repeat every half wavelength: 6.2 cm (2.4 in), 2.7 cm (1.1 in) and 2.3 cm (0.9 in).
Wall material (One wall and Standing wave): Metal (Γ 1.00), Thick concrete (Γ 0.39), Drywall (Γ 0.10), Custom; default Metal	Sets how strongly the wall reflects.	A strong reflector cancels the direct copy completely; thick concrete swings the signal by 7.2 dB; drywall barely ripples it.
Reflection strength Γ (slider, 0 to 1)	Sets the reflection yourself; moving it switches the material to Custom.	Lets the class dial the reflection down and watch the nulls fill in.
Receiver position (One wall, 0 to 100 cm) / Distance to the wall (Standing wave, 0 to 40 cm); drag the receiver in the picture or on the plot	Moves the receiver.	Watch the reflected arrow swing around the direct one as the receiver moves.

Control	What it does	Why it's there (the teaching point)
Reflectors (Many paths, 2 to 30, default 12)	Sets how many reflected copies arrive.	More copies of random phase is Rayleigh fading; the histogram shows it against the theory curve.
Reflectors up to: 10 m / 60 m / 300 m (Many paths; room, large hall, outdoors)	Sets how far away the reflectors sit.	Exists so the guard-interval lesson can fire: only the 300 m layouts put copies more than 0.8 microseconds late.
New layout (Many paths)	Places the reflectors somewhere else. Layouts are seeded, so the same settings give the same room.	A 2 m track holds only a few dozen fades, so the histogram wanders; new layouts show it averaging toward the Rayleigh curve.
Antenna A position (0 to 2 m); Antenna B offset (λ) (0 to 1 wavelength in steps of 0.05, default 0.5)	Move antenna A along the track and set how far antenna B sits from it.	Selection diversity: two antennas half a wavelength apart rarely fade at the same spot. The readout shows how often each, and both at once, are more than 10 dB below average.
Show all N paths / Show the latest N (delay list, Many paths)	Expands or shortens the list of how late each copy arrives.	Keeps the delay list short on screen while the full set is one press away. (<i>inferred</i>)

Present mode: Up and Down move the receiver a sixteenth of a wavelength (antenna A in Many paths), so eight presses go from one null to the next. There is no Space, Right or R in this tool.

A demo that works (about 3 minutes)

- 1 Set up:** Standing wave, 2.4 GHz, Metal.
- 2 Ask the room:** "I move the phone 3 cm (1.2 in). How much can the reading change?" Take guesses in dB.
- 3 Reveal:** drag the receiver onto a peak of the plot, then press Up four times (a quarter wavelength, about 3 cm): it lands in a null, where the two copies cancel. Four more presses reach the next peak, and the next null is 6.2 cm (2.4 in) from the first.
- 4 Optional follow-up:** switch to Thick concrete and the swing drops to 7.2 dB. Or go to Many paths, set Antenna B offset to 0.5λ , and compare Antenna A, Antenna B and Both at once against the 9.5% Rayleigh figure.

What it leaves out on purpose: Each wall preset is one reflection strength; a real wall reflects differently by angle and band, and thick concrete's 0.39 is for a wave hitting it head on. Many paths blocks the direct copy on purpose; with a strong line-of-sight copy the dips are shallower and rarer. The pattern along 2 m looks the same at any reflector distance; only the delay list changes.

Pairs with: Why Two Devices Disagree (it reuses this fading), Room Propagation (the close-up of the nulls near a wall), Rate vs Range. The "Why the signal jumps when you move" sequence. No lab handout yet.

Why Two Devices Disagree

devices-disagree

Teaches: two devices at the same spot report different RSSI because each adds its own offset, grip, body loss, fading and rounding, and an offset calibrates the device but not how it is held or the fade it sits in.

Controls

Control	What it does	Why it's there (the teaching point)
Reveal (button beside the question)	Answers "Laptop says -62, phone says -68, same spot. Which one is right?" with this run's numbers.	Predict, then reveal, on the question from the spec.
Re-sample	Takes the next few seconds of readings.	Fading changes from one sample set to the next while the fixed offsets do not, so a re-sample shows which part of the spread moves. (<i>inferred</i>)
Reset	Puts every setting back.	Returns to the four illustrative default devices. (<i>inferred</i>)
Apply offsets (switch, default off)	Subtracts each device's offset, as a survey tool does once you enter it per adapter.	The fixed spread disappears and fading, grip and body loss remain. This is why survey tools take an offset per adapter.
Fading (switch, default on)	Turns the Rayleigh fading off.	Off shows only the fixed parts of each reading, so Apply offsets can take the spread to zero.

Control	What it does	Why it's there (the teaching point)
Device A, B, C, D (chips)	Pick the device to edit.	Up to four generic devices, named by letter and type only; the spec forbids naming a device or vendor.
Device X offset (illustrative) (-10 to +10 dB; defaults 0, -4, -2, +1)	Sets that device's fixed offset.	The fixed part of the difference: antenna gain and how the chip calibrates.
Device X orientation and grip loss (illustrative) (0 to 10 dB)	Sets how much the way it is held costs.	How a device is held changes its reading, and no offset corrects it.
Device X held against a body (switch)	Adds the body loss set under The spot.	The person holding the device is part of the reading (Body Loss covers it in full).
Device X reporting step (1 dB / 2 dB)	Rounds that device's reports.	Two devices hearing the same power can differ by up to a step from rounding alone. <i>(inferred)</i>
Device X averages the last (1 to 10 readings)	Sets how many readings the device averages, in milliwatts.	Averaging more readings makes the fading spread smaller.
Device X type (Laptop, Phone, Tablet, Survey adapter)	Renames the device. A label only; the numbers are set per device.	Gives the class a familiar kind of device without a brand. <i>(inferred)</i>
Band (2.4 / 5 / 6 GHz, default 5)	Changes the wavelength and the true power.	Half a wavelength sets how far apart devices must be to fade separately, and it shrinks as the band rises. <i>(inferred)</i>
AP distance (1 to 50 m, default 8 m (26 ft))	Moves the AP, which changes the true power at the spot.	Shows that the spread between devices does not depend on how strong the signal is. <i>(inferred)</i>
Spacing between devices (0 to 30 cm (0 to 12 in), default 10 cm)	Sets how far apart the devices sit.	Farther apart than half a wavelength (about 2.7 cm (1.1 in) at 5 GHz), each device sees its own fade.
Devices (Two / Three / Four, default Four)	Sets how many devices are on the spot.	Two devices matches the laptop-and-phone question; four shows the full range. <i>(inferred)</i>
Body loss when on (illustrative) (0 to 10 dB, default 3)	Sets the loss the body switch adds.	Every body value is illustrative; set it to what you measure.

Present mode: Space re-samples. R resets. Up and Down move the AP one meter (one foot in imperial).

A demo that works (about 3 minutes)

- 1 **Set up:** the defaults: 5 GHz, AP 8 m (26 ft), fading on, four devices.
- 2 **Ask the room the question at the top:** laptop -62, phone -68, same spot. Which one is right?
- 3 **Reveal:** press Reveal: neither one on its own. Turn Fading off: the traces go flat and the remaining spread is the fixed parts. Turn on Apply offsets: most of it goes, but the phone (Device B) still reads low, because its grip and body loss are not an offset.
- 4 **Optional follow-up:** turn Fading back on, then raise the averaging on one device and press Space a few times: its trace steadies. Set spacing to 0 and the devices fade together.

What it leaves out on purpose: Every offset, grip and body loss, the 14 dBm AP and the exponent of 3 are illustrative; real offsets depend on the device, antennas and driver. RCPI (802.11k) is shown as a concept card only, with no values computed. Runs are seeded, so the same settings give the same readings. The fading pattern moves past the devices at an illustrative 0.1 m/s.

Pairs with: Multipath Simulator (the same fading model), Body Loss, Uplink vs Downlink. The "Why clients misbehave" and "Troubleshooting slow or unstable Wi-Fi" sequences. No lab handout yet.

Room Propagation

room-propagation

Teaches: the signal at a spot is the direct path plus reflections and diffraction past edges, and the extra loss at 6 GHz splits into free space, wall absorption and diffraction.

Controls

Control	What it does	Why it's there (the teaching point)
Floor plan (picker): Drywall offices, Concrete corridor, Open office, Glass meeting room, Empty plan; default Drywall offices	Loads a plan with its walls, doorways, AP and client. Each plan shows its one-line lesson under the picker.	Each preset carries one lesson: a doorway matters little in drywall and a lot in concrete; signal bends past each jamb; metal cabinets cast a shadow; thin glass loses less on channels 100 and 117 than on 6. Empty plan is free space to draw on.
Reset this plan	Reloads the current plan.	Undoes any wall edits in one press. <i>(inferred)</i>
Band (2.4 / 5 / 6 GHz) and Channel (default 2.4 GHz channel 6)	Set the frequency.	Watch the same spot on all three bands and see why 6 GHz loses more, part by part.
AP EIRP (0 to 36 dBm, default 20)	Raises or lowers the whole map. The receiver is a 0 dBi antenna.	Moves every 10 dB color band together, so the shape of the coverage is a property of the building, not the power. <i>(inferred)</i>
AP antenna: Upright (TE) / Flat (TM)	Sets the polarization against the walls.	Walls reflect a glancing wave differently for the two polarizations.
Reflections: None / 1 bounce / 2 bounces (default 1)	Sets how many wall bounces are added.	Switch each mechanism off and on to see what it does. None gives the straight-line picture.
Diffraction at wall ends and doorways: On / Off (default On)	Adds the ITU-R P.526 knife-edge diffraction.	Off, every shadow edge is sharp; on, signal bends into the shadow.
Fresnel zone to the client (switch, default on)	Draws the first Fresnel zone and gives its radius at the middle of the path.	Shows how wide the path to the client is; keep it clear of walls and edges.
Doorway shadow edges (switch, default on)	Draws straight lines from the AP past each door jamb.	Where a shadow would start if signal traveled only in straight lines, so the class sees the signal reaching past them.
Close-up of the ripple (switch, default on)	Shows a close-up around the client with every path added with its phase.	The map is a local average; the close-up shows the peaks and nulls every half wavelength that make RSSI jump when you move a phone a little.

Control	What it does	Why it's there (the teaching point)
Drag the AP or client on the plan; AP across, AP down, Client across, Client down (sliders)	Move the AP and the client.	The same as dragging, for exact placement and keyboard use. <i>(inferred)</i>
What a tap or drag on the plan does (picker): Move the AP and client, Draw walls, Add doorways, Select a wall	Chooses what the plan does when touched.	Draw your own plan or change a preset.
Wall (picker), Material (ITU-R P.2040 Table 3), Thickness	Pick a wall and set its material and thickness, or set them for the next wall you draw. Up to 16 walls.	Test a claim live: make one wall concrete, or thicker, and watch the map. <i>(inferred)</i>
Add a doorway, Close its doorways, Delete wall	Edit the selected wall.	A doorway is a gap in the screen; the class sees how much signal it carries past a lossy wall. <i>(inferred)</i>

Present mode: R reloads the plan. Up and Down move the AP EIRP 1 dB.

A demo that works (about 3 minutes)

- 1 Set up:** Concrete corridor, 2.4 GHz channel 6, 1 bounce, diffraction on.
- 2 Ask the room:** "The client is in a room off the corridor, behind concrete. Where is its signal coming from?"
- 3 Reveal:** set Diffraction Off: the shadows behind the door jambs go sharp and line up with the doorway shadow edges. Set Reflections to None: the straight-line picture. Turn both back on and read Where the loss comes from: free space, each wall on the line, diffraction and reflections.
- 4 Optional follow-up:** switch to 6 GHz channel 117 and read The same spot on all three bands. Free space alone is about 8.6 dB worse than channel 6, and the diffraction loss grows too.

What it leaves out on purpose: It is a 2D plan of a 3D model: no floor or ceiling bounce, no furniture, no people. Each wall is one solid uniform slab. Reflections stop at two bounces,

and reflected paths pass through walls without diffraction. The map is the average over a few centimeters; the close-up shows the exact spot.

Pairs with: Wi-Fi Through a Wall (the same wall model), Multipath Simulator (the nulls), FSPL Simulator (the free-space part). The "Why the signal jumps when you move" sequence. No lab handout yet.

Antenna Pattern

antenna-pattern

Teaches: antenna gain is not more power; the same power is reshaped, so a high-gain omni flattens into a pancake with a hole under it and a directional narrows its beam.

Controls

Control	What it does	Why it's there (the teaching point)
Antenna (picker): Half-wave dipole, Omni, set by gain, Collinear omni (array), Directional (patch or sector), Imported pattern file; default Omni at 8 dBi	Picks the antenna model.	Keith, 2026-09-25: "change from Omni to directional, change the Antenna Gain and visually see changes in the 3D antenna pattern."
Gain (Omni, 2.15 to 15 dBi)	Changes the omni's gain; the readout gives dBi and dBd.	Every antenna radiates the same power, so raising the gain visibly takes signal from somewhere else. The scale stays fixed, so a gain change reshapes the pattern instead of resizing it.
Electrical downtilt (Omni 0 to 15 degrees; Collinear 0 to 15)	Tilts the cone.	Downtilt tilts the whole cone for an omni, so the ring of strongest signal moves closer on the floor.
Elements (1 to 8) and Spacing (0.25 to 1.0 wavelength) (Collinear)	Build the array.	Stacking elements makes the nulls between lobes visible; at 1 wavelength spacing a grating lobe appears straight up and down.
Gain (sets both beamwidths) (Directional, 3 to 21 dBi)	Sets both beamwidths together with the practical 31,000 rule.	Gain and beamwidth trade directly. The readouts show what the exact shape integrates to, which is not the rule's number.

Control	What it does	Why it's there (the teaching point)
Horizontal beamwidth (15 to 180 degrees), Vertical beamwidth (10 to 120 degrees) (Directional)	Set each beamwidth.	Shape the beam one plane at a time.
Front-to-back (A_max), Side-lobe level (SLA_V) (10 to 40 dB); Mechanical downtilt (0 to 30 degrees) (Directional)	Set the rear and side-lobe floors and tilt the beam.	Front-to-back is a worst case, not the value straight back; the readout shows both, because the back lobe need not point straight back.
Rebuild from its two cuts (any built-in antenna)	Writes the antenna as an MSI file, reads it back and rebuilds the 3D from its two cuts.	A 3D view built from two 2D cuts is an estimate; this shows how far off it is against the exact shape.
Generated example (picker), Or paste pattern text (MSI or NSMA), Read pattern (Imported)	Load one of five example files the app generates, or pasted file text. There is no file picker.	Keith asked to import a two-cut pattern and see it in 3D. The examples are generated here, so the error against the exact 3D can be shown; no manufacturer file is included.
Rebuild the 3D by: Summing / Cross-weighted (Imported)	Switches the reconstruction method.	Compare two ways of building 3D from two cuts, and see that they differ behind the antenna and between the cuts.
What if: shape (x the file's dB) (0.5 to 2.0) (Imported)	Scales the file's dB values.	Keith: "change the gain to make it flatter, or wider, or have the edges fall off faster."
Mounting: Ceiling / Wall (default Ceiling)	Turns the 3D view 90 degrees against the floor and the mounting surface.	The same antenna ceiling-mounted and wall-mounted lights different parts of a room. The 2D cuts stay in the antenna's own frame, as a datasheet draws them.
Angle between the two antennas (0 to 90 degrees)	Gives the polarization mismatch loss.	Puts a number on turning one antenna sideways: 45 degrees costs 3.01 dB. <i>(inferred)</i>
Drag, pinch or scroll the 3D view; arrow keys and plus or minus with focus; Reset view; Spin / Stop (presenter bar)	Rotate, zoom and reset the view.	Look underneath a high-gain omni to find the hole, from any angle. <i>(inferred)</i>

Present mode: Space spins the view. Right turns it one 15-degree step. R resets the view. Up and Down move the main setting: 1 dB of gain (Omni, Directional), one element (Collinear), or 0.05 of shape (Imported, once a file is read); nothing on the dipole.

A demo that works (about 3 minutes)

- 1 **Set up:** Omni, set by gain at 8 dBi, Ceiling. Press Space to spin it once so the room sees the donut.
- 2 **Ask the room:** "I swap this for a 12 dBi omni on the same ceiling. Is the floor right under the AP stronger or weaker?"
- 3 **Reveal:** press Up four times to 12 dBi. The donut flattens into a pancake with a hole under it; the vertical beamwidth is near 6.8 degrees, and 30 degrees above the horizon it is weaker than a 5 dBi omni.
- 4 **Optional follow-up:** switch Mounting to Wall to see the same antenna light the room differently. Or pick Directional and press Rebuild from its two cuts to show what a planning tool's 3D is built from.

What it leaves out on purpose: Every built-in shape is normalized to 0 dBi averaged over the sphere, so the peak shown is directivity with no efficiency loss. The omni-by-gain model is the ITU-R F.1336 planning envelope, not physics (it gives the dipole's 2.15 dBi a 65.6 degree beamwidth, not 78). A 3D view from two cuts is an estimate. Mounting turns the 3D only. The NSMA azimuth direction is assumed counterclockwise from above.

Pairs with: the Antenna Fundamentals guided lesson (read it first), FSPL Simulator (antenna gain in the link), Rate vs Range. The "Antennas" and "RF math and the link budget" sequences. No lab handout yet.

Rate vs Range

rate-vs-range

Teaches: rate falls with distance in steps, one ring per MCS; a wider channel shrinks every ring, and a higher minimum basic rate shrinks the cell edge and the beacon airtime.

Controls

Control	What it does	Why it's there (the teaching point)
Drag or tap on the rings; Client (slider, logarithmic, default 20 m (66 ft))	Moves the client. The label and readout give received power, noise floor, SNR, MCS and data rate.	Read the MCS a client gets at any distance and see the step at each ring edge.
Band (2.4 / 5 / 6 GHz, default 5)	Changes the frequency path loss is taken at (2437, 5500 or 6135 MHz) and the widths on offer.	Higher bands lose more in free space, so every ring is smaller. <i>(inferred)</i>
Channel width (MHz) (20 or 40 at 2.4 GHz, up to 160 at 5 GHz, up to 320 at 6 GHz; default 20)	Changes every MCS threshold by 3 dB per doubling.	Doubling the width raises the noise floor 3 dB, so every ring shrinks. The view never rescales, so the shrink is visible.
Spatial streams (rate labels only) (1 to 4, default 2)	Changes the data rates on the labels, not the rings.	Streams multiply the rate but do not change the signal each MCS needs, so the rings stay where they are. <i>(inferred)</i>
AP EIRP (0 to 36 dBm, default 20)	Raises or lowers the downlink level everywhere.	More radiated power pushes every ring out by the same number of dB. <i>(inferred)</i>
Client antenna gain (-5 to 6 dBi, default 0)	Adds the client's antenna to the link.	A worse client antenna shrinks every ring as surely as a weaker AP. <i>(inferred)</i>
Path-loss exponent (model) (2.0 to 4.0, default 3.0)	Sets how fast the signal falls with distance.	$n = 2$ is free space, indoors is usually 3 to 4. Labeled a model with a chosen value, not a measurement.
Margin (0 to 20 dB, default 0)	Adds a safety margin to every MCS threshold.	Shows how much coverage a design margin costs, ring by ring. <i>(inferred)</i>
Minimum basic rate (picker): 6 Mbps (-82 dBm) up to 54 Mbps, the higher rates marked MCS-equivalent	Moves the dashed cell edge and the beacon airtime bar.	Raising the basic rate shrinks the cell edge and cuts beacon airtime (6 to 24 Mbps moves the edge from -82 to -74 dBm at 20 MHz). Only 6 Mbps has a published floor; 9 Mbps and the DSSS rates are not offered because they have no sourced floor.
SSIDs on this AP (1 to 16, default 4)	Scales the beacon airtime bar.	Every SSID sends its own beacons at the basic rate, so the airtime cost multiplies.

Present mode: Up and Down move the client out and in by the fourth root of 2, so four presses double or halve the distance.

A demo that works (about 3 minutes)

- 1 Set up:** the defaults: 5 GHz, 20 MHz, 2 streams, AP EIRP 20 dBm, $n = 3$, client at 20 m (66 ft). The client receives -66.3 dBm, just short of MCS 5, so it gets MCS 4.
- 2 Ask the room:** "Widen the channel from 20 to 80 MHz. What happens to the edge of coverage?"
- 3 Reveal:** set 80. The MCS 0 ring pulls in from 66.8 m (219 ft) to 42.2 m (138 ft), and every other ring shrinks with it.
- 4 Optional follow-up:** back to 20 MHz, raise the Minimum basic rate from 6 to 24 Mbps. The cell edge pulls in from 66.8 m to 36.2 m (119 ft) and four SSIDs' beacons drop from 2.19% to 0.61% of airtime.

What it leaves out on purpose: The sensitivities are conformance floors, the least a radio must manage; real radios do several dB better, so real rings are larger. The path-loss exponent is a chosen value, and walls, people and furniture make real rings uneven. The width is held fixed, where a real AP may send a narrower transmission to a far client. The beacon bar counts beacons only: no probe responses, contention or other APs.

Pairs with: FSPL Simulator, Uplink vs Downlink (the other direction of the same rings), 6 GHz Power and PSD, the SSID Airtime and MCS Index tools. The "RF math and the link budget" and "From bits to rate" sequences. No lab handout yet.

6 GHz Power and PSD

six-ghz-psd

Teaches: 6 GHz power is limited per MHz as well as in total, so a PSD-limited class keeps its SNR as the channel widens while a capped class loses 3 dB per doubling, and the client 6 dB rule applies only to Standard Power and GVP clients.

Controls

Control	What it does	Why it's there (the teaching point)
Region: US (FCC) / EU (ETSI) (default US)	Switches the rule set.	The EU has no client offset at all, which the US rules make easy to forget.
Classes to show (checkboxes; US: nine classes, EU: two; default Standard Power AP, Low Power Indoor AP, Very Low Power device)	Turn class lines on and off. Each class has its own color, line style and marker.	Compare classes on one chart; each can be told apart without color.
Chart: EIRP / SNR / Spectrum (default EIRP)	Switches the view. Spectrum draws one channel as a flat block of power per MHz.	EIRP shows the cap and the PSD limit; SNR shows the consequence; Spectrum shows the block getting wider while its height holds or drops.
Class drawn in the spectrum view (picker, Spectrum with more than one class)	Picks the class the block is drawn for.	The spectrum view draws one class at a time. <i>(inferred)</i>
Width (slider, 20, 40, 80, 160, 320 MHz, default 80); tap a column on the chart	Picks the channel width the readouts describe.	The x axis of the lesson; the readout line says whether each class is PSD-limited or cap-limited at that width.
Distance (1 to 100 m, logarithmic, default 10 m (33 ft))	Sets where SNR is read.	SNR at a chosen distance, with free-space loss at one frequency for every width so only the width changes between columns.
Extra loss (walls) (0 to 40 dB, default 0)	Subtracts a fixed loss from every class.	An optional extra-loss slider for walls, from the spec.
Noise figure (3 to 12 dB, default 7)	Raises or lowers the noise floor.	Moves every class's SNR by the same amount, so it never changes which class holds and which falls. <i>(inferred)</i>
AP authorized power: Standard Power AP (AFC) (0 to 36 dBm), GVP AP (geofencing) (0 to 24 dBm); US only, active when an SP or GVP class is shown	Sets each AP's grant. Its clients follow 6 dB lower.	The client -6 dB rule is a relative limit only for Standard Power and GVP clients; the LPI client is a flat number that happens to sit 6 dB under the LPI AP.

Present mode: Up and Down change the channel width one step.

A demo that works (about 3 minutes)

- 1 **Set up:** US, the default classes, SNR view, 10 m (33 ft), width 20 MHz.
- 2 **Ask the room:** "Double the channel to 40, then 80, then 160 MHz. Which class keeps its SNR?"
- 3 **Reveal:** press Up three times. The LPI AP holds 43.8 dB from 20 to 160 MHz: its EIRP rises 3 dB per doubling (18, 21, 24, 27 dBm) and the noise floor rises the same 3 dB. The Standard Power AP is already at its 36 dBm cap, so it loses 3 dB per doubling (61.8 dB at 20 MHz, 49.8 dB at 320).
- 4 **Optional follow-up:** switch to EIRP and turn on Standard Power client; set the Standard Power AP to 30 dBm and the client drops to 24 dBm. Then switch to EU and point out there is no client offset.

What it leaves out on purpose: These are regulatory maximums; a device may transmit less. Path loss is taken at one frequency (6105 MHz) for every width. The MCS shown is the typical figure from the Signal Thresholds table, not a guarantee. The PSD rule applies to any 1 MHz of the real emission, and an HE 20 MHz transmission fills about 18.9 MHz, a 0.2 dB difference too small to change the lesson.

Pairs with: FSPL Simulator and Wi-Fi Through a Wall (the "Why 6 GHz behaves differently" sequence), Uplink vs Downlink (the client rules in a link), Rate vs Range, the 6 GHz channel card in Course Handouts. No lab handout yet.

Uplink vs Downlink

uplink-downlink

Teaches: every Wi-Fi link is two links, the client usually transmits less than the AP, so there is a zone where the client hears the AP but the AP cannot hear the client, and turning the AP down to "match" fixes nothing on the uplink.

Controls

Control	What it does	Why it's there (the teaching point)
Drag or tap on the floor; Client (slider, default 20 m (66 ft))	Moves the client. The readout gives each direction's level, MCS and SNR, the imbalance and the width of the asymmetry zone.	Put the client inside the shaded zone and read both directions at once.
Regulatory rule (picker): Custom (no rule applied), US 6 GHz Standard Power, US 6 GHz Geofenced Variable Power (GVP), US 6 GHz Low Power Indoor (LPI), EU 6 GHz Low Power Indoor (LPI); default Custom	Applies a rule to the client's power. Every rule is a 6 GHz rule, so picking another band returns to Custom.	Regulations set the client relative to the AP (Standard Power, GVP), as a flat number (US LPI) or the same as the AP (EU). Standard Power and GVP are separate entries because their AP caps differ.
Band (2.4 / 5 / 6 GHz, default 5)	Sets the frequency path loss is taken at.	Path loss is the same both ways on any band; the imbalance comes from the radios. (<i>inferred</i>)
Channel width (MHz) (default 20)	Moves each direction's MCS thresholds 3 dB per doubling, and the per-MHz limits under the LPI rules.	Under US LPI the limits are per MHz, so at 20 MHz the AP is 18 dBm and its client 12; the flat 30 and 24 appear only at 320 MHz.
AP transmit power (0 to 30 dBm, default 20, illustrative)	Moves the downlink only.	The uplink never moves with the AP's power.
Client transmit power (0 to 24 dBm, default 14, illustrative); reads "set by the rule" under a rule	Moves the uplink only. Under a rule the client transmits at its limit.	The client is the weak end of the link; its power sets the uplink. The rule is a ceiling, and many clients transmit less.
AP antenna gain (0 to 8 dBi, default 4), Client antenna gain (-5 to 3 dBi, default -2, a phone-class antenna)	Add each antenna's gain.	Each antenna counts in both directions, so the AP's better antenna helps it hear the client as much as it helps the client hear it. The imbalance is set by the two transmit powers alone.
Path-loss exponent (model) (2.0 to 4.0, default 3.0)	Sets how fast the signal falls with distance.	Stretches or shrinks both rings without changing the dB imbalance between them. (<i>inferred</i>)

Control	What it does	Why it's there (the teaching point)
Turn AP down to match / Put the AP back to (power)	Sets the AP's transmit power to the client's, and back.	Keith's standing position: AP and client power rarely match and should not be made to. The solid ring shrinks, the dashed ring does not move, and the readout says what changed and what did not. The tool never recommends matching.
Reveal the answer / Hide the answer; Put the client in the zone	Answers "The client shows four bars. Can the AP hear it?" and moves the client into the asymmetry zone.	Bars come from the downlink only and cannot tell you whether the AP hears the client.

Present mode: Up and Down move the client out and in (four presses double or halve the distance). R resets. P shows or hides the answer. M turns the AP down to match, and back.

A demo that works (about 3 minutes)

- 1 Set up:** the defaults: Custom, 5 GHz, 20 MHz, AP 20 dBm with 4 dBi, client 14 dBm with -2 dBi, client at 20 m (66 ft).
- 2 Ask the room the question on screen:** the client shows four bars. Can the AP hear it?
- 3 Reveal:** press P. At 20 m the client hears the AP at -64.3 dBm (MCS 6) and the AP hears the client at -70.3 dBm (MCS 3), a 6 dB imbalance. Press Put the client in the zone: between 49 m (161 ft) and 78 m (256 ft) the client still decodes the AP while the AP cannot decode the client.
- 4 Optional follow-up:** press M. The solid ring pulls in from 78 m to 49 m and the uplink stays at -70.3 dBm: the AP gave up its reach and the uplink gained nothing. Then pick US 6 GHz Standard Power on 6 GHz and move the AP slider: the client follows the AP's authorized power, 6 dB lower.

What it leaves out on purpose: All radio values are illustrative. Both ends use the same conformance receiver floors, where many APs hear better with extra antennas and receive chains. Under a rule the client is set to the most it may transmit. No walls or bodies; path

loss is the log-distance model. For one direction in detail, with cable losses and fade margin, use Link Budget (the tool has a button).

Pairs with: Rate vs Range (the same thresholds and path loss), 6 GHz Power and PSD (the same client rules), Body Loss, Link Budget. The "RF math and the link budget" and "Why clients misbehave" sequences. No lab handout yet.

Body Loss

body-loss

Teaches: people cost decibels, the person holding the device often costs the most when they turn their back to the AP, and a crowd on the line is why an empty-building survey reads better than the building in use.

Controls

Control	What it does	Why it's there (the teaching point)
Band (2.4 / 5 / 6 GHz, default 5)	Sets the frequency and applies that band's body-loss multiplier.	Higher bands generally lose more to the body, shown as an illustrative trend.
Holder facing (slider, 0 to 360 degrees, 0 up the page and 90 to the right); Face the AP, Back to the AP (buttons); tap the floor to turn the holder toward that spot	Turns the holder. The holder loss stays at zero until the holder has turned more than 120 degrees away from the AP, then ramps in to the full value with the back square to it.	Orientation matters: facing the AP with the device in front costs little; turning your back puts your body in the path.
Drag the holder; Holder (slider, distance from the AP)	Moves the holder.	Changes the path loss and which people stand on the line, so the class can find a clear spot. <i>(inferred)</i>
Drag any person	Places a person on or off the line.	Shows that only the people on the straight line count in this model. <i>(inferred)</i>
Holder body loss at 2.4 GHz (illustrative) (0 to 20 dB, default 8)	Sets the holder's full loss with the back square to the AP.	No primary source was read for any body-loss figure, so every loss is a labeled, adjustable setting; set it to match what you measure.

Control	What it does	Why it's there (the teaching point)
Loss per person on the line at 2.4 GHz (illustrative) (0 to 10 dB, default 4)	Sets what each person on the line costs.	Same reason: illustrative, adjustable, labeled so.
5 GHz body-loss multiplier (illustrative) (1.0 to 2.0, default 1.2), 6 GHz body-loss multiplier (illustrative) (1.0 to 2.0, default 1.3)	Scale the losses at 5 and 6 GHz.	The band trend applies to the crowd as well as the holder, because a crowd is bodies (a build decision past the spec).
Building: Empty / Occupied (default Occupied)	Removes or restores the crowd. Nothing else changes.	The Saturday survey against the Monday room.
People in the room (0 to 50, default 30)	Sets the crowd size.	Crowds add up, one body on the line at a time.
Scatter the crowd	Places the people in a new arrangement from a new seed.	The same number of people costs more or less depending on how many land on the line. <i>(inferred)</i>
Reveal the answer / Hide the answer; Show Saturday, empty / Show Monday, occupied	Answers "You surveyed the auditorium empty on Saturday. What happens Monday at 9 a.m.?" and switches between the two.	Predict, then reveal, on the question from the spec.

Present mode: Left and Right turn the holder 15 degrees. Space empties or fills the room. Up and Down change the crowd by 5 people. P shows or hides the answer. R resets.

A demo that works (about 3 minutes)

- 1 **Set up:** the defaults: 5 GHz, Occupied with 30 people, the holder 14 m (46 ft) from the AP, facing it. Two people stand on the line.
- 2 **Ask the room the question on screen:** surveyed empty on Saturday, what happens Monday at 9 a.m.?
- 3 **Reveal:** press P, then Space to empty the room: -61.8 dBm (MCS 7). Space again to fill it: -71.4 dBm (MCS 3), 9.6 dB worse, from two people at 4.8 dB each.
- 4 **Optional follow-up:** press Right one step at a time. The first eight presses change nothing; after that the holder loss ramps in, and with the back square to the AP the level is -81.6 dBm, MCS 0. Then compare 2.4 and 6 GHz.

What it leaves out on purpose: Every body-loss value is an illustrative setting with no measured figure behind the defaults, so use it for the shape of the effect, not as a design value. Only people on the straight line count; real signals also reach around a crowd by reflection. Bodies are 0.5 m wide seen from above, and the device is held 0.3 m in front of the holder. Path loss is the log-distance model at $n = 3$.

Pairs with: Why Two Devices Disagree (body as one part of a reading), Uplink vs Downlink, Wi-Fi Through a Wall and RF Attenuation for walls rather than people. No lab handout yet.

Signals and PHY

This shelf is the physical layer, from the bottom up. Modulation turns bits into a point on a constellation. Fourier and FFT shows how a receiver pulls those points back out of a symbol, and how an analyzer measures the band. OFDMA splits the channel into resource units, MIMO and Beamforming explains streams and spare chains, and the PHY Preamble Reference opens up the first 20 μ s and what follows it. The Number on the Box vs the Number in Your Hand adds all of that up into the rate one client gets, and A Frame's Journey follows one frame across the air and back as an ACK.

The Teacher's Guide sequence "From bits to rate" runs Modulation Simulator, then Rate vs Range, then PHY Preamble Reference. "Wi-Fi 6 and 7 features" opens with OFDMA Resource Units.

Every tool on this shelf has a **Present** button in its title bar when the window is at least 720 by 480 pixels. In presenter mode F switches full screen, ? lists the keys for that tool and Esc leaves. The keys listed under each tool below are the ones it adds or uses on top of those three. A tool that has nothing to play or step ignores Space and the Right arrow.

Where a "why" is our reading of what the control lets the class see, rather than a recorded design reason from a spec, brief or build log, it is marked (*inferred*).

Modulation Simulator

modulation-simulator

Teaches: a group of bits picks one point on the I/Q plane, higher orders pack more bits into points that sit closer together, and so they need a cleaner signal to decode.

Controls

Control	What it does	Why it's there (the teaching point)
Modulation (select: BPSK, QPSK, 16-QAM, 64-QAM, 256-QAM, 1024-QAM, 4096-QAM, each with its bits per symbol)	Picks the constellation. Default 16-QAM. The line under it gives the point count, bits per symbol and the amendment that first used it.	The spec's lesson is that higher orders carry more bits per symbol and need a cleaner signal. Seeing BPSK next to 4096-QAM makes that plain (help, whyHere).

Control	What it does	Why it's there (the teaching point)
Play / Pause	Sends one symbol per tick at the chosen speed. The simulator always opens paused.	Paused by default so nothing moves until you ask (spec, reduced motion). Play lets the cloud build up while you talk.
Step	Sends exactly one symbol.	One symbol is one point, one amplitude and one phase on the carrier strip. Step is how you show that mapping slowly, and it is the suggested path when reduced motion is on.
+100	Sends 100 symbols at once.	Measured EVM from a handful of symbols wanders, and a few hundred settle close to theory (help, <code>fieldNotes</code>). +100 gets you there in two presses. (<i>inferred: the button's purpose is not recorded, the reason follows from that note</i>)
Reset	Clears the received points and counts. Disabled when nothing has been sent.	Starts a fresh reading. Every other change clears the points too, so each reading belongs to one setting (help, <code>fieldNotes</code>).
Signal-to-noise ratio (SNR) slider, 0 to 45 dB in 0.5 dB steps, default 25 dB	Sets the Gaussian noise added to each received point.	The spec's one noise control: the received cloud spreads, points cross decision boundaries and are marked wrong.
Speed (symbols per second): 1/s, 4/s, 20/s. Default 4/s.	Sets how fast Play sends.	(<i>inferred</i>) 1/s is slow enough to narrate each symbol on the carrier strip; 20/s fills the constellation fast enough to see the cloud's shape.
Bits to send: Random, Your text	Random bits, or the UTF-8 bits of a typed message.	The spec calls for both. Typed text makes the errors land in something the class can read, and shows the zero-padding of the last symbol.
Message (text field, up to 120 characters, default "Hello, Wi-Fi"; appears with Your text)	The message to send. Shows bytes, bits, symbols and padding, then Progress, Received so far and Last full copy.	Lets the room watch a readable message break as the SNR drops. (<i>inferred</i>)

Control	What it does	Why it's there (the teaching point)
Tap or hover a point on the constellation (64-QAM and above)	Shows that point's bits.	Up to 16-QAM every point is labeled. Above that the labels would overlap, so the spec asks for labels on hover or tap only.
Present (title bar)	Opens the presenter layout over the same state.	See the shelf intro.

The readout cards (bits per symbol, levels per axis, K_MOD, current bits, I and Q, amplitude and phase, decided as, symbol and bit errors, EVM measured and theory, and the 802.11 transmit EVM limits with Meets or Misses) are not controls, but point at the limits table. The spec requires it to be labeled as the transmitter's required accuracy, not a receiver sensitivity threshold, and the screen says so.

Present mode: Space plays or pauses, Right arrow steps one symbol, R resets, Up and Down (or] and [) move the SNR 1 dB per press.

A demo that works (about 3 minutes)

- 1 Set up:** 16-QAM, SNR 25 dB, Random bits. Press +100 twice. The cloud sits tight on each point, EVM measured is close to -25 dB, and there are no symbol errors.
- 2 Ask the room:** "Same SNR, but I switch to 256-QAM. How many errors?"
- 3 Reveal:** switch to 256-QAM and press +100. Errors appear at once, because the same noise now crosses boundaries that are four times closer on each axis. Press Up and watch the errors thin out; by our arithmetic they become rare somewhere past 30 dB.
- 4 Optional follow-up:** switch to Your text, drop the SNR until "Hello, Wi-Fi" arrives garbled, then open the EVM card. 16-QAM at 25 dB meets both 16-QAM limits and the 64-QAM rate 3/4 limit of -25 dB, and misses 64-QAM rate 5/6 at -27 dB (help, example).

What it leaves out on purpose: no coding (the coding rates appear only in the EVM limits table), no OFDM (one carrier, one symbol at a time), and only white Gaussian noise: no phase noise, fading or interference. The carrier strip draws 3 cycles per symbol; a real

carrier runs billions per second. EVM limits are transmitter accuracy, not receiver sensitivity.

Pairs with: Rate vs Range (where each MCS stops working), the MCS Index card in Course Handouts, and Fourier and FFT mode 4, which puts these same constellations on every subcarrier. The Teacher's Guide lists a Modulation Simulator lab under "From bits to rate". It is not built yet: [Deliverables/2026-09-27-teachers-guide-labs/](#) holds only the FSPL and Airtime Fairness labs.

Fourier and FFT

fourier-fft

Teaches: any signal is a sum of sines, an analyzer computes its spectrum from a finite chunk of samples so its settings change what you see, and a Wi-Fi receiver is itself an FFT analyzer whose bins are the subcarriers.

Controls

Control	What it does	Why it's there (the teaching point)
Mode (select): Waves, FFT, Swept vs FFT race, OFDM is an inverse FFT	Switches between the four modes. Default Waves.	Spec parts 1 and 2 build the lesson in that order: signal, measurement, analyzer, Wi-Fi. The selector became a dropdown at four modes.
<i>Waves mode</i>		
Start from (select): One sine, Square wave (odd harmonics), Two close tones, Tone plus weak neighbor. Your own mix appears once you edit.	Loads a preset signal.	The spec's presets. The square wave shows a shape built from sines; the last two set up the FFT lessons.
Show each sine behind the sum (switch)	Draws each component under the summed trace.	(inferred) Lets the class see the sum is only its parts.
Sine 1 to Sine 5 chips, Add a sine, remove (per sine)	Picks the sine you are editing, adds one (up to five), removes one.	Spec: "Add up to five sines; each gets a row with the three knobs."
Frequency slider, 100 to 5,000 Hz in 10 Hz steps	Frequency of the picked sine.	A sine has three knobs (spec lesson 1). The time trace and the spectrum move together.

Control	What it does	Why it's there (the teaching point)
Amplitude slider, 0 to 1.00 (1.00 = 0 dB full scale)	Amplitude of the picked sine.	Same, the second knob.
Phase slider, -180 to 180 degrees	Phase of the picked sine.	Same, the third knob. The time trace moves and the magnitude spectrum does not, which is a point worth making aloud. (<i>inferred</i>)
Play sound / Stop sound	Plays the sum through the speaker.	The spec allowed optional sound at audio frequencies. The frequencies are real, so the sound is what the plots show. It does not follow the phase sliders.
<i>FFT mode</i>		
Lesson (select): Free play, Weak neighbor 40 dB down, Tone between bins, Halve N	Sets the signal and analyzer for one lesson.	The spec's three lessons: rectangular leakage hides the weak tone and Blackman-Harris reveals it; a tone between bins reads low until flat top; two tones merge when you halve N.
Window (select): Rectangular, Hann, Hamming, Blackman-Harris (4-term), Flat top. Default Hann.	Applies the window before the FFT.	The window trades a clean floor against how sharp each peak is (spec lesson 3).
Sample rate (Fs) (select): 12.8, 25.6, 51.2, 102.4 kHz. Default 25.6 kHz.	Sets Fs.	Bin spacing is F_s / N and capture time N / F_s , so their product is always 1 (help).
FFT size (N) (select) 64 to 4,096, powers of two, default 256; Halve N, Double N	Sets N.	More samples give finer bins and a longer capture (spec lesson 3). The two buttons make it a one-press comparison.
Disclosures: ENBW, RBW, sidelobes, scalloping and The Wi-Fi receiver is an FFT analyzer	Extra readouts and the closing card.	The spec's "Wi-Fi bridge": 20 MS/s with $N = 64$ gives bins exactly 312.5 kHz apart and a 3.2 μ s frame.
<i>Swept vs FFT race mode</i>		
Span (select): 20, 50, 100 MHz, centered on 2450 MHz. Default 100.	Width the swept analyzer covers.	Sweep time is $k \times \text{Span} / \text{RBW squared}$ (spec, Keysight AN 150).

Control	What it does	Why it's there (the teaching point)
RBW (select): 3 MHz, 1 MHz, 300 kHz, 100 kHz, 30 kHz, 10 kHz. Default 100 kHz.	Resolution bandwidth.	The one knob that makes the swept analyzer slow: at 100 MHz, 0.25 ms at 1 MHz RBW, 25 ms at 100 kHz, 2.5 s at 10 kHz (spec tests).
Run length (select): 50, 200, 1,000 ms. Default 200.	How much time the race covers.	<i>(inferred)</i> Longer runs give more hops and pulses to count, so the caught/missed contrast holds up.
Run the race / Pause / Resume, Step	Replays the run slowed down, or steps it one fortieth at a time.	The spec calls this mode "the highest-value lesson". Replaying it slowly lets the room watch the lime diagonal miss hops. <i>(inferred for the replay itself)</i>
Scene: mains, oven sweep, camera disclosure: Mains frequency (60 Hz, 50 Hz), Microwave sweep width (near CW, 10, 20, 40 MHz; default 20), Video camera bandwidth (4, 8, 18 MHz; default 8), New scene	Changes the synthetic scene and draws new hops.	Sources conflict on how far an oven sweeps, and the camera width is not pinned, so the spec made both parameters. 50 Hz is for classes outside North America. <i>(inferred for the 50 Hz reason)</i>
<i>OFDM is an inverse FFT mode</i>		
View: Teaching (16 subcarriers), Real Wi-Fi (20 MHz)	Small N for visibility, then the full channel.	Spec: a small N such as 16 to start, then real Wi-Fi.
Subcarrier chips (-8 to +7), All on, All off (Teaching view)	Turns subcarriers on and off. Default 1, 2 and 3 on.	The student builds the symbol one subcarrier at a time (spec).
Modulation on every subcarrier (select): BPSK to 1024-QAM. Default QPSK.	The constellation each subcarrier carries.	Reuses the Modulation Simulator's math, so the two tools agree (spec).
New data	Picks new random points for every subcarrier.	<i>(inferred)</i> Shows the recovered constellation matches whatever was sent, not one lucky pattern.
Highlighted subcarrier slider	Picks the subcarrier whose sinc is drawn bold.	The orthogonality lesson: every sinc is zero at every other subcarrier's center (spec).

Control	What it does	Why it's there (the teaching point)
Numerology: Legacy, HE; Guard interval (select, HE: 0.8, 1.6, 3.2 μs)	Switches legacy 312.5 kHz / 3.2 μ s to HE 78.125 kHz / 12.8 μ s.	Spec: the symbol stretches four times, and the screen must say plainly that the carrier frequency does not change.
Same time scale for legacy and HE (switch, under Time scale and the receiver)	Keeps both numerologies on one time axis. Default on.	<i>(inferred)</i> So the 4x longer HE symbol looks 4x longer instead of being rescaled to fit.

Present mode: 1 to 4 switch modes in every mode. The other keys follow the mode.

- Waves: Space plays or stops the sound, R goes back to One sine, Up and Down move the picked sine's frequency.
- FFT: R resets the analyzer, Up doubles N and Down halves it, W steps to the next window.
- Race: Space runs or pauses, Right steps, R rewinds, Up and Down step the RBW.
- OFDM: Right moves the highlight to the next subcarrier, R resets, Up and Down step the modulation.

A demo that works (about 3 minutes)

- 1 Set up:** FFT mode, Lesson "Weak neighbor 40 dB down". The rectangular window is on and the spectrum shows one tone.
- 2 Ask the room:** "There are two tones in this signal. Where is the second one?"
- 3 Reveal:** press W (or pick Blackman-Harris). The leakage skirt drops and the weak neighbor appears 40 dB down.
- 4 Optional follow-up:** press 3 for the race at its defaults (100 MHz span, 100 kHz RBW, 25 ms sweep). Ask which analyzer will catch more Bluetooth hops, then Space to run it and read the Caught and Missed counts.

What it leaves out on purpose: nothing aliases, because every tone stays below half the lowest sample rate. The race scene is synthetic, from published timings: Bluetooth hops are uniform random (the real sequence is pseudo-random), and the FFT analyzer is ideal and gapless, where a real one has a real-time bandwidth limit. Real swept analyzers with digital RBW filters sweep faster than the formula. In the OFDM teaching view the same 16

subcarriers are kept when you switch to HE, so they span a quarter of the width. Real HE fills the channel with four times as many.

Pairs with: the Spectrum Analysis guided lesson (the race reuses its interferer fingerprint wording), Modulation Simulator for mode 4, and PHY Preamble Reference for where the symbols sit in a PPDU. The Teacher's Guide lists a Fourier and FFT lab under "Spectrum analysis". It is not built yet.

PHY Preamble Reference

phy-preamble

Teaches: every PPDU since 802.11a starts with the same 20 μ s legacy preamble so older radios defer; what follows is where each PHY differs, and it is where a capture's PHY details come from.

Controls

Control	What it does	Why it's there (the teaching point)
Mode: Explore, Which PHY?	Explore opens the fields; Which PHY? walks a receiver's decision.	Spec lessons 1 to 3 are Explore; lesson 4 (how a receiver tells the PHYs apart) is Which PHY?.
PPDU type (select): Legacy (non-HT), HT mixed (802.11n), VHT (802.11ac), HE SU, HE ER SU, HE MU, HE TB (802.11ax), EHT MU, EHT TB (802.11be). Default HE SU.	Draws that PPDU's preamble to scale in μ s.	Spec: the preamble for each PHY as labeled blocks, sand for the legacy part, blue for what the PHY adds.
Blocks on the bar, and the field list under it; Close	Opens a block's bit table (bit range, field, width, meaning, evidence tag).	Spec lesson 3: what is inside each SIG field, bit by bit, with the evidence level shown so single-source fields are never taught as settled.
Spatial streams (N_STS) slider: 1 for Legacy, 1 to 4 for HT, 1 to 8 for VHT, HE and EHT. Default 1.	Adds or removes LTFs.	The preamble grows with streams, one LTF per stream group (spec controls list).

Control	What it does	Why it's there (the teaching point)
LTF size and guard interval (select, HE and EHT only): 2x 0.8 μs GI, 2x 1.6 μs GI, 4x 0.8 μs GI, 4x 3.2 μs GI	Changes each HE or EHT LTF's length.	Spec: GI and LTF size for HE. 1x is not offered, because the brief's EHT list does not include it (help).
HE-SIG-B / EHT-SIG symbols slider, 1 to 12, set by the AP (HE MU and EHT MU only). Default 2.	Lengthens the SIG-B or EHT-SIG field.	The count depends on the users and the SIG MCS, so it is a setting, not computed (help).
Channel width (select): 20, 40, 80, 160 MHz	Changes only the VHT-SIG-B layout.	Spec: bandwidth "where it changes" the preamble. Durations per field do not change with width, and this shows that. <i>(inferred for the second half)</i>
Next step / Walk complete, Back, Reveal, Reset (Which PHY?)	Asks the receiver's next question, takes one back, shows every question and the verdict, starts over.	Spec: walk the decision tree step by step (BPSK vs QBPSK, the RL-SIG repeat, LENGTH mod 3).
Mystery (Which PHY?)	Picks a PPDU at random and hides its name until the walk ends.	Built for predict, then reveal (help, <code>howToUse</code>).
L-SIG LENGTH calculator disclosure: True PPDU duration, TXTIME (μs) field (default 200), 2.4 GHz: 6 μs signal extension after the data switch, This preamble + 100 μs of data	Shows the spoofed LENGTH, m, LENGTH mod 3, and how long a legacy radio counts and defers.	Spec lesson 5: how L-SIG LENGTH is set so a legacy radio defers for the whole newer PPDU.
Readouts: L-SIG and BSS color and Sources and evidence tags disclosures	L-SIG RATE, LENGTH mod 3, where the BSS color sits for this type; the evidence key.	The BSS color moves with the HE PPDU type and is fixed in EHT (help). <i>(inferred that this is why it gets its own readout)</i>

Present mode: Right opens the next block (in Which PHY?, asks the next question), R closes the block (in Which PHY?, starts the walk again), Up and Down change the PPDU type. No Space.

A demo that works (about 3 minutes)

- 1 **Set up:** Which PHY?, then press **Mystery**.
- 2 **Ask the room:** "The first symbol after L-SIG is a BPSK copy of L-SIG. What have we narrowed it to?"
- 3 **Reveal:** press Right through the questions. An RL-SIG repeat means HE or EHT, and LENGTH mod 3 splits them: 1 is HE SU or TB, 2 is HE MU or ER SU, 0 is EHT (help, algorithm).
- 4 **Optional follow-up:** Explore, HE SU, 2 streams, 2x LTF with 0.8 μ s GI: 50.4 μ s of preamble. Open the L-SIG LENGTH calculator at 200 μ s: LENGTH 130, mod 3 = 1, and a legacy radio defers exactly 200 μ s.

What it leaves out on purpose: bit positions come from open-source driver and receiver code and secondary sources, not a reading of the standard, and every field carries its tag. Unsettled fields are marked on screen (HE TB HE-SIG-A, EHT-SIG positions, LTF counts above 4 streams and others listed in the help). The packet extension and the 2.4 GHz signal extension come after the data and are not drawn. SIG-B symbol counts are a setting, not computed.

Pairs with: Airtime Anatomy (the same preamble durations, in a whole TXOP) and A Frame's Journey (what comes after the preamble). No lab handout yet.

The Number on the Box vs the Number in Your Hand

box-vs-hand

Teaches: a router's class number adds every radio at its maximum, one client uses one link with its own streams, and distance takes the rate lower again.

Controls

Control	What it does	Why it's there (the teaching point)
Show the next step / Back a step (on the stage)	Walks the three bars: On the box, Best case for this client, At a stated distance and width. Each new bar starts at the length of the one above and shrinks.	Pax's brief: "Visible change: a bar shrinks from the box number to what that client can reach at best."
Client: phone and laptop are 2x2: Phone, Laptop, 4-stream. Default Phone.	Picks the client's stream count.	The brief's one control, and Keith's standing ruling that throughput defaults to a 2x2 client (Apple and Google publish 2x2 for current phones). Phone and laptop give identical bars on purpose: the stream count sets the ceiling (Felix build log). The 4-stream reference still reaches only the one fastest link.
Distance from the router slider, 1 to 60 m (3 to 197 ft), default 5 m (16 ft)	Sets step 3's distance.	Step 3 of the lesson: the signal weakens, the MCS steps down, the rate follows. The 5 m default is illustrative, and the screen says so.
Channel width, 6 GHz (MHz): 20, 40, 80, 160, 320. Default 320.	Sets step 3's width.	A wider channel needs a stronger signal for the same MCS, so far enough out a narrower one wins (help).
Reset	Back to the box, the phone, 5 m and 320 MHz.	Starts the walk over.
Open Throughput Calculator, Open MCS Index, Open Rate vs Range (under Why the numbers differ)	Opens the tools whose math this one reuses.	The build added no table of its own. Rates come from the Throughput Calculator and level and MCS from Rate vs Range (Felix build log), so the class can check every number there.
Metric / imperial switch (title bar)	Changes the distance units.	Metric first, imperial for US rooms. <i>(inferred)</i>

Present mode: Right shows the next step, Left the step before, Space jumps between the box and the hand, Up and Down change the client, R resets.

A demo that works (about 3 minutes)

- 1 **Set up:** the defaults. The top bar reads 18,656 Mbps, the sum of the three radios on a BE19000-class box.
- 2 **Ask the room:** "My phone is Wi-Fi 7. How much of that does it get, standing next to the router?"
- 3 **Reveal:** Right. The best case for a 2x2 phone is 5,765 Mbps, 31% of the box. Right again: at 5 m on 320 MHz, MCS 7 and an estimate of 2,306 Mbps, 12% of the box (help, example).
- 4 **Optional follow-up:** slide to 20 m and compare 320 MHz with 160 MHz. The narrower channel wins (help: MCS 2 at 346 Mbps against MCS 0 at 231 Mbps). Then switch to Laptop and ask why nothing moved.

What it leaves out on purpose: the best case assumes the client supports 320 MHz and 4096-QAM. The 0.80 factor is the Throughput Calculator's favorable estimate, so a speed test usually reads lower. Other clients sharing airtime, walls, interference and Multi-Link Operation are left out, and the client keeps all its streams at a distance, so step 3 is an upper limit. The router vendor is not named, per the house rule on vendor names (Pax's brief).

Pairs with: MIMO and Beamforming (why the smaller side sets the streams), Rate vs Range, the MCS Index card in Course Handouts, and Multi-Link Operation. No lab handout yet.

OFDMA Resource Units

ofdma-simulator

Teaches: OFDMA splits one channel into resource units so the AP serves several clients in one transmission, and the saving for small frames comes from paying for contention, the preamble and the acknowledgment once, not from a faster PHY.

Controls

Control	What it does	Why it's there (the teaching point)
Direction: Downlink, Uplink. Default Downlink.	Picks which OFDMA timeline the readouts compare with SU.	Spec: DL OFDMA and UL OFDMA cost different overhead (UL adds a trigger frame and a Multi-STA BlockAck).
Channel width (select): 20, 40, 80, 160 MHz. Default 20.	Sets how many RUs fit.	The spec's RU table: counts per width.
Clients (select): 1 to 9 at 20 MHz, up to 18 at 40 MHz and wider. Default 4.	Number of clients, each with one frame.	Spec lesson 2 needs several clients; spec lesson 3 needs one.
Frame size per client (select): 64, 100, 200, 500, 1,000, 1,500 bytes. Default 200.	The frame each client gets.	Small frames are where OFDMA wins; large frames to one client gain little (spec lessons 2 and 3).
MCS (every client) (select): MCS 0 to 11 with modulation and coding rate. Default MCS 7.	One MCS for every client.	The defaults reproduce the brief's worked example (model comment: "The research brief's worked example"). MCS 10 and 11 need a 242-tone RU or larger.
Client (select: All clients, or Client A, B ...) and RU size (select: 26, 52, 106, 242, 484, 996, 2x996 tones, with data + pilot and how many fit). Default four 52-tone RUs.	Sets one client's RU, or every client's.	Spec: RU size per client, with the tone plan enforcing valid placements.
Largest equal RUs	Gives every client the biggest RU that fits.	<i>(inferred)</i> A one-press fair split, so the room is not placing RUs by hand.
Re-pack	Moves every RU to the low edge.	<i>(inferred)</i> Tidies the strip after drags.
Move left, Move right	Moves the selected client's RU along the channel.	The keyboard and button route for the spec's RU puzzle.
Drag a client on the channel strip, or tap it then tap a dashed outline	Places an RU. The dashed outlines are the free spots the tone plan allows.	Spec: "the user drags clients into RUs across the channel; the tone plan enforces valid placements".
Show the arithmetic / Hide the arithmetic	Shows every segment's formula. Estimates carry an Assumption tag.	Spec: every INF estimate is labeled on screen, never shown as a standard value.

Present mode: Right selects the next client, Up and Down change the number of clients. No Space, no R.

A demo that works (about 3 minutes)

- 1 **Set up:** the defaults. Four clients, 200 bytes each, 20 MHz, MCS 7, four 52-tone RUs.
- 2 **Ask the room:** "OFDMA is sold as faster. Is the data itself going faster?"
- 3 **Reveal:** read the timelines. SU takes 915.6 μ s for four TXOPs; DL OFDMA takes 403.3 μ s, about 2.2 times less. The data takes 13.6 μ s *longer* on the narrower RUs. The saving is three fewer contentions, three fewer preambles and fewer acks (help, example).
- 4 **Optional follow-up:** press Down to 1 client, pick 1,500 bytes and a 242-tone RU. OFDMA comes out slightly slower than SU: the longer preamble and triggered ack cost time and there is no one to share them with.

What it leaves out on purpose: one frame per client, no collisions or retries, the average backoff, best effort, one spatial stream, control frames at 24 Mbps. The HE-SIG-B length, the block-ack rates, the trigger frame size and the Multi-STA BlockAck size are estimates, each labeled. The strip is in tone-plan order with every 26-tone slot the same width, without guard, DC or leftover tones. Scheduling, buffer status reports and sounding are not modeled.

Pairs with: Airtime Anatomy (the SU timing comes from its service), Medium Access Simulator (what contention costs), then Spatial Reuse and Multi-Link Operation in the "Wi-Fi 6 and 7 features" sequence. The Teacher's Guide lists an OFDMA lab there; it is not built yet.

A Frame's Journey

frame-journey

Teaches: one wrong bit anywhere in a frame fails the FCS, the receiver drops it without a reply, and the sender learns about the loss only from the ACK that never comes. Radiotap is added by the capturing receiver and never travels on the air.

Controls

Control	What it does	Why it's there (the teaching point)
Play / Pause, Back, Step, Reset	Moves through the hop one step at a time. The top card says what happens at each step.	<i>(inferred)</i> One step per event (bits to RF, the FCS check, SIFS, the ACK, the retry) so you can stop and ask at each one.
Step slider (1 of n)	Jumps to any step.	<i>(inferred)</i> For going straight back to the FCS check.
Ask the class, answer buttons An ACK, A request to send it again, Nothing, then Reveal (card: Predict, then reveal)	Loads the question "One bit of the frame arrives wrong. What does the receiver send back?" with a bit corrupted, records the room's pick, and jumps to just after the FCS check.	The predict prompt proposed for this tool in the build questions to Keith (C16). "A request to send it again" is there because 802.11 has no negative acknowledgment to send. <i>(inferred for that distractor)</i>
Band: 2.4 GHz, 5 GHz, 6 GHz. Default 5 GHz.	Sets channel 6 (2437 MHz), 36 (5180 MHz) or 37 (6135 MHz).	Channel and signal level in the radiotap header follow the band. The SIFS differs at 2.4 GHz (10 μ s plus a 6 μ s signal extension) and is 16 μ s at 5 and 6 GHz (Pax's source pins).
Distance slider, 1 to 30 m (3 to 98 ft), default 8 m (26 ft)	Moves the receiver. The wave's height and the signal level drop; the wavelength drawn never changes.	Shows distance changes amplitude, never frequency (help).
Corrupt one bit on the first attempt (switch) and Which bit slider, 0 to the frame's last bit, default bit 133	Flips one bit and names the field it lands in.	Any single bit fails the check, even one inside the FCS itself. The build tests flip every bit of a frame one at a time to prove it (Felix build log).
Receiver is capturing (radiotap) (switch), default on	Adds or removes the radiotap header on the receiving side.	Radiotap is a pseudo-header added by the capturing driver and not sent over the air (Pax's source pins). The on-air frame is the same either way.
Transport: TCP, UDP. Default TCP.	Changes the transport header.	<i>(inferred)</i> Changes the frame's size and bytes, so the FCS changes with it.
Data (select): 100, 500, 1460 bytes. Default 1460.	Sets the payload.	<i>(inferred)</i> Same, and 1460 bytes is a full TCP segment.

Control	What it does	Why it's there (the teaching point)
Model notes	What is real and what is illustrative.	Answers "is it really like that?"
Metric / imperial switch (title bar)	Changes the distance units.	<i>(inferred)</i>

Present mode: Space plays or pauses, Right steps, R resets, Up and Down move the receiver 1 m (about 3 ft), C corrupts a bit or sends the frame clean, B picks the next band.

A demo that works (about 3 minutes)

- 1 Set up:** the defaults (5 GHz, 8 m, TCP, 1460 bytes). Step through once, clean: the frame is 1538 bytes, it arrives at -44.8 dBm, the FCS matches, and 16 μ s later the ACK goes back.
- 2 Ask the room:** press **Ask the class**. "One bit arrives wrong. What does the receiver send back?" Take a show of hands for each answer and tap the winner.
- 3 Reveal:** press Reveal. "Nothing." The receiver cannot trust any part of a frame that fails its FCS, not even the sender's address, so it has no one to reply to. Step on to the retry: the Retry bit is set, the FCS is new, and it gets through.
- 4 Optional follow-up:** turn **Receiver is capturing** off and ask what changed on the air. Nothing did. Point out the trip across takes 26.7 ns and the SIFS is 16,000 ns.

What it leaves out on purpose: the FCS is a real CRC-32 over this frame's bytes, but the Duration, sequence number, MAC addresses and data are illustrative, as are the radiotap timer and MCS. Not modeled: the preamble, how bits are coded onto subcarriers, the ACK timeout, the backoff before the retry, retry limits, block acknowledgment for aggregated frames, and encryption. The timeline is not to scale.

Pairs with: Down the Stack (the same air frame, built by the same function, opened layer by layer), PHY Preamble Reference (what precedes the frame), and Airtime Anatomy (what each retry costs in airtime). No lab handout yet.

MIMO and Beamforming

mimo-beamforming

Teaches: streams stop at the smaller side, the spare chains still add gain in each direction, beamforming costs sounding airtime, and a nearby sniffer is not the device the AP is aiming at, so it misses beamformed frames the client received fine.

Controls

Control	What it does	Why it's there (the teaching point)
AP chains (select): 1, 2, 3, 4, 8. Default 4.	AP antenna chains.	Keith's lesson 1 (spec): a 4x4 AP and a 2x2 client talk at 2 streams. 8 is the spec's optional large AP.
Client chains (select): 1 to 4. Default 2.	Client antenna chains.	Same. Clients stop at 4.
Swap	Swaps the two counts.	Keith's "vice versa" case (spec lesson 2): flip to a 2x2 AP with a 4x4 client, and now the client has the spare chains.
Downlink / Uplink (on the stage, above the streams picture)	Shows one direction. Spare chains that help are drawn dashed into the lanes.	Spec lesson 2: spare AP chains give transmit beamforming gain downlink and receive combining gain uplink. The table under the stage shows both side by side.
AP transmit beamforming: On, Off. Default On. Disabled with 1 AP chain.	Steers the beam pattern's main lobe at the client.	Spec: "beamforming on/off". A 1-chain AP cannot beamform.
Client angle and Sniffer angle sliders, 80 degrees left to 80 right (default client 20, sniffer -35); or drag either in the pattern	Moves the client and the sniffer around the AP.	Spec lesson 4: the sniffer's level is the pattern value at its angle compared with the client's.
Sniffer chains (select): 1 to 4. Default 1.	The sniffer's receive chains.	Spec lesson 4: a sniffer with fewer chains than the streams sent cannot separate them.
Channel width and sounding interval disclosure: Channel width (select: 20, 40, 80, 160 MHz; default 80) and Sounding interval slider (2, 5, 10, 20, 50, 100, 200 ms; default 10)	Draws the NDPA, NDP and report exchange to scale and its share of airtime. Disabled when beamforming is off.	Spec lesson 3: beamforming is not free. Width is there to show it barely matters here: the report grows, but it also travels over the wider channel (help).
Both directions side by side, Sounding, in detail, Our measurement, in full disclosures	The full tables and the capture figures.	Spec: the capture numbers are shown as a cited measurement, not a model output.

Present mode: Up and Down steer the client 5 degrees per press, R resets. No Space, no Right.

A demo that works (about 3 minutes)

- 1 **Set up:** the defaults (4x4 AP, 2x2 client, beamforming on, sniffer at 1 chain).
- 2 **Ask the room:** "Four antennas on the AP, two on the phone. How many streams, and are the other two AP antennas wasted?"
- 3 **Reveal:** 2 streams both ways. Downlink, the AP's 2 spare chains give up to +3.0 dB of beamforming gain; uplink, the same 2 chains give up to +3.0 dB of combining gain. Press Swap and ask who has the spare chains now.
- 4 **Optional follow-up:** press Up a few times to swing the beam toward the client and watch the sniffer's relative level fall. Point at "cannot decode" for the 1-chain sniffer, then open **Our measurement, in full:** beamformed downlink frames failed their FCS 50.8% of the time against 5.1% for non-beamformed frames at matched signal strength.

What it leaves out on purpose: the gains are ideal upper bounds. The beam pattern is line of sight, one steered stream, from a straight line of elements; indoors reflections make real patterns less clean. The client does not beamform (in our capture, 93% of the client's frames were not beamformed). Sounding figures are estimates for one client, without the wait for the medium and at a fixed report rate.

Pairs with: The Number on the Box vs the Number in Your Hand (the streams ceiling, as a rate), OFDMA Resource Units (sounding is one of the things it does not model), and the beamforming capture study in [Deliverables/2026-08-19-beamforming-blinds-the-sniffer/](#) for the figures behind the capture lesson. No lab handout yet.

Airtime and Access

This shelf is about time on the air: who gets to talk, how long each turn takes, and what quietly eats the turns before any of your data moves. The twelve simulators run from the contention rules themselves (Medium Access Simulator) through the cost of one frame (Airtime Anatomy) to the things that waste airtime in the field: slow clients, lost QoS marks, interferers, multicast, old 802.11b devices, sleeping clients, and the Wi-Fi 6 and 7 features meant to win some of it back. Every tool also has Present and Copy in its title bar, and in Present mode F switches full screen, ? lists the keys and Esc exits; the tables below list only what each tool adds.

Entries follow the order of the shelf in the app.

Medium Access Simulator

medium-access-simulator

Teaches: Wi-Fi is half duplex and shared, so stations take turns by counting down a random backoff; collisions double the contention window, more stations mean more waiting, and EDCA lets voice win because it waits less.

Controls

Control	What it does	Why it's there (the teaching point)
Play / Pause	Runs or freezes the simulated clock. The tool always opens paused at time zero.	The spec requires it to open paused and keep Step working (reduced motion), so the instructor decides when the clock starts.
Step 1 slot	Advances one 9 μ s slot.	Backoff counts down one slot at a time and freezes when the medium goes busy (spec rules 2 and 3). Stepping lets the room call each count and catch the freeze and resume.
Reset	Replays the same run from time zero.	Runs are seeded, so the same settings give the same run. A demo you rehearsed plays the same collisions live. (<i>inferred</i>)

Control	What it does	Why it's there (the teaching point)
Speed (air time per real second)	0.1, 0.5, 2, 20 or 200 ms per second; opens at 0.1 ms per second.	The slow end shows single slots; the fast end lets the throughput and collision numbers settle. <i>(inferred)</i>
Zoom: Slots / Frames / Wide	Changes the timeline's time scale. While paused, drag the timeline sideways to look back.	Slots shows the countdown cells; Wide shows many frames so the pattern of turns is visible. <i>(inferred)</i>
Add station / Remove	1 to 10 stations; opens with 3. Each station row also has its own remove button.	"More stations mean more waiting and more collisions" is a stated learning goal of the spec.
Access category (per station)	Voice, Video, Best effort or Background. Only live in EDCA.	The spec's EDCA table gives Voice AIFSN 2 and CW 3 to 7 against Best effort's AIFSN 3 and CW 15 to 1023: priority traffic wins because it waits less.
Offered load (per station)	Always has a frame, or 50, 200, 500, 1,000 or 2,000 frames/s with random arrivals.	Saturated stations show contention at its worst; light loads show how rarely it bites when the channel is quiet. <i>(inferred)</i>
Per-station table	CW, backoff, delivered and dropped for every station.	The spec asks for current CW per station, so the class can watch 15 become 31, then 63, after each collision.
Channel access: Legacy DCF / EDCA	DCF gives everyone DIFS and CW 15 to 1023; EDCA applies each station's access category.	The spec builds a DCF mode for everyone as the baseline, then EDCA to show voice getting through first.
Hidden node	Stations A, C, E cannot hear B, D, F; everyone still reaches the AP. Needs two or more stations.	Shows collisions nobody could hear coming: the stations never freeze on each other, so frames overlap at the AP (spec).
RTS/CTS	Every exchange becomes RTS, CTS, data, ACK; anyone who hears the CTS sets its NAV.	The spec's fix for the hidden node.

Control	What it does	Why it's there (the teaching point)
Frame size	100, 500, 1,000, 1,500 or 2,304 bytes; default 1,500.	The spec makes frame size and PHY rate user controls so the class sees airtime per frame change. The help says the same.
PHY rate	6 to 54 Mbps (legacy OFDM); default 54.	Same as frame size. The Frame card underneath shows the airtime formula with your numbers in it.

Present mode: Space plays or pauses, Right arrow steps one slot, R resets, Up and Down change the speed. In Present, Zoom moves onto the stage and the per-station table folds behind "Per-station table".

A demo that works (about 3 minutes)

- 1 Set up:** the defaults. Three stations, Legacy DCF, 1,500 bytes at 54 Mbps, speed 0.1 ms per second.
- 2 Ask the room:** "Two stations reach zero in the same slot. What happens to both frames, and what does each station do next?"
- 3 Reveal:** press Right until a red collision block appears, then open the per-station table: both colliding stations' CW has gone from 15 to 31. Press Up a few times to speed up, Play, and read the collision rate and throughput. Add stations up to 10 and watch the collision rate climb and throughput fall. The help puts one station alone at about 31 Mbps delivered on a 54 Mbps link.
- 4 Optional follow-up:** switch to EDCA, set station A to Voice and the rest to Best effort, and compare average access delay. Or turn on Hidden node, watch the collisions, then turn on RTS/CTS.

What it leaves out on purpose: timing is legacy OFDM (802.11a/g, 9 μ s slots, 20 μ s preamble); newer PHYs have longer preambles and aggregate frames, and the contention lesson is the same. Uplink only: the AP sends only ACK or CTS. No rate adaptation, no aggregation, no NAV reset rule. RTS/CTS removes most hidden-node collisions, not all: a hidden station whose RTS overlaps the AP's CTS never hears that CTS.

Pairs with: Voice Priority, End to End (runs on this engine), Airtime Anatomy, Channel Utilization Meter, Rate Adaptation. Teacher's Guide sequences "Why is the Wi-Fi slow?"

and "QoS and contention".

Voice Priority, End to End

voice-priority

Teaches: a call gets the Voice queue on Wi-Fi only if its EF marking survives every hop before the AP and the AP maps it to UP 6; turning on WMM alone does not do it.

Controls

Control	What it does	Why it's there (the teaching point)
Send the packet / Pause	Moves the voice packet along the path toward the phone.	Lets the class watch the mark each hop passes on. <i>(inferred)</i>
Next hop	Moves the packet one hop.	Stops the packet at the hop that says "Marking lost here" so you can talk about that device. <i>(inferred)</i>
Reset	Back to the defaults, packet at the caller, no question showing.	Returns to the starting state between demos. <i>(inferred)</i>
Where the marking is lost: Nowhere / At the AP mapping / At the tunnel / At the internet provider	Picks the hop that resets EF to 0. A dropdown on a phone, where four segments truncate.	Pax's research brief made this "the one control": the visible change is the call dropping from Voice to Best effort while a download runs. It corrects "turning on WMM makes my calls good" and "QoS works across the internet".
How the AP maps DSCP to user priority: RFC 8325 table (recommended) / Top three bits (older default)	RFC 8325 maps EF to UP 6 (Voice); top three bits turns EF (46) into UP 5 (Video).	RFC 8325 notes that the common default of copying the top three bits puts EF in the Video queue, and recommends UP 6. This shows the call landing in the wrong queue with nothing on the path at fault.
A download is running	On by default.	With the download off, the four queues come out within a tenth of a millisecond of each other (help). Priority only matters when something else is queued.

Control	What it does	Why it's there (the teaching point)
Download frames ahead in Best effort (illustrative)	0 to 256 in steps of 8; default 64. Live only when the call lands in Best effort with the download on; locked while the class predicts.	The queue depth depends on the AP's buffer, so the help labels it illustrative. The builder's default is 64 frames $\times$ 0.398 ms = 25.9 ms against 0.48 ms in Voice.
Ask the class / Voice / Video / Best effort / Reveal / close	Loads the question (the provider resets the mark while a download runs), hides the answer, takes a guess, reveals.	The house "predict, then reveal" pattern, built into the tool.
Open Medium Access Simulator / Open DSCP / QoS Markings	Links to the related tools.	The brief said not to rebuild EDCA (the Medium Access Simulator shows it) and that the DSCP / QoS card holds the RFC 8325 table.

Present mode: Space sends the packet or pauses, Right arrow moves it one hop, R resets, Up and Down move where the marking is lost, D starts or stops the download, M switches the AP mapping, P asks, reveals and closes the question. "Model notes" folds away in Present.

A demo that works (about 3 minutes)

- 1 Set up:** the defaults. Marking lost Nowhere, RFC 8325 mapping, download running.
- 2 Ask the room:** press P. "The phone has WMM on and the app marks its voice EF. Your provider resets the mark to 0. While a download runs, which queue does the call wait in?" Take votes on Voice, Video and Best effort.
- 3 Reveal:** press P again. Best effort, behind the download: about 25.9 ms of waiting at the default 64 frames, against 0.48 ms in Voice.
- 4 Optional follow-up:** press P to close, set the loss back to Nowhere and press M. The call lands in Video with nothing on the path losing the mark. Then press D to stop the download and show the queues come out nearly equal.

What it leaves out on purpose: the Wi-Fi hop uses the Medium Access Simulator's simplifications (legacy 54 Mbps timing, one frame size, no aggregation, default client EDCA settings). The download and the call contend as two transmitters, not two queues inside one AP. Whether a tunnel copies the mark outward depends on the device (RFC 4301).

Wired hops add delay the tool does not number. The upstream direction, where the phone picks the UP itself, is not modeled. The voice packet every 20 ms and the download queue depth are illustrative.

Pairs with: Medium Access Simulator, the DSCP / QoS Markings card, Power Save (U-APSD and the voice-call traffic pattern).

Airtime Anatomy

airtime-anatomy

Teaches: a PHY rate is only the speed of the data symbols; every transmission also pays for waiting, a preamble and an acknowledgment, and aggregation recovers the airtime that a faster MCS cannot.

Controls

Control	What it does	Why it's there (the teaching point)
Time / Structure (on the stage)	Switches between the TXOP drawn to scale in μ s and the inside of the PSDU.	The help says "aggregation is not one thing": the Structure view shows why an A-MSDU and an A-MPDU fail differently.
Compare with scenario B	Stacks a second TXOP on the same scale. On by default.	The spec: "Up to two scenarios stacked for comparison (A and B), because comparing is the lesson."
Edit scenario: A / B	Picks which scenario the inputs change.	Needed once there are two scenarios.
Preset buttons: Legacy 6 Mbps, VHT one frame, VHT 32 aggregated, HE 32 aggregated	Loads one of the four scenarios. Opens with A = Legacy 6 Mbps and B = HE 32 aggregated.	These are the four defaults of the WLAN Pros Airtime Calculator the tool ports (spec 05), and its tests reproduce their totals: 2,242.5, 214.5, 666.5 and 548.9 μ s.
Band: 2.4 / 5 / 6 GHz	Applies the band rules: 2.4 GHz adds a 6 μ s signal extension; 6 GHz is HE only; VHT is 5 GHz only.	Band is one of the things the spec says the student should see change the share of air that carries data.
PHY	Legacy (a/g), HT (n), VHT (ac), HE (ax, single-user).	The generations differ in preamble length and in data bits per symbol. The spec lists PHY as a per-scenario control.

Control	What it does	Why it's there (the teaching point)
Channel width	20, 40, 80 or 160 MHz, limited by PHY.	Named in the spec's learning goal.
Legacy data rate / MCS (per stream)	6 to 54 Mbps for Legacy; MCS 0 to 11 otherwise, limited by PHY.	Named in the spec's learning goal: a faster MCS shrinks only the data block.
Spatial streams	1 to 8 (HT up to 4).	More streams shrink the data block and leave the fixed costs alone, the same lesson as MCS. <i>(inferred)</i>
Guard interval	HT and VHT 0.4 or 0.8 μ s; HE 0.8, 1.6 or 3.2 μ s.	Named in the spec's learning goal.
Payload per frame	64 to 2,304 bytes; default 1,500.	Small frames pay the same fixed costs for less data. <i>(inferred)</i>
Frames aggregated (A-MPDU)	1 to 64; Legacy sends 1.	Named in the spec's learning goal, and the headline of the tool: aggregation recovers airtime.
More settings / Fewer settings	Shows or hides the five less-used settings below.	The spec: "Group the less-used ones behind More."
Encryption overhead: 0 / 16 bytes	Adds CCMP/GCMP bytes to every frame.	Ported from the calculator sheet. Shows that security costs a little airtime too. <i>(inferred)</i>
Access category	Voice, Video, Best effort, Background (default client EDCA values).	Changes AIFS and the average backoff, the wait at the front of every TXOP. The Teacher's Guide QoS sequence uses it. <i>(inferred)</i>
RTS/CTS protection	Reserves the medium before the data frame.	Shows what the handshake costs on a clean channel. <i>(inferred)</i>
Control frame rate: 6 / 12 / 24 Mbps	Rate for RTS, CTS, ACK and Block Ack.	Shows how much of the fixed cost is control frames sent at a legacy rate. <i>(inferred)</i>
HE packet extension	0, 4, 8, 12 or 16 μ s, HE only.	Ported from the calculator sheet; shows a padding cost that carries no data. <i>(inferred)</i>

Control	What it does	Why it's there (the teaching point)
Tap or hover a segment / duration table	Shows the segment's duration, its share of the TXOP and the formula with your numbers in it.	The spec asks for each segment's duration and the formula that produced it.
Arrangement (Structure view)	Single MPDU, A-MSDU, A-MPDU or A-MPDU of A-MSDUs; Legacy offers only a single MPDU.	An A-MSDU shares one header and one FCS, so one bad bit loses them all; an A-MPDU gives each subframe its own, so Block Ack names the failed one (help).
MSDUs per A-MSDU: 2 / 3 / 4	How many packets share one header.	A teaching choice, not a limit from the standard (help).
Corrupt one MSDU, the arrow buttons, or tap a lime block	Damages one MSDU and shows the Block Ack bitmap, what is resent in bytes and time, and all four arrangements side by side.	Makes the A-MSDU versus A-MPDU difference visible as a resend cost. (help)

Present mode: Right arrow walks the TXOP one segment at a time (in Structure, moves the damage to the next MSDU), Up and Down change the rate of the scenario being edited, V switches Time and Structure, A steps through the arrangements, C corrupts one MSDU or repairs it. No Space and no R: nothing plays. The segment table and all readouts fold behind one disclosure.

A demo that works (about 3 minutes)

- 1 Set up:** scenario A on "VHT one frame", scenario B on "VHT 32 aggregated", compare on.
- 2 Ask the room:** "One 1,500-byte frame at 867 Mbps. What throughput does it deliver?"
- 3 Reveal:** press Right to walk scenario A one segment at a time. The lime data block is a sliver between the wait, the preamble and the ACK. The help puts it at about 56 Mbps. Scenario B carries 32 frames in 666.5 μ s against A's one frame in 214.5 μ s.
- 4 Optional follow-up:** press V for Structure, A to reach A-MSDU, C to corrupt one MSDU: the whole A-MSDU is resent. Press A to reach A-MPDU: only the one subframe is resent.

What it leaves out on purpose: one station, no contention, no retries; backoff is the average of a draw from 0 to CWmin. HT and VHT use one BCC encoder; HE uses LDPC with pre-FEC padding ignored, so the symbol count can be one short. Single-user HE only, not OFDMA. The Check line misses some VHT MCS exclusions. EHT (Wi-Fi 7) is not modeled. A resend is timed as preamble plus data only, without the new wait or the larger contention window.

Pairs with: Airtime Fairness, Multicast at the Basic Rate, Legacy Protection Cost and Channel Utilization Meter (all built on this timing), the MCS Index tool for unusual combinations, Multi-Link Operation (its frame airtime option). Teacher's Guide sequence "Why is the Wi-Fi slow?".

Airtime Fairness

airtime-fairness

Teaches: under plain contention every client gets the same number of turns, not the same amount of time, so one slow client drags every fast client down; airtime fairness gives each client an equal share of time instead.

Controls

Control	What it does	Why it's there (the teaching point)
Sharing rule: Packet / Airtime / Compare	Packet fairness (equal turns), airtime fairness (equal time), or both side by side. Opens on Compare.	The spec's fairness switch and compare view: the performance anomaly (Heusse et al., 2003) against its fix.
Replay round	Runs the round animation again.	The spec's animated round: each transmission drawn as wide as its airtime, so the slow client's block visibly dominates under packet fairness.
Play / Pause, Step, Reset (Present mode)	Plays the round, steps one transmission, empties it.	Same round, paced by the presenter. <i>(inferred)</i>
Payload per frame	64 to 1,500 bytes; default 1,500.	The spec's first lesson: every transmission pays a fixed overhead, so small frames waste most of the air.
Add client	Up to 8 clients. Opens with three at 867 Mbps and one legacy at 6 Mbps.	The spec's default cell, chosen to show the anomaly on sight.
PHY rate (per client)	Presets 6, 24, 54 (legacy); 150, 300 (HT); 433, 867 (VHT); 600, 1201 (HE); or Custom rate.	The spec's preset list, one per generation, so the class can build a realistic mixed cell.
Custom rate and Preamble: Legacy / HT or newer	Any rate from 1 to 10,000 Mbps with a 20 μ s or 40 μ s preamble.	The spec uses 20 μ s legacy and 40 μ s as a single teaching value for HT, VHT and HE.
Frames per turn (aggregation)	1 to 64; legacy clients always 1.	The spec's aggregation count: fast clients recover airtime by sending many frames per turn.
Remove (per client)	Removes that client.	Build the cell down as well as up. <i>(inferred)</i>

Present mode: Space plays or pauses the round, Right arrow steps one transmission, R empties the round, Up and Down change the sharing rule. The per-client detail folds behind "Each client's turn, and alone on the air".

A demo that works (about 3 minutes)

- 1 **Set up:** the defaults, then set the sharing rule to Packet.
- 2 **Ask the room:** "If one client drops to 6 Mbps, what happens to everyone else?" (the Teacher's Guide's own example question).
- 3 **Reveal:** play the round. The 6 Mbps block fills most of the round. Read the takeaway line under the switch, which states the slow client's share of the air and what every client gets. Press Up or Down to switch to Airtime: the fast clients recover, the slow client gets less, and the total rises.
- 4 **Optional follow-up:** set every client to 867 Mbps. Packet and airtime fairness give identical results (one of the spec's unit tests), which shows the anomaly needs mixed rates.

What it leaves out on purpose: one AP sending downlink to every client; every client always has data; no collisions, no retries, no rate adaptation. The 40 μ s preamble is one teaching value for HT, VHT and HE. Aggregate size limits are not enforced. Airtime Anatomy does the exact per-PHY timing.

Pairs with: the finished lab handout, *Airtime Fairness: why one slow client slows everyone* ([Deliverables/2026-09-27-teachers-guide-labs/airtime-fairness/:airtime-fairness-lab.html](#), [Airtime-Fairness-Lab-Student.pdf](#), [Airtime-Fairness-Lab-Instructor.pdf](#)), four exercises from "A PHY rate is not a throughput" to "Modern clients, same problem". Also Airtime Anatomy, Legacy Protection Cost (a slow device that costs airtime while silent), Rate Adaptation. Teacher's Guide sequences "Why is the Wi-Fi slow?" and "Troubleshooting slow or unstable Wi-Fi".

Rate Adaptation

rate-adaptation

Teaches: a radio does not know its best rate; it learns it by trying rates and keeping statistics, and every retry costs more airtime than the first try.

Controls

Control	What it does	Why it's there (the teaching point)
Play / Pause	Runs the link frame by frame. Opens paused with every rate untried.	The spec's play control; opening untried shows the learning from zero.
Step	Sends one frame with all its retries.	Shows one retry chain in full: which rates it tried and in what order. (<i>inferred</i>)
Restart	Starts a new link that has learned nothing.	The help pairs it with sampling at 0%: restart to see a link that never finds its higher rates.
Speed (air time per second)	1/50 (watch single frames), 1/10, 1/4, Real time; default 1/4.	The spec lists speed with play and step. The slow end shows single frames, the fast end the trend. (<i>inferred</i>)
Path: Walk / Steady / Fading	Walk goes out to 60 m and back to 2 m every 20 s; Steady holds 4 m; Fading holds 12 m with up to about 8 dB of swell and fade.	The spec's fourth lesson: as the client walks away the rate steps down, and on the way back sampling finds the higher rates again.
SNR offset	-20 to +20 dB on top of the path; default 0.	The spec's SNR slider: move the whole link up or down the rate table.
Sampling share	0 to 30% of frames; default 10%.	The spec: a small share of frames sample other rates. At 0% the link never discovers a better rate.
EWMA weight on history	50 to 95%; default 75%.	The spec's smoothing, 75% weight on history: lower it and the table reacts faster and noisier.
Retry chain: On / Off, same rate	On uses best throughput, second best, best probability, then lowest rate; Off keeps every attempt at the first rate.	The spec's retry chain order. Off shows why falling back to a safer rate matters.
Retry limit (attempts): 7 (short) / 4 (long)	Attempts before the frame is dropped.	The spec's two retry limits.
ACK timeout: 45 μs / 50 μs	How long the sender waits for an ACK that never comes.	The sources disagree on the exact constant, so the help makes it a setting.

Present mode: Space plays or pauses, Right arrow steps one frame, R restarts, Up and Down move the SNR offset. The four rarer settings fold behind "EWMA, retry chain, retry limit, ACK timeout".

A demo that works (about 3 minutes)

- 1 **Set up:** the defaults. Walk path, 10% sampling, speed 1/4.
- 2 **Ask the room:** "This link has never sent a frame. How does it find the right rate, and when the client walks to 60 m, does the rate slide down smoothly or in steps?"
- 3 **Reveal:** Play. The chart shows the chosen rate stepping down against the rate the SNR supports, and the table shows what the radio has learned about each rate, updated every 50 ms. On the walk back, sampling finds the higher rates again.
- 4 **Optional follow-up:** set Sampling share to 0% and press Restart. The link stays low. Or press Down several times to cut the SNR and watch retries pile up.

What it leaves out on purpose: the success curve is a teaching model anchored to the conformance-floor sensitivity table, not a measured error curve; real radios do better. It follows Minstrel's published outline; minstrel_ht samples on a timer and sizes chain stages by a time budget. No A-MPDU or Block Ack, so delivered throughput sits far below the PHY rate. One station, no collisions, mean backoff instead of a random draw.

Pairs with: Medium Access Simulator (contention and the doubling CW), Airtime Anatomy (why a lower rate takes longer), What an Interferer Costs (failed frames that rate control can mistake for a weak link). Teacher's Guide sequence "Why is the Wi-Fi slow?".

What an Interferer Costs (and how a NIC hears the air)

interferer-cost

Teaches: for the common interferers, the bigger cost is usually waiting, not lost frames: your radio defers to Wi-Fi it can decode at -82 dBm but to anything else only at -62 dBm, 20 dB (100 times the power) higher.

Controls

Control	What it does	Why it's there (the teaching point)
Opening question: The oven / The neighbor's AP / The same, then Reveal, then Ask again	"A microwave oven and a neighbor's AP on your channel, both at -70 dBm. Which one costs your Wi-Fi more airtime?" Reveal loads that scene into the tool.	Keith's lesson for this tool: waiting is the cost, not frame loss.
Source: Wi-Fi AP / Microwave / Bluetooth / Video	Picks the interferer on the meter. A dropdown on a phone.	Four sources side by side. The analog video sender is the counterexample the research brief required: it stays under energy detect, so nobody waits, and it ruins most frames anyway (Airshark).
Level slider (under the meter)	-100 to -30 dBm in 1 dB steps; the label names the source ("Neighbor AP at your radio", "Oven at its strongest frequency"). The meter's scale never moves.	Watch the marker cross preamble detect and energy detect. Defaults are illustrative: Wi-Fi -78, oven -50, Bluetooth -42, video -70.
Mains power: 60 Hz / 50 Hz	The oven's on-off cycle in the 60 ms sample. Oven only.	Airshark measured the oven's cycle at 16.66 ms and a duty of 0.5. The switch lets a 50 Hz country see its own cycle. (<i>inferred</i>)
Your AP's channel: Ch 1 / Ch 6 / Ch 11 / 5 GHz	The oven peaks in channel 11 and is weaker on 6 and 1 (illustrative); 5 GHz has no oven and no Bluetooth. Default 11.	The research brief notes that Keith's field observation (an AP on a running oven, high rates, no retries) fits an AP on channel 1 or 5 GHz.
Channel width (MHz): 20 / 40 / 80 / 160	On 5 GHz, preamble detect moves to -79, -76, -73 dBm while energy detect stays at -62.	The builder scaled preamble detect only, so the 20 dB gap narrows to 17, 14 and 11 dB: the same -82 per 20 MHz.
Neighbor airtime (illustrative)	10 to 80%; default 30%.	The research brief's illustrative default for a co-channel neighbor: 30% airtime at -78 dBm, above -82, so you defer every time it talks.

Control	What it does	Why it's there (the teaching point)
Path-loss exponent, for the distances: 2.0 free / 3.0 indoor / 3.5 walls	Changes only the distance bars.	Turns 20 dB into distance: a Wi-Fi neighbor defers you from 10 times farther in free space and 4.6 times farther indoors (help).
Reset to defaults	Restores every setting.	Back to the start between demos. <i>(inferred)</i>

Present mode: Up and Down move the selected source's level 1 dB, N picks the next source, C the next channel, R resets. No Space and no Right arrow: nothing plays. Neighbor airtime, path loss and Reset fold together, and the opening question folds behind "Opening question".

A demo that works (about 3 minutes)

- 1 Set up:** open the tool fresh. The question sits at the top.
- 2 Ask the room:** the question as written. Oven or neighbor's AP, both at -70 dBm? Take a show of hands for The oven, The neighbor's AP, The same.
- 3 Reveal:** pick the room's answer and press Reveal. The neighbor's AP: at -70 it is above -82, so your radio waits every time it transmits. The oven at -70 is under -62, so your radio sends into it, and at that level it barely touches your frames.
- 4 Optional follow-up:** press N to reach Video at -70. Nobody waits and most frames die, so non-Wi-Fi interference can do real harm. Then press C to reach 5 GHz, open the width and pick 160 to show the gap narrow to 11 dB.

What it leaves out on purpose: -82 and -62 dBm are the standard's minimum requirements, not what chips do; real chips often detect preambles near -91 dBm. The 40, 80 and 160 MHz values come from one source, and 6 GHz is not modeled. The corrupted-frame shares and the oven's weaker levels on channels 6 and 1 are illustrative. Waiting and frame survival are both computed at your radio. Left out: simultaneous-start collisions, Bluetooth's adaptive hopping, rate adaptation after failures, secondary channels, the 802.11ax per-20 MHz rules.

Pairs with: Spatial Reuse (BSS coloring relaxes the -82 dBm rule taught here), the Spectrum Analysis lesson's interferer gallery, the Swept vs FFT race in Fourier and FFT, Medium Access Simulator.

Spatial Reuse

spatial-reuse

Teaches: two BSSs on one channel take turns whenever they hear each other above -82 dBm; 802.11ax BSS coloring and OBSS_PD let a radio ignore a neighbor, but every 1 dB of threshold costs 1 dB of its own transmit power, so spatial reuse buys concurrency with range.

Controls

Control	What it does	Why it's there (the teaching point)
BSS coloring on (off = legacy radios); in Present, "BSS coloring (802.11ax)"	On by default. Off applies legacy rules and disables the color and OBSS_PD controls.	The spec's color on/off toggle, where off equals legacy behavior.
BSS A color / BSS B color	1 to 63; defaults 6 and 26.	The spec: a 6-bit color lets a radio tell its own BSS from a neighbor. Give both the same number and -82 dBm applies again whatever OBSS_PD is (help).
OBSS_PD threshold, per 20 MHz	-82 to -62 dBm in 1 dB steps; default -82.	The spec's third lesson: raise the threshold and AP B may ignore the neighbor, at TX_PWRref minus the amount raised.
TX_PWRref (single source): 21 dBm / 25 dBm	Client or AP with 1 or 2 streams, or AP with 3 or more.	The spec's two reference classes, labeled single-source.
Reset to the starting line	Restores the defaults.	Back to the start between demos. <i>(inferred)</i>
Link A MCS / Link B MCS	MCS 0 to 13 each must hold; default MCS 7.	Each link card says whether it still holds that MCS, so the cost of reuse shows up as a verdict, not just a dB number. (help)
Channel width, both BSSs	20, 40, 80 or 160 MHz.	The spec: OBSS_PD is per 20 MHz and rises 3 dB per width doubling (single-source).
AP transmit power, both	5 to 25 dBm; default 20.	Sets the starting power the restriction is taken from. <i>(inferred)</i>
Path-loss exponent n	2.0 to 4.0; default 3.0.	The spec's log-distance model; lets you match the room to the building. <i>(inferred)</i>
Positions (AP A, Client A, Client B, AP B), or drag a radio on the stage	0 to 60 m on a line; defaults 5, 15, 45 and 55 m.	The spec's draggable positions. The help: move the APs closer until energy detect (-62 dBm) fires, which no color or threshold gets past.

Present mode: Up and Down move OBSS_PD 1 dB, R resets. No Space and no Right arrow. MCS, width, power and path loss fold behind one disclosure, positions behind another, and "The rules and their sources" behind a third.

A demo that works (about 3 minutes)

- 1 **Set up:** the defaults. AP B hears AP A at about -77.7 dBm, above -82, so it waits and the pair needs two frame-times.
- 2 **Ask the room:** "How far do we have to raise OBSS_PD before AP B can send at the same time, and what does AP B give up?"
- 3 **Reveal:** press Up once per dB. At -77 dBm AP B stops waiting, both frames go at once, and AP B's power has dropped 5 dB. Watch each client's SINR and MCS verdict fall as you keep going.
- 4 **Optional follow-up:** set both colors to the same number and show AP B waiting again at any OBSS_PD. Or drag the APs together until energy detect fires.

What it leaves out on purpose: only AP B decides; AP A already holds the air, and uplink and clients' own deferral are left out. SRG thresholds and parameterized spatial reuse are out of scope. The MCS sensitivities are conformance floors, so a link that fails here may still work. Airtime counts one equal-length frame per BSS, with no backoff or ACKs. TX_PWRref and the 3 dB per width doubling come from one source.

Pairs with: What an Interferer Costs (the -82 and -62 dBm rules this tool relaxes), Channel Planner (same thresholds), OFDMA Resource Units and Multi-Link Operation (Teacher's Guide sequence "Wi-Fi 6 and 7 features").

Power Save

power-save

Teaches: a sleeping client wakes for beacons and reads the TIM to learn whether the AP holds frames for it; each power save mode trades time awake for time waiting, so every saving costs latency.

Controls

Control	What it does	Why it's there (the teaching point)
Mode	Always awake, Legacy PS (PS-Poll), U-APSD or Target Wake Time (TWT); default Legacy PS.	The spec's four modes: PS-Poll fetches one frame per poll, U-APSD a service period per trigger, TWT lets the client sleep through beacons.
Compare with	A second mode, or Nothing; default TWT.	The spec: two modes side by side show that every saving costs latency.
Timeline span: 0.3 s / 1 s / 10 s / 60 s	Zooms the timeline.	0.3 s shows single beacons and polls; 60 s shows the long TWT sleeps. (<i>inferred</i>)
Window start slider	Slides the view along the run.	Lets you find a moment worth pointing at. (<i>inferred</i>)
Beacon interval	50, 100, 200, 300, 500 or 1,000 TU; default 100 TU (102.4 ms).	The spec: 100 TU is the common interval, not a mandated value.
DTIM period	1 to 10 beacons; default 3.	The spec's second lesson: DTIM decides how often broadcast and multicast go out; longer saves battery and adds latency.
Listen interval	1 to 10 beacons; default 3.	The spec's legacy PS wake rule. When it does not line up with DTIM, the tool says the client wakes for both.
U-APSD access categories (Voice, Video, Best effort, Background checkboxes)	Which ACs are trigger-enabled; all on by default. The QoS Info byte is worked out as you go.	The spec's U-APSD flags (QoS Info B0 to B3).
Max SP Length	All buffered, 2, 4 or 6 frames per service period.	The spec's codes 0 to 3 (QoS Info B5 and B6).
Agreement: Individual / Broadcast	TWT agreement type.	Listed in the spec's TWT model.
Wake interval mantissa (text field)	1 to 65,535; default 62,500.	The spec: the mantissa is a 16-bit field, and the interval is mantissa $\times 2^{\text{exponent}}$ μs .

Control	What it does	Why it's there (the teaching point)
Wake interval exponent	0 to 31; default 4 ($62,500 \times 2^4 \mu s = 1 s$). Intervals over 5 minutes are refused.	Same formula as the mantissa.
1 s / 30 s buttons	Set $62,500 \times 2^4$ or $58,594 \times 2^9$.	Quick jumps to a sensible interval and to the 30 s TWT in the one vendor battery example the tool cites. (<i>inferred</i>)
Minimum wake duration	1 to 255 units; default $16 \times 256 \mu s = 4.1 ms$.	The spec's wake duration field.
Duration unit: 256 μs / 1 TU (1024 μs)	Unit for the wake duration.	The spec's two units.
Beacons: Sleep through / Wake for DTIM	Whether a TWT client still wakes for DTIM beacons.	The spec's third lesson: TWT lets the client sleep through beacons entirely.
Traffic pattern	Phone, mostly idle; Sensor; Voice call; Web page; Custom.	The spec lists traffic pattern as a control; each preset carries its own one-line description.
Downlink bursts, Frames per burst, Uplink frames, Broadcast and multicast, Access category	The traffic detail behind the pattern.	The spec's downlink rate and burstiness plus uplink traffic.
Arrivals: Regular / Random, and New random pattern	Evenly spaced or random arrivals, and a new draw.	Random arrivals show worst-case latency that regular ones hide. (<i>inferred</i>)
Awake current, Doze current, Battery	Awake 20 to 200 mA, doze 5 to 200 μA , battery 250 to 10,000 mAh.	The spec: energy is awake time $\times$ awake current plus doze time $\times$ doze current, with currents as labeled parameters.

Present mode: Right arrow slides the window along the run, R returns it to the start, Up and Down change the DTIM period. No Space. Listen interval, U-APSD and TWT fold together, traffic detail and currents fold together, and all readouts fold behind a third disclosure.

A demo that works (about 3 minutes)

- 1 **Set up:** the defaults. Legacy PS compared with TWT (1 s wake interval, sleeping through beacons), traffic "Phone, mostly idle".
- 2 **Ask the room:** "The TWT client skips every beacon and wakes once a second. What does that buy in battery, and what does it cost?"
- 3 **Reveal:** read the readouts for both modes: time awake, average current, battery estimate, mean and worst latency. The TWT client is awake far less and waits far longer for its frames.
- 4 **Optional follow-up:** switch traffic to "Voice call" and compare Legacy PS with U-APSD, the case U-APSD was made for. Or press Up to lengthen DTIM and watch broadcast latency grow. Keep the listen interval a factor or multiple of DTIM (3 with 6 or 9), because when they do not line up the client wakes for both.

What it leaves out on purpose: frame-exchange durations, traffic patterns and currents are illustrative. Real clients stay awake after traffic, retry, scan and roam, and APs limit how long they hold frames. The TIM is set whenever any frame waits; the finer U-APSD rules for which ACs the TIM covers are not modeled. The Silicon Labs battery figures in the Currents card are one vendor-commissioned example and the simulator does not use them.

Pairs with: Multicast at the Basic Rate (DTIM holding multicast), Voice Priority, End to End, Medium Access Simulator. Teacher's Guide sequence "QoS and contention".

Multicast at the Basic Rate

multicast-basic-rate

Teaches: multicast and broadcast go at a basic rate so every client can decode them, with no acknowledgment and no retry, so the same bytes can take many times the airtime of unicast; converting to unicast often wins when few clients listen.

Controls

Control	What it does	Why it's there (the teaching point)
Play the second / Pause	Sweeps the one-second timeline six times slower than real time.	Shows the stream's busy blocks, the beacons and the DTIM beacons arriving in time. <i>(inferred)</i>
Next beacon	Moves one beacon at a time.	Walks the DTIM release one beacon at a time with power save on. <i>(inferred)</i>
Reset	Clears the sweep.	Back to the start of the second. <i>(inferred)</i>
Presets (illustrative): Voice paging / Video / Service discovery chatter	Loads a stream's bit rate and packet size.	The spec's preset list: a paging stream, a video stream and mDNS-style chatter.
Stream bit rate (illustrative)	0.064 to 20 Mb/s in fixed steps; default 4.	The spec's slider, labeled illustrative.
Packet size (illustrative)	200, 400, 576, 1,000, 1,316 or 1,500 bytes; default 1,316.	The spec's packet size, labeled illustrative.
Band: 2.4 GHz / 5 GHz	Default 5 GHz. The 802.11b rates exist only on 2.4 GHz.	Lets you show the 1 Mb/s worst case, which 5 GHz cannot have. <i>(inferred)</i>
Basic rate	1, 2, 5.5 or 11 (802.11b, 2.4 GHz only), 6, 12 or 24 Mb/s; default 6.	The spec's fourth lesson: raising the basic rate helps, at the cost of range for multicast.
Show: Multicast / Converted to unicast / Compare both	Default Compare both.	The spec compares the stream with the same stream sent as one unicast copy per listener.
Listening clients	1 to 30; default 5.	The spec: unicast conversion grows with the listener count, and the readout reports the break-even count.
Listener rates (illustrative)	Spread, MCS 1 to 11; All close, MCS 9; All far, MCS 1.	Unicast copies go at each listener's own rate, so far listeners make conversion expensive. <i>(inferred)</i>
DTIM period	1 to 10 beacons; default 3.	The spec's third lesson: with a client in power save, multicast waits for the DTIM beacon.

Control	What it does	Why it's there (the teaching point)
A client is in power save	Off by default.	Same lesson: turn it on to see multicast held and released in bursts.
Ask the class / Under 10% / 10 to 30% / 30 to 60% / Over 60% / Reveal / close	Loads the question and hides the airtime answers until Reveal.	The spec's predict-then-reveal question: "A 4 Mb/s video stream on a Wi-Fi 6 AP. How much of the channel does it use?"

Present mode: Space plays or pauses the second, Right arrow moves to the next beacon, R resets it, Up and Down change the basic rate. Stream presets and all readouts fold away.

A demo that works (about 3 minutes)

- 1 Set up:** the defaults. 4 Mb/s video, 1,316-byte packets, 5 GHz, 6 Mb/s basic rate, five listeners.
- 2 Ask the room:** press Ask the class. "A 4 Mb/s video stream on a Wi-Fi 6 AP. How much of the channel does it use?" Take votes on the four ranges.
- 3 Reveal:** Over 60%. At a 6 Mb/s basic rate it uses 73.8% of the channel (the tool's own screen test). Press Up to raise the basic rate to 24 Mb/s: 21.9%.
- 4 Optional follow-up:** raise Listening clients past the break-even readout (8 listeners at the defaults, per the same test) and watch unicast become the more expensive choice. Or turn on "A client is in power save" and play the second to see the stream held until each DTIM beacon.

What it leaves out on purpose: one AP, no other traffic, no collisions, no retries. Beacons are drawn as markers; their own airtime is SSID Airtime's job. The stream presets, default rates and listener rates are illustrative. A unicast copy to a dozing client would wait for that client; the power save switch holds multicast only. The 802.11b timing uses the standard's well-known values, not a fresh read of IEEE 802.11. RFC 9119 describes the problem; the numbers come from the airtime model.

Pairs with: SSID Airtime (beacon cost), Airtime Anatomy, Power Save (DTIM). Teacher's Guide sequences "Where the airtime goes" and "Troubleshooting slow or unstable Wi-Fi".

Channel Utilization Meter

channel-utilization

Teaches: the channel utilization an AP puts in its beacon is the share of time its radio found the channel busy, averaged over seconds; busy is not the same as used for your data, and idle is not the same as available.

Controls

Control	What it does	Why it's there (the teaching point)
Run the channel / Pause	Scrolls the strip of channel time.	The spec's play control.
Next beacon interval	Runs one 102.4 ms interval.	Each bar under the meter is one beacon interval (spec).
Skip one window	Fills a whole averaging window at once.	Saves waiting 5 seconds for the average to settle in front of a class. <i>(inferred)</i>
Add a 1-second non-Wi-Fi burst	Adds a burst now.	A spec test: one 1-second burst in a 5.12 s window moves the reading by at most about 20%, so a spike barely moves the average.
Reset	Starts over.	Back to an empty window. <i>(inferred)</i>
Senders	1 to 50; default 1.	The spec's many-sender lesson: with every sender saturated, busy stays near its ceiling while your data's share falls, drawn beside Bianchi's 2000 figures.
Offered load, each	5 to 100% in steps of 5; default 100 (saturated).	A lightly loaded channel reads low for the right reason. <i>(inferred)</i>
Data rate	6 to 54 Mb/s (802.11g); default 54.	The spec's rate control.
Frame payload	100, 500, 1,000, 1,500 or 2,304 bytes; default 1,500.	The spec's frame size control.
Associated but idle	0 to 60 stations; default 0.	The spec's fourth lesson: Station Count counts associated stations, not active ones.

Control	What it does	Why it's there (the teaching point)
Averaging window, beacon intervals (default 50: one secondary source)	1 to 255; default 50 (5.12 s).	The spec: the default is the standard's according to one secondary source, and the label says so.
Listener view (counts reservations)	Off: what the AP reports. On: what a device outside the exchange sees, counting the NAV across the SIFS.	The spec's physical-versus-virtual carrier sense toggle. The same traffic reads 73% as the AP reports it and 76% from outside.
A neighbor network on this channel, and Neighbor's airtime (illustrative)	Adds another network's busy time, 5 to 60%; default 20%.	The spec: busy time the AP's own station count cannot explain, and it pushes clients away whether or not it is the AP's own.
Non-Wi-Fi bursts (illustrative)	Adds 4 ms bursts about 20% of the time.	The spec: busy counts non-Wi-Fi energy loud enough to read busy.
Speed	1 ms per second, 20 ms per second, Real time or 5 s per second; default Real time.	The slow end shows single frames and gaps in the strip; the fast end fills the averaging window. (<i>inferred</i>)
Ask the class / About 25% / About 50% / About 75% / 100% / Reveal / close	Loads one saturated laptop and hides the meter.	The spec's question: "One laptop sends as fast as it can. What will the meter read?"

Present mode: Space runs or pauses the channel, Right arrow runs one beacon interval, R resets, Up and Down change the number of senders, B adds a 1-second burst, W skips ahead one window. The formula, readouts and many-sender chart fold behind one disclosure, the rarer settings behind "More settings".

A demo that works (about 3 minutes)

- 1 **Set up:** the defaults (one saturated sender at 54 Mb/s, 1,500 bytes).
- 2 **Ask the room:** press Ask the class. "One laptop sends as fast as it can. What will the meter read?" Most rooms say 100%.
- 3 **Reveal:** About 75%. The meter reads 73%, because every frame waits a DIFS and a random backoff first, and that required silence is idle time. Only about 56% of the cycle is payload (spec).
- 4 **Optional follow-up:** press B during a run and show the 5-second average barely move. Or raise "Associated but idle" to 40: Station Count jumps and utilization does not move. Then turn on the neighbor network: utilization rises and Station Count does not.

What it leaves out on purpose: the formula is the definition, not any product's code; real APs may compute it differently. Only the primary 20 MHz channel is modeled. Every station hears every other (no hidden nodes); after a collision everyone waits the ACK timeout instead of EIFS; beacons' own airtime is left out. The neighbor's airtime and the burst pattern are illustrative. Bianchi's points use 1 Mbps parameters: the shape transfers, the exact numbers do not. Available Admission Capacity shows as not in use.

Pairs with: Airtime Anatomy, Medium Access Simulator, the BSS Load tool (macOS, reads the real element), SSID Airtime. Teacher's Guide sequences "Troubleshooting slow or unstable Wi-Fi" and "Where the airtime goes".

Legacy Protection Cost

legacy-protection

Teaches: one associated 802.11b device costs a 2.4 GHz network airtime even while it sends nothing: a slow protection frame in front of every modern frame, and a longer slot time.

Controls

Control	What it does	Why it's there (the teaching point)
Play / Pause	Sweeps the two-lane window about two thousand times slower than real time.	Shows modern-only and this network side by side, repeating send cycles. <i>(inferred)</i>

Control	What it does	Why it's there (the teaching point)
Next cycle	Moves one send cycle.	Lets you stop on one cycle and point at each cost. (<i>inferred</i>)
Reset	Clears the sweep.	Back to the start. (<i>inferred</i>)
802.11b device associated	On by default. Sets NonERP_Present and Use_Protection and forces the 20 μ s slot.	The spec: protection must turn on when an 802.11b station is associated, and the long slot is the second cost.
802.11b network heard nearby	Off by default. Sets Use_Protection only, never the long slot.	The spec: protection may turn on when an old network is only heard.
The old network is: On our channel / On another channel	Where the heard network sits.	The spec's "own channel only / adjacent channels too" question, labeled because sources disagree.
This AP reacts to: Own channel only / Adjacent channels too	Whether a heard network on another channel triggers protection.	One open-source AP reacts only to its own channel; a 2004 lab report saw protection spread between channels 1 and 11. The standard allows both (spec).
The 802.11b device can use short preamble	Off by default, which sets Barker_Preamble_Mode.	The spec's third ERP bit. Off is illustrative: an old scanner is assumed long-preamble only.
Protection frame: CTS-to-self / RTS/CTS	The kind of protection frame.	The spec's picker: CTS-to-self costs one slow frame, RTS/CTS costs two.
Rate and preamble grid	1 (long only), 2, 5.5 or 11 Mb/s, long or short preamble; each cell shows its cost. 1 Mb/s short is grayed out. Default 1 Mb/s long.	The spec's default is the worst case the standard allows. 1 Mb/s short is disabled because 802.11b defines short preamble only for 2, 5.5 and 11 Mb/s.
Contrast: 6 Mb/s OFDM	A protection frame at 6 Mb/s OFDM, which an 802.11b device cannot decode.	The spec's contrast option: most of the cost is the old preamble, not protection itself.
CWmin with an 802.11b device associated: 15 / 31 (unverified)	Minimum contention window while the device is associated.	"31 when an 802.11b station is present" comes from one unverified source, so the spec makes it a labeled setting.

Control	What it does	Why it's there (the teaching point)
Frame size	64, 256, 512, 1,000 or 1,500 bytes; default 1,500.	The spec's sender. Smaller frames make the fixed protection cost a bigger share. <i>(inferred)</i>
Data rate	6 to 54 Mb/s (802.11g); default 54.	The spec's sender.
Ask the class / Nothing / 5 to 25% / 25 to 40% / 40 to 60% / Over 60% / Reveal / close	Loads the silent-scanner question and hides the answers.	The spec's question: "An old 802.11b scanner sits in a drawer, associated but silent. How much does it cost the 54 Mb/s laptop?"

Present mode: Space plays or pauses, Right arrow moves to the next send cycle, R resets, Up and Down make the protection frame faster or slower. The sender, readouts and ceiling table fold together, and the heard-network settings fold behind "Where the old network is heard (sources disagree)".

A demo that works (about 3 minutes)

- 1 Set up:** the defaults. 802.11b device associated, CTS-to-self at 1 Mb/s long preamble, 1,500 bytes at 54 Mb/s.
- 2 Ask the room:** press Ask the class. The scanner in the drawer, associated but silent: how much does it cost the laptop? Take votes on the five ranges.
- 3 Reveal:** 40 to 60%. The laptop's ceiling falls from about 30.5 to about 14.8 Mb/s. Point at the two costs in the lower lane: the 304 μ s CTS, longer than the 254 μ s data frame it protects, and the stretched slots.
- 4 Optional follow-up:** press Up until the protection frame reaches 11 Mb/s short (19.5 Mb/s in the spec's ceiling table). Pick the 6 Mb/s OFDM contrast. Then turn "associated" off and "heard nearby" on: the protection frame stays and the long slot goes.

What it leaves out on purpose: everything is 2.4 GHz; there is no DSSS at 5 GHz. Which rate an AP uses for protection is up to the implementation; 1 Mb/s long is the slowest legal case. The HT Protection value is a readout only; mode 2 is not modeled. One sender, no

collisions, no retries, and the old device sends nothing. The network is open; with 16 bytes of encryption the data frame takes one more symbol.

Pairs with: Airtime Fairness (the cost of a slow device's own traffic), Airtime Anatomy. Teacher's Guide sequences "Where the airtime goes" and "Troubleshooting slow or unstable Wi-Fi".

Multi-Link Operation

mlo-simulator

Teaches: MLO's latency win comes from sending on whichever link frees up first, so it is real when links are about equally busy, shrinks when one is much busier, and can become a loss; EMLSR pays a switch cost and real drivers often turn it off.

Controls

Control	What it does	Why it's there (the teaching point)
Lesson	Two equal links: MLO wins; One link much busier: the win shrinks; A slow 2.4 GHz link: MLO can be worse; EMLSR switch cost: one clean link. Shows "Your own settings" once you edit.	Each lesson sets up one of the spec's learning points in one pick. <i>(inferred)</i>
New random traffic	Draws a fresh sample with the same settings.	Shows that a result holds across samples, not just one lucky draw. <i>(inferred)</i>
2.4 GHz / 5 GHz / 6 GHz link switches	Each link on or off; at least one stays on. Default 5 and 6 GHz.	The spec: a client holds links on two or three bands.
<band> busy	0 to 95% per link in steps of 5; default 50%.	The spec's busy fraction per link: the lesson turns on whether the links are equally busy.
<band> mean busy period	200, 500, 1,000, 2,000 or 5,000 μ s; default 1,000.	The spec's busy-period length per link.
Frames per second	100 to 3,000 in steps of 100; default 1,000.	The spec's arrival rate.

Control	What it does	Why it's there (the teaching point)
Frame airtime: Fixed / Airtime Anatomy	Fixed μ s per frame, or computed from HE MCS and aggregation.	The spec: airtime fixed per frame, or from the Airtime Anatomy service at a chosen MCS.
Airtime per frame, every link	100 to 2,000 μ s; default 300.	Fixed-airtime option.
HE MCS and Frames per A-MPDU (1, 8, 32)	Airtime Anatomy option; default MCS 7.	Lets a slow 2.4 GHz link cost real airtime, as in the "slow link" lesson. (<i>inferred</i>)
STR / NSTR / EMLSR switches	Which modes to compare; all on. Single link is always shown.	The spec: compare single link, STR, NSTR and EMLSR on the same random traffic.
Padding delay / Transition delay	EMLSR switch delays, 0 to 256 μ s; defaults 64 and 64.	The spec takes these values from the Linux kernel's list. EMLSR pays them before each exchange on a newly chosen link.
EMLSR disabled by driver	Falls back to one link. Needs two or more links.	The spec: real drivers may disable EMLSR. Intel's Linux driver leaves it for many reasons, some for 300 or 600 s (help).
Show the lanes for, Time shown (2 / 5 / 20 ms), View starts at	Which mode's frame placement the lanes draw, how much time, and where.	Lets you put one mode's choices on the lanes at a scale where individual frames read. (<i>inferred</i>)

Present mode: R goes back to the lesson, Up and Down slide the lanes half a window along the run, N draws new random traffic. No Space and no Right arrow: nothing plays. Busy periods and frame airtime, EMLSR delays, the latency table and "The four modes, and the one study" each fold away.

A demo that works (about 3 minutes)

- 1 **Set up:** Lesson "Two equal links: MLO wins".
- 2 **Ask the room:** "Wi-Fi 7 adds a second link. Does latency always drop? What if one link is much busier, or slow?"
- 3 **Reveal:** read the verdict line and the histograms. With equal links, STR beats the best single link. Switch the lesson to "One link much busier: the win shrinks", then to "A slow 2.4 GHz link: MLO can be worse", and watch the verdict line say so.
- 4 **Optional follow-up:** pick "EMLSR switch cost: one clean link" to see EMLSR pay its delays, then turn on "EMLSR disabled by driver". Press N between lessons to show the result is not one lucky draw.

What it leaves out on purpose: the one published MLO latency study (Carrascosa et al., IEEE ICC 2022) is a model driven by real traffic traces, not a field measurement, and no independent Wi-Fi 7 hardware measurement was found. EMLMR is described but not simulated. No collisions, retries or uplink; one frame or A-MPDU per exchange; HE timing stands in for EHT. With EMLSR off the model keeps the least busy link, while a real driver picks by its own rules.

Pairs with: Airtime Anatomy (frame airtime), Spatial Reuse and OFDMA Resource Units (Teacher's Guide sequence "Wi-Fi 6 and 7 features").

Network Design and Security

This shelf is where the class stops looking at one radio and starts designing and securing a network: who shares a channel, when a client roams or changes band, how a survey and a heat map are built, what DHCP and PoE do to an event or a new AP, and what every frame of an association and an 802.1X exchange is for. Most of these tools open with a predict-then-reveal question already built in, so the demo pattern is to load the question, take guesses, and let the tool show the answer. In Present mode every tool also answers F (full screen), ? (list the keys) and Esc (leave), so the key lists below give only the keys that matter for each tool.

Controls, labels, ranges and keys were read from the screen code on the `wifi-lab/preview` branch (`fa091dce`) on 2026-09-28. A "why" marked *(inferred)* is reasoned from what the control lets the class see; every other "why" comes from the build spec, the in-app help, or one of Keith's recorded rulings. No lab handout exists yet for any tool on this shelf: `Deliverables/2026-09-27-teachers-guide-labs/` holds only the FSPL and Airtime Fairness labs.

Channel Planner

channel-planner

Teaches: APs on overlapping channels that hear each other share one channel's airtime, and a wider channel leaves fewer channels to reuse, so more APs end up sharing.

Controls

Control	What it does	Why it's there (the teaching point)
Band: 2.4 GHz / 5 GHz	Switches the floor between the two bands. A new band starts with every AP on one channel.	The two planning problems differ: 2.4 GHz is about masks and 1, 6 and 11; 5 GHz is about how many channels survive each width. 6 GHz planning is left to the 6 GHz Power tool (code comment).
Region: US / EU	Swaps the channel plan. The EU plan has no U-NII-4 and stops at 140.	Channel counts differ by region (US 25/12/6/2 at 20/40/80/160 MHz, EU 19/9/4/2), and the class should see that the count is a rule, not physics.

Control	What it does	Why it's there (the teaching point)
DFS channels (5 GHz)	Adds 52 to 64 and 100 to 144.	Shows how much DFS buys: without it the US has 9 channels at 20 MHz and none at 160 MHz.
U-NII-4 (US) (5 GHz)	Adds 169 to 177. Disabled in the EU.	Lets the class see what the newest US channels add at each width (<i>inferred</i>).
Transmit mask: OFDM (g/n/ax) / DSSS (802.11b) (2.4 GHz only)	Picks the mask used for adjacent-channel energy. DSSS has no 40 MHz channel.	This is where 1, 6 and 11 comes from: the OFDM mask is about -34 dBr at 25 MHz spacing but -23.6 dBr at 15 MHz (1 and 4).
Channel width, every AP (Up/Down in Present)	Sets every AP's width at once.	The core lesson: at 20 MHz six APs share nothing; at 80 MHz only two non-DFS channels exist, so three APs share each.
Auto-plan	Greedy assignment that shrinks the largest contention domain at the current width.	The floor opens with all six APs on one channel, "the plan Auto-plan is there to fix" (code comment). It is labeled a teaching heuristic, not any vendor's RRM or DCA.
Reset to the starting floor (R)	Six APs on 36 at 40 MHz, US, DFS off.	Returns to the worst case so the fix can be shown again.
Selected AP, Add AP, Remove AP	Picks an AP; adds or removes one (2 to 12).	More APs on the same floor means more sharing at the same width (<i>inferred</i>).
AP n channel, AP n width	Sets one AP's primary channel and width.	Lets you put one AP on a wide channel and show it colliding with narrower neighbors (<i>inferred</i>).
AP n across, AP n down (in "Move AP n by the numbers")	Moves the selected AP by the numbers. Tap and drag on the floor do the same.	A keyboard way to do what dragging does (<i>inferred</i>).
Wall tool: drag on the floor, Add wall top to bottom, Add wall side to side, Remove wall n, Clear walls	Draws or removes walls.	A wall can drop a neighbor below -82 dBm and split one contention domain into two (<i>inferred</i>).
Loss per wall (0 to 30 dB, default 10)	One loss for every wall.	Kept to one wall type and one number on purpose; real materials live in the RF Attenuation tool.

Control	What it does	Why it's there (the teaching point)
AP EIRP (0 to 30 dBm, default 20), Path-loss exponent n (2.0 to 4.0, default 3.0)	Sets how far each AP is heard.	Contention is decided by what each AP hears, so lower power or a lossier building shrinks the groups (<i>inferred</i>).
Floor width, Floor depth	20 to 100 m (66 to 328 ft) by 10 to 60 m (33 to 197 ft).	The same APs on a smaller floor hear each other more (<i>inferred</i>).

Present mode: Up and Down step the channel width for every AP; R resets the floor.
No Space or Right arrow: there is no clock.

A demo that works (about 3 minutes)

- 1 Set up:** press R. Six APs, 5 GHz, US, DFS off, all on 36 at 40 MHz: one domain of six, about 17% airtime each.
- 2 Ask the room:** "Press Auto-plan at 40 MHz. How many APs still share a channel? Now what happens at 80 MHz?"
- 3 Reveal:** Auto-plan at 40 MHz uses all four channels and leaves pairs sharing (largest domain 2). Press Up to 80 MHz and Auto-plan again: two channels, three APs on each. Press Down to 20 MHz and plan: nobody shares.
- 4 Optional follow-up:** switch DFS on at 80 MHz and plan again, or switch to 2.4 GHz and open "Why they defer, and why 1, 6 and 11" to compare 1 and 6 against 1 and 4.

What it leaves out on purpose: 6 GHz. The mask is a ceiling, not what a radio emits, so the 2.4 GHz figures are the worst case the mask allows. The -82, -72 and -62 dBm thresholds are the commonly cited values. Deferring on a secondary channel is counted as sharing to keep the picture simple. 1/N airtime assumes an even split among APs that all hear each other; clients, retries and load change the real share. Auto-plan knows only the APs on this floor.

Pairs with: Adjacent Channels and AP Stacking (why a clean channel still hurts up close), DFS and Radar, the 2.4 and 5 GHz channel cards in Course Handouts. Lesson sequences "Channels and the rules" and "Designing a cell plan". No lab handout yet.

Adjacent Channels and AP Stacking

adjacent-channel

Teaches: a neighbor on a channel that does not overlap yours still hurts when it is close, because its transmit mask leaks energy into your 20 MHz where no receive filter can remove it, and distance is what saves you.

Controls

Control	What it does	Why it's there (the teaching point)
Opening question: Yes, a problem / No, they do not overlap; Reveal; Ask again	Records the class's answer; Reveal loads the scene (36 and 44, 30 cm apart) and explains it.	The spec's predict-then-reveal: "Two APs on channels 36 and 44, 30 cm apart on the same ceiling. Any problem?"
Band: 2.4 / 5 / 6 GHz (default 5)	Changes the channel plan and the separations offered.	2.4 GHz brings the 1/6/11 against 1/4/8/11 comparison; 5 and 6 GHz show gaps between 20 MHz channels.
Neighbor transmit mask: OFDM (802.11a/g) / HE/EHT (802.11ax/be)	Picks the neighbor's mask shape. Shown only when both apply (not in 6 GHz).	Lets the class compare the older and newer masks' skirts (<i>inferred</i>).
Neighbor width (MHz)	20 in 2.4 GHz, up to 160 in 5 GHz and 320 in 6 GHz. You always listen on 20 MHz.	A 20 MHz neighbor's mask is flat past 1.5 channel widths, but a wider neighbor's keeps falling, so an extra empty channel helps there (Pax's correction, 2026-09-27).
Channel separation: Next channel, One channel gap, Two channel gaps; in 2.4 GHz 1 and 6 (25 MHz), 4 and 8 (20 MHz), 1 and 4 (15 MHz)	Moves the neighbor's channel. Center frequencies never move on screen.	Separation and distance both matter; for a 20 MHz neighbor the second empty channel buys nothing more from the mask.
Neighbor distance (slider under the floor line, 0.3 to 30 m (1 to 98 ft), default 3 m; Up/Down in Present)	Moves the neighbor along the line.	The main control: leakage falls with distance at the log-distance slope, and this is what separates "stacked on one ceiling" from "down the hall".
Who is listening: Your client (downlink) / Your AP (uplink)	Relabels the receiver and the wanted transmitter. Default your AP.	Covers both cases the help names: two APs stacked on a ceiling, and a client sitting next to someone else's AP.

Control	What it does	Why it's there (the teaching point)
Neighbor power (0 to 30 dBm, default 20)	The neighbor's power, antenna gain included.	Leakage rises dB for dB with the neighbor's power (<i>inferred</i>).
Your AP power / Your client power; Your AP distance / Your client distance (1 to 60 m, default 10)	The wanted signal's power and distance (label follows Who is listening).	A stronger wanted signal survives more leakage, so the same neighbor can be harmless to a close client and fatal to a far one (<i>inferred</i>).
Path-loss exponent (model) (2.0 to 4.0, default 3.0)	Sets the propagation slope.	How fast both signals fall with distance (<i>inferred</i>).
Selectivity, next channel / one gap or more (illustrative) (20 to 60 dB; defaults 35 and 51)	How far your receiver's filter pushes down the neighbor's own channel. One slider, for the current separation.	Pax's correction: rejection does not change with rate (ACR plus minimum sensitivity is -66 dBm at every MCS), so one value replaces the four per-MCS sliders the spec first asked for. Illustrative: no measured selectivity was read.
CCA energy-detect threshold (-82 to -52 dBm, default -62)	The level at which the receiver calls the air busy.	Leakage can stop your AP from transmitting at all, not just lower its rate.
Reset to defaults (R)	Back to the opening defaults.	Returns to the question's starting point (<i>inferred</i>).

Present mode: Up and Down move the neighbor away and closer; R resets. No Space or Right arrow.

A demo that works (about 3 minutes)

- 1 **Set up:** press R, then read the opening question aloud.
- 2 **Ask the room:** "Channels 36 and 44 don't overlap. 30 cm (12 in) apart on one ceiling: problem or not?" Click the room's answer.
- 3 **Reveal:** press Reveal. From the help's worked example: -56.7 dBm leaks into channel 44, above the -62 dBm energy-detect threshold, so the AP hears the air as busy; its client 10 m (33 ft) away would get MCS 8 alone, but with the neighbor the SINR is -0.8 dB and the link is lost. Then press Up a few times and watch the link come back as the neighbor moves away.
- 4 **Optional follow-up:** switch to Two channel gaps and show that a 20 MHz neighbor gains almost nothing over one gap; then 2.4 GHz and compare 1 and 6 with 1 and 4.

What it leaves out on purpose: the mask is the worst case the rule allows; real radios run at or below it. Selectivity values are illustrative. Below 1 m the path is taken as free space, so the 30 cm numbers are rough. No walls, antenna patterns or fading.

Pairs with: Channel Planner (which channels overlap on paper, who shares airtime). Lesson sequence "Channels and the rules". No lab handout yet.

Roaming Walk

roaming-walk

Teaches: the client decides when to roam: it holds its AP until the signal falls below its trigger, then moves only to an AP at least delta stronger, and most of each roam's gap is scanning.

Controls

Control	What it does	Why it's there (the teaching point)
Play / Pause / Play again, Step (Step 1 s), Restart (R), Walk time, Speed: 1x (walking pace), 2x, 5x, 10x	Runs, steps and scrubs the walk.	Lets you stop the client just before a roam and ask the room what happens next (<i>inferred</i>).

Control	What it does	Why it's there (the teaching point)
Client: Phone, transmitting (-70/8); Phone, idle (-70/12); Laptop (-75/12); Illustrative sticky (-85/12); Illustrative jumpy (-65/2); Custom	Loads a trigger (dBm) and delta (dB).	The phone and laptop values are Apple's published rules; sticky and jumpy are illustrative because other client platforms publish no roam thresholds. Generic names on screen, the real source in the source line (Keith, 2026-09-27).
Trigger (-90 to -55 dBm; Up/Down in Present)	Where the client starts looking.	Set too low, the client is sticky and clings to a distant AP at a poor rate.
Delta (how much stronger a new AP must be) (0 to 20 dB)	The hysteresis.	Set too small, the client ping-pongs between two APs of similar strength.
802.11k neighbor list, PMK caching, 802.11r fast transition (FT)	Shorten the scan (k) or the authentication (PMK caching, FT).	Scanning dominates a slow roam, so 802.11k shortens it more than FT does.
Illustrative timings: Dwell per channel, Channels in a full scan, Time per auth frame, Authentication server time (802.1X)	Set what a roam costs.	Every timing is a labeled parameter: the only measured figures (Mishra et al. 2003, one FT capture of 14 ms) are shown as context, not used.
Path: Along the corridor, Under the APs, Corner to corner, Across at 20 m, Down and back, Draw your own (Use this path, Undo, Cancel)	Picks or draws the walk.	Different paths put the client between APs of similar strength, where ping-pong shows (<i>inferred</i>).
APs (2 to 6), Move, AP n across, AP n down (or drag on the floor)	Sets AP count and positions. Changing the count spaces them evenly.	Closer APs mean more overlap and earlier candidates (<i>inferred</i>).
Band: 2.4 / 5 / 6 GHz; AP power (EIRP, illustrative) (5 to 30 dBm, default 14); Path-loss exponent n	Sets the signal model. The -67 and -70 dBm reach readouts update.	Apple's -70 dBm trigger means a -67 dBm overlap design still leaves a phone on its AP down to -70 dBm (spec point 4).
Shadowing (sigma) (0 to 6 dB, default 2), New shadowing	Adds smooth, seeded shadow fading along the path.	Shadowing is what makes a jumpy client ping-pong between two similar APs.
Show all / Show the last N (roam log)	Expands the roam log.	Keeps the log short on screen (<i>inferred</i>).

Present mode: Space plays or pauses; Right steps 1 s; R restarts the walk; Up and Down move the roam trigger.

A demo that works (about 3 minutes)

- 1 **Set up:** Restart, Client "Phone, transmitting", three APs, Along the corridor.
- 2 **Ask the room:** "The phone walks right past AP 2. Does it roam to it the moment AP 2 is stronger?"
- 3 **Reveal:** play it. The phone holds its AP until it falls below -70 dBm, then roams. In the help's example (shadowing off) it roams twice and spends 0.2 s below -70 dBm; the Laptop preset spends 11.8 s there, and Illustrative sticky never roams.
- 4 **Optional follow-up:** pick Illustrative jumpy with shadowing on and count the PP (ping-pong) marks; then turn on 802.11k and FT and watch a 351 ms full 802.1X roam drop to 32 ms (help example).

What it leaves out on purpose: APs radiate equally in every direction and there are no walls; contours show the average signal without shadowing. Every timing is illustrative. 802.11v is not simulated. Real clients average their readings, scan on their own schedule and weigh more than signal strength.

Pairs with: Band Steering (bands on one AP, where this is APs on one band), 802.1X and EAP Ladder in Roam mode (the frames of a roam), Association, Frame by Frame. Lesson sequences "Association and roaming" and "Designing a cell plan". No lab handout yet.

Band Steering

band-steering

Teaches: the client chooses the band; the AP can only hide, refuse, suggest or disconnect, and the same AP sounds weaker on 5 GHz, so a client that joined 2.4 GHz at the edge stays there by its own published rules.

Controls

Control	What it does	Why it's there (the teaching point)
Walk / Pause, Step 1 m, Reset, Walked slider (on the stage)	Moves the client along its path 1 m (3 ft) at a time.	Walking toward the AP only makes the 2.4 GHz signal better, and the class needs to see that the client still does not move (<i>inferred</i>).
Walking path: Edge to AP / AP to edge	Direction of the walk (55 m to 2 m by default).	Edge to AP is the question's scene: joining 2.4 GHz at the edge, then never looking again.
Predict, then reveal: Ask the class; Yes, it moves to 5 GHz / No, it stays on 2.4 GHz; Reveal; Close the question	Loads the question (steering off, edge to AP), then walks it in.	The spec's question: "The client walks from the edge of coverage right up to the AP. Does it move to 5 GHz?" Under Client A's and Client B's rules, no.
AP steering: Off; Probe suppression on 2.4 GHz; Authentication refusal on 2.4 GHz; Transition request (802.11v amendment); Deauthentication on 2.4 GHz	Picks what the AP does to a client it has heard on 5 GHz.	The AP's tools, weakest to strongest, and none of them hides the 2.4 GHz beacons. Deauthentication was added 2026-09-27 as the fourth method; it only disconnects the client, which then chooses again.
Client: Client A / Client B / Client C	Loads one published rule set, drawn as threshold lines on the gauges.	Each is one platform owner's published rules, with no vendor or OS name on screen (Keith, 2026-09-26). Client C has no published band rule.
Client uses a random address while scanning	The AP's tracker loses the client.	The AP recognizes a dual-band client by its MAC on both radios; a random scan address stops probe suppression and refusal from applying.
Client C's driver supports transition requests (Client C only)	Whether Client C answers an 802.11v request.	Client C's maker publishes transition-request support and no band rule.
Extra 5 GHz wall loss (illustrative) (0 to 10 dB, default 3)	Adds loss on 5 GHz only.	Free space alone makes 5 GHz 7.1 dB weaker than 2.4 GHz, and walls usually add more, so 5 GHz drops below the entry level first at the edge.
Refusals a client tolerates (illustrative) (1 to 10, default 3)	Used only with authentication refusal.	How many refusals a client tolerates is not published, so it is a labeled slider.

Control	What it does	Why it's there (the teaching point)
Client rescan time (illustrative) (1 to 10 s, default 3); AP retry interval (illustrative) (1 to 30 s, default 5)	Used only with deauthentication.	No source gives deauthentication timings; Keith kept 3 s and 5 s (2026-09-27), and ruled that only a 5 GHz-capable client on 2.4 GHz gets deauthenticated.

Present mode: Space walks or pauses; Right steps 1 m; R resets the walk; Up and Down move the client 5 m (16 ft) along the path.

A demo that works (about 3 minutes)

- 1 Set up:** Reset, Client A, steering Off, Edge to AP.
- 2 Ask the room:** press Ask the class and read the question.
- 3 Reveal:** press Reveal and let it walk. At 55 m (180 ft) 5 GHz is below the floor, so it joins 2.4 GHz; above -70 dBm on 2.4 GHz (about 29 m) Client A stops looking, and at 2 m it is still on 2.4 GHz with 5 GHz at -45.3 dBm (help example).
- 4 Optional follow-up:** pick Deauthentication and walk again. Client A is thrown off at 33 m, comes back to 2.4 GHz three times, and lands on 5 GHz at 6 m after 4 deauthentications and 12 s without traffic (help example).

What it leaves out on purpose: probe suppression and refusal follow one open-source AP's documented behavior; commercial thresholds are not published. Client rules are one published set each; device makers can change every number. Deauthentication steering is vendor behavior, not a mechanism the standard defines, and its timings are illustrative. A PMF client ignores an unprotected deauthentication from anyone but its AP.

Pairs with: Roaming Walk (APs on one band), Uplink vs Downlink, Why Two Devices Disagree. Lesson sequence "Why clients misbehave". No lab handout yet.

Survey Walk

survey-walk

Teaches: walking never distorts a measurement, but walking speed sets how far apart each channel's samples land, through the scanner's revisit time, and Keith's Rule 4 says the longest revisit allowed is the guess range divided by the pace.

Controls

Control	What it does	Why it's there (the teaching point)
Predict, then walk: Yes, it matters / No, light is too fast; Reveal now; Ask again	Opening question; the answer appears after the first full walk.	The spec's opening card: "RF travels at the speed of light, so does walking speed matter?" Both answers are half right.
Walk / Pause / Walk again, Step, Back to the start (R), Walk time, Speed (1x to 10x)	Runs and scrubs the walk.	Lets you stop and point at a white gap as it forms (<i>inferred</i>).
Survey type: Passive / Active / Hybrid	Active shows its five limits and greys out every AP but the associated one. Hybrid needs 2 NICs.	Active surveys are taught for their limits, in Keith's words (2026-09-26): one client, one point in time, one path, one set of traffic, a single AP.
Data-collection device, by network interface cards (NICs): 1 NIC (a laptop's built-in radio), 2, 3 or 4 NICs	Same channels and dwell on 1 to 4 radios.	Keith's ruling: a generic device, no product name, 1 to 4 NICs. At the defaults 1 NIC fails Rule 4 and 2 NICs only just pass.
Channels scanned: 2.4 GHz 1, 6, 11 (3); + US 5 GHz (28); + 6 GHz PSCs (43); every 20 MHz channel, US (95); 6 GHz only (59); Custom count (Channel count slider)	The channel list the scanner cycles.	Revisit time grows with the list: 95 channels at 250 ms is almost 24 s per cycle on one radio.
Dwell per channel (20 to 500 ms, default 250); Channel switch per hop (illustrative) (0 to 50 ms)	Time on each channel, and switching overhead.	250 ms is labeled a common survey default; dwell times the channel count is the cycle.
Hopping algorithm: Sequential / Band per NIC / Priority; Channels in use (priority) chips; Priority every k-th slot (2 to 6, default 3)	How NICs share the list.	Keith's ruling that pace "works in tandem with how many NICs ... and the channel changing algorithms". Priority channels get short spacing and the rest get longer.
Capture method: Continuous / Line / Stop and go; Stop every (2 to 20 m, Stop and go only)	Where samples land between clicks.	Continuous spreads samples evenly in time, so a pause slides them; stop and go stacks them at a point in one fade.
Timestamps: Per channel / Per cycle	One stamp per channel or per cycle.	Per-cycle stamping moves early channels up to one revisit distance off; which one products use is not published.

Control	What it does	Why it's there (the teaching point)
Pause at the door; Pause (1 to 10 s, default 3)	Stops the walker halfway along the first leg with no click.	Shows placed samples sliding away from where they were taken (rings mark the true spots).
Walk path: Straight corridor, Loop, Perimeter, Draw your own	Picks or draws the walk.	Longer paths and turns add clicks and pauses (<i>inferred</i>).
Walking pace (0.5 to 2.0 m/s, default 1.4; Up/Down in Present)	Walker speed.	Spacing = pace x revisit time. Comfortable gait is 1.27 to 1.46 m/s (4.2 to 4.8 ft/s).
Guess range (accuracy distance) (1 to 20 m, default 5)	How far the map may guess from a sample.	Keith's indoor preference is 5 m. Any stretch with no sample within half the guess range turns white: no data, never no coverage (Rule 6).
Show this channel's samples: All channels (shape = band) or one channel	Filters the dots.	Each channel has its own spacing, so one channel's gaps are the lesson (<i>inferred</i>).
Signal along the path; New fading	Shows each sample's level, with a new fading draw.	A moving scanner draws a new fade at every sample; a stationary one repeats one fade.

Present mode: Space plays or pauses; Right steps 1 s; R restarts the walk; Up and Down change the walking pace.

A demo that works (about 3 minutes)

- 1 Set up:** Back to the start. Passive, 1 NIC, 28 channels at 250 ms, 1.4 m/s (4.6 ft/s), guess range 5 m (16 ft), Straight corridor.
- 2 Ask the room:** the opening question. Record the answer.
- 3 Reveal:** walk it. 1 NIC revisits a channel every 7.0 s, 9.8 m (32 ft) apart, and Rule 4 fails (3.57 s allowed). Switch to 2 NICs: 3.5 s and 4.9 m, which only just passes (help example).
- 4 Optional follow-up:** turn on Pause at the door in Continuous (samples slide up to 2.0 m), then switch Timestamps to Per cycle (up to 9.45 m).

What it leaves out on purpose: the walls are drawn but not modeled. Whether real survey apps stamp per channel or per cycle, and how they place samples between clicks, is not published. No product names or vendor timings anywhere. No active-survey direction-skew mode (Keith's answer to Pax).

Pairs with: Heat Map Builder, then Predict, Then Measure. Lesson sequence "Surveys and heat maps". No lab handout yet.

Heat Map Builder

heat-map-builder

Teaches: a heat map is mostly guesses drawn in the same colors as the measurements; the interpolation method, the guess range and the sample spacing decide how wrong it is, and white means no data, not no coverage.

Controls

Control	What it does	Why it's there (the teaching point)
A tap on the floor will: Add a sample / Inspect a cell	A tap adds a sample (snapped to 0.5 m) or shows which samples made a cell, drawn as lines as thick as their weights.	Inspecting a cell shows the map is a weighted average of dots, not a measurement.
Remove the last sample, Clear every sample	Undo and clear.	Lets you rebuild the three-dots scene by hand (<i>inferred</i>).
Grid, Corridor walk; Grid and walk spacing (1 to 10 m, default 3)	Places samples on a grid or along a corridor.	Closer samples make the map more accurate, up to the point where noise on single readings dominates.
Method: IDW / Nearest neighbor	Inverse distance weighting, or copy the closest sample.	One documented method from the interpolation literature; no survey vendor publishes its algorithm, so the tool never says which a product uses.
Power (p) (1 to 6, default 2; P steps 1, 2, 4, 6 in Present)	How local the IDW guess is.	High power approaches nearest neighbor and turns the map blocky; the worked example shows -60.5, -58.1 and -55.8 dBm for $p = 1, 2$ and 4.

Control	What it does	Why it's there (the teaching point)
Average the samples in: dB / Milliwatts (mW)	Averaging domain.	The answer moves (-58.1 against -56.2 dBm at $p = 2$), and which domain products use is not published.
Guess range (1 to 20 m, default 5; Up/Down in Present)	Cells farther than this from every sample are white.	Keith's Rule 6: no data is not no coverage. The fix is to go get the data, not to widen the range.
Beyond the guess range: Off / Flat IDW / Path-loss fill	What fills past the guess range.	IDW can never go below its lowest sample, so it goes flat past the last point; a path-loss model can extrapolate distance but cannot see an unmeasured wall.
Map view: Heat map / Truth / Error map (E in Present)	Switches what the floor shows.	The error map is where the hidden wall shows up.
Start: three dots, then Reveal: widen the guess range, Reveal: the hidden wall, End the lesson	The built-in three-step lesson (N in Present).	The spec's question: "Three dots can paint the whole floor green. Should they?"
Reveal wall / Hide wall	Shows or hides the 12 dB wall no sample crossed (W in Present).	A model cannot see a wall no one measured across: about 17 dB of error on the far side.
Spacing experiment	Re-runs the grid at several spacings and plots RMSE against spacing.	Shows accuracy improving with spacing, then flattening once noise dominates.
Noise on one reading, sigma (0 to 8 dB, illustrative); Readings averaged per point (1 to 36); Take the samples again (Space in Present)	Adds fading noise and averaging.	Averaging N readings shrinks the noise by the square root of N.
Inspected cell, across / down	Moves the inspected cell by the numbers.	A keyboard way to inspect a cell (<i>inferred</i>).
APs (1 to 4, default 2); Path-loss exponent (n) (2.0 to 4.0)	Sets the truth the samples read.	Changes the floor under the same method so the class sees the method's error, not the building's (<i>inferred</i>).

Present mode: Space takes the samples again with new noise; R resets; Up and Down move the guess range 1 m; P steps the power through 1, 2, 4 and 6; E switches heat map, truth and error map; W shows or hides the hidden wall; N is the next step of the three-dots lesson.

A demo that works (about 3 minutes)

- 1 Set up:** R, then press Start: three dots (N).
- 2 Ask the room:** "Three dots can paint the whole floor green. Should they?"
- 3 Reveal:** N widens the guess range and the floor fills with confident color from three readings; N again reveals the hidden wall on the error map, about 17 dB wrong on its far side.
- 4 Optional follow-up:** press P to step the power and watch the map turn blocky, or run the Spacing experiment.

What it leaves out on purpose: the truth is a log-distance model with wall losses, and the noise is illustrative; real maps also carry antenna, adapter and position errors. IDW uses the 8 nearest samples: plain IDW over every sample rises past the last point and would fail the "goes flat" lesson (build notes). The spec says Keith's own teaching write-up, when it arrives, becomes the primary model.

Pairs with: Survey Walk before it, Predict, Then Measure after it. Lesson sequence "Surveys and heat maps". No lab handout yet.

Predict, Then Measure

predict-then-measure

Teaches: every wall loss in a predictive design is a claim, and only an AP-on-a-stick walk with samples on both sides of a wall can test it (Keith's Rule 5).

Controls

Control	What it does	Why it's there (the teaching point)
Scenario (illustrative): Office / School / Warehouse	Loads a floor, walls and a hidden truth.	Different buildings, same lesson (<i>inferred</i>).

Control	What it does	Why it's there (the teaching point)
Reset scenario	Resets the design, the AP, the walk and the reveal; keeps the hidden truth.	Lets you walk the same building again.
A tap or drag on the floor will: Draw the walk / Move the AP	Drag draws a walk leg (a tap continues it); or moves the AP on a stick.	The student designs the validation walk.
One side only; Both sides of every wall (O and B in Present)	Loads a corridor walk that stays on one side, or short legs across every wall.	The Rule 5 contrast: one side leaves walls untested; both sides tests them.
Remove the last leg of the walk; Clear the walk (C in Present)	Undo and clear.	Rebuild the walk (<i>inferred</i>).
Noise on each sample (illustrative) (0 to 8 dB; Up/Down in Present)	Adds noise to each reading.	Shows what real readings do to the wall estimates.
Map: Predicted / Measured / Difference / Truth (Truth only after reveal; M in Present)	Switches the map.	The Difference map is each sample minus the design at that spot, which is where a wrong wall shows.
Update model (U in Present)	Replaces each tested wall's design loss with its measured loss.	After correcting the model, the design may need an AP moved, added or removed. Untested walls keep what the design said.
Reveal the truth / Hide the truth (Space in Present)	Shows the true wall losses and the Truth map.	The reveal for "The design shows green everywhere. What would you check before signing it off?"
Wall, "Wall n design loss" (0 to 45 dB), "<Material> default, X dB"	Edits a wall's predicted loss; resets it to its material default.	Every loss is illustrative: drywall 3, glass 4, brick 10, concrete 15, elevator shaft 25 dB.
AP across, AP down	Moves the AP on a stick.	Moving the AP walks the same path again from the new spot.
Instructor: the hidden truth: "Wall n true loss", New hidden truth	Sets any wall's true loss, or draws a new seeded truth.	The instructor can plant the wrong wall where the lesson needs it. By default two walls are much worse than the design and one is better.

Present mode: Space reveals or hides the truth; M steps through the maps; U updates the model; C clears the walk; B walks both sides of every wall; O walks one side only; Up and Down change the noise; R resets. Tab is not a map key: it moves between controls.

A demo that works (about 3 minutes)

- 1 **Set up:** Office, Reset scenario. Show the Predicted map: green, every wall "untested".
- 2 **Ask the room:** "The design shows green everywhere. What would you check before signing it off?"
- 3 **Reveal:** press O (one side only) and show the walls still untested. Press B (both sides): walls turn "tested" with measured losses, and the Difference map lights up the wrong ones. Press Space to reveal the truth and show which wrong walls the first walk missed.
- 4 **Optional follow-up:** press U and show the Predicted map change, then ask whether the AP needs to move.

What it leaves out on purpose: straight-line wall losses only: no doors, reflections or floors above and below. Wall losses are illustrative values, not material data; no primary source for multi-wall coefficients was used. An AP-on-a-stick walk tests the walls it crosses and nothing else.

Pairs with: Heat Map Builder (the same sample-and-map method), Survey Walk, Room Propagation and Wi-Fi Through a Wall for how one wall attenuates. Lesson sequence "Surveys and heat maps". No lab handout yet.

Where Am I? Signal Strength vs Round-Trip Timing

location-rssi-ftm

Teaches: signal strength gives distance badly, as a factor that grows with distance, while fine timing measurement (FTM) gives meter-class distance, and a blocked direct path makes timing read long.

Controls

Control	What it does	Why it's there (the teaching point)
Method: Signal strength / Timing / Both (Tab or M in Present)	Shows one method's circles and scatter, or both side by side. Default Both.	Side-by-side scatters make the difference visible at a glance.
Re-sample (Space in Present)	Draws new measurement errors.	The 50 repeated estimates show the spread, not one lucky answer.
Ask: -70 dBm, how far?; Reveal the answer; End the question (P in Present)	The built-in question.	The spec's question: "Your phone sees an AP at -70 dBm. How far away is it?"
Path-loss exponent (n) (2.0 to 4.0, default 3.0)	The model the distance is inverted with.	The one-sigma error factor is $10^{(\sigma / 10n)}$.
Shadowing sigma (illustrative) (0 to 10 dB, default 6; Up/Down in Present)	Random loss on each signal reading.	A few dB of shadowing becomes meters: at 6 dB and $n = 3$ the factor is 1.585, so 10 m reads 6.3 to 15.8 m.
Timing error, one standard deviation (vendor-documented 1 to 2 m) (0.5 to 3 m, default 1.5)	FTM ranging noise.	Keeps the timing claim tied to its source, a vendor developer document.
Blocked AP chips (AP 1 to AP n; B toggles AP 1 in Present); Blocked-path extra distance (illustrative) (0 to 10 m, default 4)	Blocks an AP's direct path; its circle is dashed and reads long.	The first path to arrive is a reflection, so timing reads long, never short.
APs (3 to 6, default 4); Device position, across / down (or drag the device)	Sets the geometry.	Three or more distances give a position; bad distances give a bad, and sometimes impossible, position (spec point 4).

Present mode: Space re-samples; Tab switches signal strength, timing and both while no control has focus (Shift+Tab moves into the controls); M also switches the method; R resets; Up and Down move the shadowing sigma 1 dB; B blocks or clears AP 1's direct path; P steps the -70 dBm question.

A demo that works (about 3 minutes)

- 1 **Set up:** R. Method Both, four APs, sigma 6 dB.
- 2 **Ask the room:** press P. "Your phone sees an AP at -70 dBm. How far away is it?" Take numbers.
- 3 **Reveal:** press P again. The model says 21.1 m (69 ft), and one sigma of shadowing makes it 13.3 to 33.5 m (help example). Point at the two scatters: timing is tight, signal strength is wide.
- 4 **Optional follow-up:** press B to block AP 1 and watch the timing position pull off while its circle turns dashed.

What it leaves out on purpose: the log-distance model, shadowing, 17 dBm per AP and the blocked-path distance are illustrative. The 1 to 2 m timing figure is a vendor document, not a measurement here. Real systems also face AP position errors, clock offsets and floors above and below.

Pairs with: the Find My, Explained lesson (where the dot on the map comes from), Roaming Walk for the same log-distance model. No lab handout yet.

Repeaters and Mesh Backhaul

repeater-mesh

Teaches: a relay with one radio on one channel makes every frame cross the air twice, so the hops take turns ($1/T = 1/T1 + 1/T2$); a dedicated backhaul radio or a cable lets them run at once, and a weak hop back to the AP limits everything behind it.

Controls

Control	What it does	Why it's there (the teaching point)
Predict, then reveal: Ask the class; three reasons; Reveal; Done / Close the question	Loads the question's scene (repeater 30 m down the corridor, laptop 2 m from it) and hides the throughputs.	The spec's question: "Your repeater shows full bars to the laptop. Why is it slower than before?"
Relays: 1 / 2 / 3 (Up/Down in Present)	Number of relays, spaced evenly.	Each extra same-channel hop costs airtime again and adds delay. At the default spot two relays beat one, and three lose to two.
Backhaul: Same channel, one radio; Dedicated backhaul radio, another channel; Wired backhaul	How the relays reach the AP.	The core contrast: turns on one channel against the slowest hop (dedicated) against the last hop alone (wired).
Band: 2.4 / 5 / 6 GHz	Changes path loss and channel width.	Shows the same relay lesson at each band's range (<i>inferred</i>).
Efficiency (illustrative) (0.40 to 0.80, default 0.60)	Turns PHY rate into what a hop carries.	Labeled illustrative so no one reads the Mb/s as a measurement.
Forwarding delay per hop (illustrative) (0 to 10 ms, default 1)	Adds per-hop delay.	Relaying adds delay as well as cutting throughput.
Relay position / Client position (or drag on the corridor)	Moves a node along the 60 m (197 ft) corridor.	The spec's point 5: where the relay stands matters more than where the client is. The sliders are the keyboard way to do what dragging does (code comment).
Play the frames / Pause, Reset	Animates frames taking turns, or running in parallel with a dedicated backhaul.	Makes the shared air visible.

Present mode: Space plays or pauses the frames; Up and Down add or remove a relay; R resets.

A demo that works (about 3 minutes)

- 1 **Set up:** Reset. One relay, same channel, 5 GHz.
- 2 **Ask the room:** press Ask the class and read the question. Pick the class's reason.
- 3 **Reveal:** press Reveal. The laptop has MCS 11 to the repeater, but the repeater hears the AP at MCS 1: 77.2 Mb/s through the repeater against 86.5 Mb/s the laptop got straight from the AP (help example).
- 4 **Optional follow-up:** drag the repeater toward the AP (at 12 m the total rises to 115.3 Mb/s), then switch Backhaul to Dedicated and press Space to show both hops sending at once.

What it leaves out on purpose: the formula is for one radio relaying on one channel, worked out from airtime; no published multihop measurement was checked. With several relays sharing one dedicated backhaul channel, those hops would take turns again. Traffic flows one way, with no neighbors, collisions or retries. The delay counts only forwarding, not waiting for the channel.

Pairs with: Airtime Fairness, the Mesh, Extenders and Wired Access Points explainer. No lab handout yet.

Why a Busy Line Lags

latency-under-load

Teaches: lag is delay, not speed: an upload fills the line's queue on any plan and the call waits behind it, and smart queue management keeps the queue short where a faster plan does not.

Controls

Control	What it does	Why it's there (the teaching point)
Smart queue management (SQM): Off / On (Up turns on, Down turns off, Q switches in Present)	Switches the router's queue management. The other setting stays as a dashed line.	The research brief's "one control": the latency trace drops from hundreds of milliseconds back toward idle. It corrects "more Mbps fixes lag".
The home line: Fiber / Cable / DSL (default Cable)	Loads the FCC's figures for that technology and rescales the chart.	Keith kept this picker beside the one switch (2026-09-27) so the class sees DSL suffer most.
Play the call / Pause, 1 s on, Reset	Runs the 20 s in real time or steps it.	Lets you stop while the upload runs and read the delay (<i>inferred</i>).
Time into the run (0 to 20 s), or drag across the chart	Moves the playhead; the queue picture shows what the call's packet waits behind.	The queue picture is the mechanism behind the number (<i>inferred</i>).
Open Network Quality, Open Test My Connection	Opens the two tools that measure a real line under load.	Moves the class from the model to their own line.

Present mode: Space plays or pauses; Right steps one second; Up turns SQM on and Down turns it off; Q switches it; R resets.

A demo that works (about 3 minutes)

- 1 Set up:** Reset (Cable, SQM off, whole run drawn).
- 2 Ask the room:** "Someone starts a big upload during your video call. Does a faster plan fix the lag?"
- 3 Reveal:** play it. The call goes from 18 ms idle to 225 ms, 12.5 times idle. Press Up: with SQM on the same upload leaves it at 23 ms (help example).
- 4 Optional follow-up:** switch to DSL with SQM off: 29 ms to 665 ms.

What it leaves out on purpose: the SQM-on curve is illustrative (idle plus FQ-CoDel's 5 ms target); no consumer measurement of it was found. The busy figures are read by eye from

the FCC's chart. Left out: the Wi-Fi hop, queues inside the provider and at the far end, jitter and loss. There is no built-in predict question on this tool; ask it yourself.

Pairs with: Test My Connection and Network Quality, the Home Internet explainer. No lab handout yet.

DFS and Radar

`dfs-simulator`

Teaches: before transmitting on a DFS channel an AP must listen for radar (60 s, or 10 minutes on EU channels 120 to 128); after radar it must leave within 10 s and stay off for 30 minutes, and the only real choice is where it goes next.

Controls

Control	What it does	Why it's there (the teaching point)
Play / Pause / Play again, Step 10 s, Restart, Clock, Clock speed: 1x (real time), 10x, 60x (1 min per second), 300x (5 min per second)	Runs the one-hour clock. Default 10x.	A 30-minute block and a 10-minute CAC need a fast clock to fit a class (<i>inferred</i>).
Radar now (D in Present)	The AP detects radar now; unavailable on a non-DFS channel.	The spec's radar-event button: shows the closing, the move and the 30-minute block.
Region: US (FCC) / EU (ETSI)	Swaps the channel plan and the rules.	The 10-minute CAC is ETSI only; channel 144 is DFS in the US and not in the EU plan.
Width (20, 40, 80, 160 MHz)	Channel width. Default 20.	Wider channels touch more DFS channels (<i>inferred</i>).
Starting channel (default 100)	Where the AP starts; the list says which are DFS and how long each CAC takes.	Starting on DFS shows the CAC before the first frame.
After radar: Prefer non-DFS / Another DFS	Where the AP moves.	The only real choice: a non-DFS channel is available at once; another DFS channel means another CAC.
Random radar: Off, 2, 6 or 12 per hour; New random pattern	Adds seeded random radar.	Shows repeated moves over an hour.
Timeline: Whole hour / 3 minutes	Zooms the timeline.	A short move is invisible at the hour scale.
AP EIRP (17 to 30 dBm, default 23)	Feeds the detection-threshold readout only.	Detection thresholds are shown for context, not modeled.

Present mode: Space plays or pauses; Right steps 10 s; D fires radar now; R restarts; Up and Down move the clock one minute.

A demo that works (about 3 minutes)

- 1 **Set up:** Restart. US, 20 MHz, channel 100, Prefer non-DFS.
- 2 **Ask the room:** "The AP reboots on channel 100. When does it send its first frame? And if radar hits at 5:00, how long are clients without service?"
- 3 **Reveal:** play. First transmission at 1:00. Press D near 5:00: it moves to 36, a 1.0 s outage, and 100 is blocked until 35:00 (help example).
- 4 **Optional follow-up:** switch After radar to Another DFS (104, a 61.0 s outage), then EU, starting on 116: the move lands on 120 and waits 10 minutes.

What it leaves out on purpose: the FCC's 60 ms of control signals comes from one secondhand source. The AP's 1 s move and client rejoin delays are illustrative. Not modeled: EU off-channel CAC, a second radio listening while the first serves, radar signal levels. Some real APs block only the part of a wide channel that saw radar.

Pairs with: Channel Planner (DFS on or off), the 5 GHz channel card in Course Handouts. Lesson sequences "Channels and the rules" and "Designing a cell plan". No lab handout yet.

Conference Wi-Fi Runs Out of Addresses

dhcp-exhaustion

Teaches: the DHCP pool has to cover everyone who arrived within one lease time, not the people in the room, so an event network runs dry with full bars, and adding APs adds no addresses.

Controls

Control	What it does	Why it's there (the teaching point)
Predict, then reveal: Ask the class; Add more APs / A shorter lease or a bigger pool / A faster internet line; Reveal; Close the question (P in Present)	Loads the defaults and stops at the first device with no address.	Corrects "the Wi-Fi is down, add APs" (research brief).
Play the morning / Pause, +10 min, Reset, Time	Runs 07:00 to 13:00.	The dry time is a moment in the morning, so the class watches it arrive (<i>inferred</i>).
Lease time (5 min to 8 days, default 1 day; Up/Down in Present)	The server's lease.	The brief's one control: the pool runs dry mid-morning with long leases and holds with short ones.
Devices rotate their private address on this open network (M in Present)	Rotating devices come back as new clients.	Phones that rotate on open networks take a second address. One platform maker documents rotating as the default on open networks.
Share of devices that rotate (assumption) (0 to 100%, default 50); A rotating device comes back as new every (assumption) (15 to 240 min, default 60)	The rotation assumptions.	No device maker publishes a rotation interval, so both are labeled assumptions.
Subnet: /24 to /19	Pool size.	The other fix: a /23 holds even with the 1-day lease.
Reserved: gateway, servers, printers (illustrative) (0 to 100, default 10)	Addresses taken off the top.	Set to your event (<i>inferred</i>).
People over the morning (50 to 5,000, default 250); Devices each person puts on the Wi-Fi (1.0 to 3.0, default 1.5); Average stay (15 to 360 min, default 90)	The crowd. All illustrative.	Keith kept the example defaults (2026-09-27); the help says set them to your event.
Open Association, Frame by Frame; Open Subnet Planner	Links out.	One DHCP exchange frame by frame, and subnet sizing (<i>inferred</i>).

Present mode: Space plays or pauses the morning; Right moves 10 minutes; R resets; Up and Down change the lease time; M turns rotation on or off; P asks, reveals and closes the question.

A demo that works (about 3 minutes)

- 1 **Set up:** Reset, then P (Ask the class).
- 2 **Ask the room:** "Mid-morning, full bars, nothing loads. What fixes it?"
- 3 **Reveal:** P again. The pool (244 addresses) ran dry at 10:18 with only 113 devices in the hall: 132 addresses are still held for devices that left (help example). Press Down until the lease reads 30 min and the same morning holds.
- 4 **Optional follow-up:** set a 2-hour lease (it holds), then press M for rotation: it runs dry at 09:21.

What it leaves out on purpose: nothing is released early, though some devices do send a release when they leave on purpose. Not modeled: short leases for new clients, per-network leases, captive portals, IPv6. The crowd, arrival shape and stay are illustrative.

Pairs with: Association, Frame by Frame (one DHCP exchange), Subnet Planner. No lab handout yet.

Down the Stack, Across the Air, Up the Other Side

down-the-stack

Teaches: data gains a port, an IP address and a MAC address on the way down; the IP addresses stay the same end to end while the MACs change on every hop, and Address 3 on the air is the router's MAC, not the server's.

Controls

Control	What it does	Why it's there (the teaching point)
Predict, then reveal: Ask the class; The server's / The router's / The AP's; Reveal; Close the question	Loads laptop to a server on another subnet; Reveal jumps to where the laptop fills in the 802.11 header.	The predict prompt "Whose MAC is in Address 3?" is final (Keith kept it).
Play / Pause, Back, Step, Reset, Journey step (Up/Down in Present)	Moves the data one layer or hop at a time.	Each header added, read and replaced is one step, so the class can watch each change (<i>inferred</i>).
Show: The journey / Address fields (V in Present)	Switches the drawing and the address table.	Separates the path from the four address fields (<i>inferred</i>).
To DS / From DS buttons: 0 0, 0 1, 1 0, 1 1 (D in Present)	Shows each case's worked scene and Addresses 1 to 4.	The fourth address exists for a hop where both radios are relays, such as a mesh or WDS backhaul.
Direction: To the server / The reply	Reverses the journey.	Shows the MACs change on every hop in both directions (<i>inferred</i>).
Server: Another subnet / Same subnet	Adds or removes the router.	On the same subnet there is no router and Address 3 becomes the server.
Transport: TCP / UDP	20-byte or 8-byte header.	Shows the transport header's size in the stack (<i>inferred</i>).
Data: 100, 500 or 1460 bytes (fills 1500 with TCP)	Payload size.	1460 bytes fills a 1500-byte IPv4 packet, so the class can add up 1538 bytes on the air (<i>inferred</i>).
QoS (quality of service) Data header, 26 bytes	26-byte QoS Data header or the textbook 24-byte header.	Nearly every data frame on a modern network is a QoS Data frame (on-screen note).

Present mode: Space plays or pauses; Right steps; Up and Down move one step forward or back; D shows the next To DS / From DS case; V switches between the journey and the address fields; R resets.

A demo that works (about 3 minutes)

- 1 **Set up:** Reset. To the server, Another subnet, TCP, 1460 bytes.
- 2 **Ask the room:** press Ask the class. "On the air, whose MAC address is in Address 3?"
- 3 **Reveal:** Reveal jumps to the 802.11 header: Address 1 is the AP's BSSID, Address 2 the laptop, Address 3 the router's LAN interface. Step on and show the router lower the TTL from 64 to 63 and build a new frame, while 192.0.2.10 and 198.51.100.20 never change (help example).
- 4 **Optional follow-up:** switch Server to Same subnet and show Address 3 become the server; press D through the four DS cases.

What it leaves out on purpose: no NAT (most home routers change the source IP). MACs, port 51000 and TTL 64 are illustrative; IPs come from the RFC 5737 documentation ranges. Layers 5 and 6 are not drawn. Left out: ARP, the ACK, the PHY preamble, encryption, aggregation, VLAN tags, IPv6, and 802.11s six-address mesh frames.

Pairs with: A Frame's Journey (on Signals and PHY, the air hop, FCS, SIFS and the ACK), Association, Frame by Frame. No lab handout yet.

Association, Frame by Frame

join-ladder

Teaches: Open System authentication and association are bookkeeping, not security; the keys come in the 4-way handshake, and the client still has no IP address until DHCP and an address check finish, which is often where a slow association spends its time.

Controls

Control	What it does	Why it's there (the teaching point)
Play / Pause / Play again, Back, Step, Reset, Show all, Speed: Slow / Normal / Fast	Sends one frame at a time, takes one back, or draws the whole ladder.	One frame at a time is how the class reads a capture (<i>inferred</i>).
Tap any sent message	Shows what it carries; the Association Request lists SSID, rates, RSN element with the PMF bits, and HT to EHT capabilities.	Association is where client and AP agree on capabilities (spec point 2).

Control	What it does	Why it's there (the teaching point)
Security: Open, OWE (Enhanced Open), WPA2-Personal (PSK), WPA3-Personal (SAE), 802.1X (Enterprise) (default WPA2-Personal)	Changes the middle of the ladder.	SAE replaces Open System authentication before association; 802.1X hands off to the EAP ladder's middle.
EAP method, Inner method (EAP-TTLS), Certificate size (802.1X only)	The EAP middle, shared with the 802.1X ladder.	Reuses the ladder's engine by import, not a copy (spec 21b).
PMF (802.11w): Off / Optional / Required	The RSN element's MFPC and MFPR bits; forced where the standard requires it.	PMF protects management frames only once keys exist; beacons, probes, authentication and association are never protected.
Band: 2.4 / 5 / 6 GHz (default 5)	The AP is on 6, 44 or 37. In 6 GHz, Open is drawn as OWE and WPA2 as WPA3.	6 GHz requires WPA3 or OWE with PMF.
Scan: Active / Passive	Probe requests, or listening for beacons.	A passive scan waits for beacons every 102.4 ms, a default, not a mandate.
6 GHz discovery: PSC scan / Known via RNR (6 GHz only)	Probe the 15 PSCs, or learn the AP from a 2.4 or 5 GHz beacon.	The spec's point 1: 6 GHz discovery differs.
Address check: ACD / DNav4	Address Conflict Detection (three ARP probes 1 to 2 s apart) or Detecting Network Attachment (one unicast ARP).	Keith asked to show both options (2026-09-26); ACD is drawn as the long bar it is.
Timing (illustrative): Active dwell per channel (default 30 ms), Passive dwell per channel (default 111 ms; Up/Down in Present), Time per frame over the air, RADIUS round-trip time, Crypto time (public-key work), Wired LAN round trip (DHCP, gateway, DNS), ACD wait before the first probe, ACD time between probes	Every phase time.	No published measurement breaks a join down by phase, so every phase is a labeled setting; the dwell defaults are Linux mac80211 as one real example.

Present mode: Space plays or pauses; Right sends the next frame; Left takes one back; R resets; Up and Down change the passive dwell by 10 ms.

A demo that works (about 3 minutes)

- 1 **Set up:** Reset. WPA2-Personal, 5 GHz, Active, ACD.
- 2 **Ask the room:** "After association, is the client on the network?" Then: "Of the 7.6 seconds this join takes, where does the time go?"
- 3 **Reveal:** Step through: keys arrive only in the 4-way handshake, and the timeline shows scan 2.05 s and ACD 5.50 s against a few milliseconds of Wi-Fi security (help example). Switch Address check to DNav4: the join drops to 2.08 s.
- 4 **Optional follow-up:** switch Scan to Passive and press Down until the dwell is below 51 ms: the scan misses the beacon and the AP is not found (at 40 ms per the help).

What it leaves out on purpose: every phase time is a setting. Not drawn: channel changes, retries, the DHCP client's start-up delay, IPv6, 802.11ai (FILS). With DNav4 a real client often skips the four DHCP messages; the ladder keeps them so the two checks compare side by side. The Block Ack setup at the end is there to show the PMF shield and is not counted.

Pairs with: 802.1X and EAP Ladder (Roam mode for the roam), Why a Long Wi-Fi Password Matters More on WPA2 (links in to the 4-way handshake and SAE), Conference Wi-Fi Runs Out of Addresses. Lesson sequences "Association and roaming" and "Wi-Fi security modes". No lab handout yet.

802.1X and EAP Ladder

eap-ladder

Teaches: in 802.1X the AP relays and the RADIUS server authenticates; EAP rides in EAPOL over the air and inside RADIUS on the wire, the PMK arrives in the Access-Accept, and nothing is encrypted until the 4-way handshake completes.

Controls

Control	What it does	Why it's there (the teaching point)
Mode: Authenticate / Roam	Three-lane ladder, or four lanes with a target AP.	Roam mode compares a full 802.1X reconnect with the fast roams, and puts the scan back in the total.
Play / Pause / Play again, Back, Step, Reset, Show all, Speed: Slow / Normal / Fast	Sends one message at a time, takes one back, or draws the whole ladder.	Show all lets you switch methods with the whole ladder drawn and compare them (help).
Method: EAP-TLS, PEAP (MSCHAPv2), EAP-TTLS, WPA2-Personal (PSK), WPA3-Personal (SAE) (Authenticate)	Changes the middle of the ladder.	Keith's original ask (2026-09-25): "change for different types, like TLS, MS-CHAP". PSK and SAE are there for contrast: no RADIUS at all.
Inner method: MSCHAPv2 / PAP (EAP-TTLS only)	The method inside the TTLS tunnel.	Only EAP-TTLS has a choice; PEAP here always carries MSCHAPv2.
Roam mode: Full / PMK caching / FT (802.11r) (Authenticate)	What a return or roam skips.	PMK caching skips EAP; FT folds the handshake into four frames.
Certificate size (fragments per certificate message) (1 to 6; Up/Down in Present)	Splits certificates into more EAP messages.	Round trips depend on certificate size and fragmentation, so it is a setting, not a fixed count.
RADIUS round-trip time (illustrative) (1 to 200 ms, default 10)	Time per trip to the server.	A distant server slows every round trip.
Roam method: Full 802.1X, PMK caching, OKC, FT over the air, FT over the DS (Roam)	Which roam to play.	OKC is labeled a vendor extension, not IEEE; FT over the DS starts with an Action frame to the current AP.
Band, Scan, PMF (802.11w), 6 GHz discovery, EAP method, Timing sliders (Roam)	The same settings as Association, Frame by Frame, plus AP to AP round trip, over the DS.	The timeline splits scan, authentication and key handshake: FT shrinks the middle while the scan bar stays.

Present mode: Space plays or pauses; Right sends the next message; Left takes one back; R resets; Up and Down change the certificate size. The Mode toggle sits at the top of the panel.

A demo that works (about 3 minutes)

- 1 **Set up:** Reset. Authenticate, EAP-TLS, Full, 1 fragment.
- 2 **Ask the room:** "Who checks the client's certificate, the AP or the server? And when is the air encrypted?"
- 3 **Reveal:** step through: every EAP message crosses the air and then the wire; Keys available lights after the Access-Accept, Traffic protected only after message 4. 19 frames over the air, 8 RADIUS messages, 4 round trips (help example).
- 4 **Optional follow-up:** switch Mode to Roam with the full ladder shown, then pick FT over the air: authentication drops from 103 ms to 2 ms while the 2.05 s scan does not move.

What it leaves out on purpose: TLS follows the TLS 1.2 shape; TLS 1.3 and session resumption are not modeled. Authenticate mode does not count the scan; Roam mode does. The TLS tunnel protects the inner exchange, not the Wi-Fi link. The FT captures (14 ms and 88 ms) are single captures, shown as context only; vendor roam-time ranges are not shown.

Pairs with: Association, Frame by Frame, the EAP Types reference, Roaming Walk. Lesson sequences "Association and roaming" and "Wi-Fi security modes". No lab handout yet.

Why a Long Wi-Fi Password Matters More on WPA2

wpa2-password

Teaches: on WPA2-Personal one recorded join lets an attacker test guesses offline, so length is the whole defense; on WPA3 every guess is a live exchange the AP can see, and transition mode keeps the WPA2 door open.

Controls

Control	What it does	Why it's there (the teaching point)
Network security: WPA2 / WPA3 / Transition (1, 2, 3 in Present)	Changes where a guess is checked. The password stays the same; the counts start over.	The research brief's one control, with the same short password on all three.
Length (8 to 63; Up/Down in Present)	Password length.	Each extra character multiplies the possibilities by the characters in play.
Characters: Lowercase letters (26), Lowercase letters and digits (36), Upper and lowercase letters and digits (62), Any keyboard character (95)	The character set.	Possible passwords = characters ^ length.
Keep guessing / Stop guessing (Space), Try one guess (Right), Reset (R)	Runs the guessing dot. On WPA2 and transition it loops inside the attacker's computer; on WPA3 it travels to the AP, which logs it.	The visible difference: the AP's failed-attempt count stays at 0 on WPA2. The pace is slowed to follow and is not a speed.
Reveal the answer / Hide the answer (P in Present)	The built-in question, from the Wi-Fi Alliance's example of 5,000 possible passwords.	Uses the source's own example rather than invented crack times.
The WPA2 4-way handshake, The WPA3 SAE exchange, Open WPA Security	Links to the frames.	Shows the frames behind the guess (<i>inferred</i>).

Present mode: Space keeps guessing or stops; Right tries one guess; 1, 2 and 3 choose WPA2, WPA3 or transition; Up and Down change the length; P reveals the answer; R resets.

A demo that works (about 3 minutes)

- 1 **Set up:** R. WPA2, 8 lowercase letters.
- 2 **Ask the room:** "The password is one of 5,000 and the attacker has the list. How much of the list can they try on WPA2? On WPA3?"
- 3 **Reveal:** P. On WPA2 they test all 5,000 offline and find it for certain; on WPA3 the chance reaches 50% only after 2,500 live attempts at the AP (help example). Press Space on WPA2, then 2 and Space on WPA3, and watch the AP's count.
- 4 **Optional follow-up:** press 3 for transition mode and show the dot back inside the attacker's computer.

What it leaves out on purpose: no time to crack anywhere: it depends entirely on the attacker's hardware (research brief: no crack-time numbers). The count assumes a random password; words and patterns fall long before it runs out. Enterprise (802.1X) is left out, along with attacks on particular products, tools and steps.

Pairs with: Association, Frame by Frame (the 4-way handshake and SAE), 802.1X and EAP Ladder, WPA Security reference. Lesson sequence "Wi-Fi security modes". No lab handout yet.

PoE: Why the New AP Runs at Half Strength

poe-half-strength

Teaches: a Wi-Fi 7 AP on a port that cannot supply its full power still boots and lights up, then quietly runs with fewer streams or fewer radios.

Controls

Control	What it does	Why it's there (the teaching point)
Switch port type: 802.3af / 802.3at / 802.3bt (Up, Down, Right, or 1, 2, 3 in Present; default 802.3at)	Changes the power reaching the AP; the stream marks and power bar update. The power light stays on in all three.	The research brief's one control, which corrects "the light is on, so power is fine".
What the AP gives up on 802.3at: Three at 2x2 / Two at 4x4 (Space in Present)	6 of 12 streams, or 8 of 12 with 2.4 GHz off. Applies only on 802.3at.	The vendor guide the model follows lists both; vendors cut different things.
Reveal the answer / Hide the answer (P in Present), Show it on 802.3at	The built-in question, then a jump to the at port.	"The new Wi-Fi 7 AP goes on last year's 802.3at switch and its power light comes on. Is it running at full strength?"
Reset (R)	Back to the defaults.	Returns to the question's port (<i>inferred</i>).
Open PoE Budget, Open PoE Reference	Links out.	For adding up a whole switch and every PoE class (<i>inferred</i>).

Present mode: Up and Down or Right change the port type; 1, 2 and 3 pick 802.3af, 802.3at and 802.3bt; Space switches what the AP gives up on 802.3at; P reveals the answer; R resets.

A demo that works (about 3 minutes)

- 1 Set up:** R, then press 3 (802.3bt): 51.0 W reaches the AP and all 12 streams run.
- 2 Ask the room:** read the question. "The light is on. Full strength?"
- 3 Reveal:** press 2 (802.3at): 25.5 W, 3.5 W short of the about 29 W it needs, and it runs 6 of 12 streams with the light still on (help example). Press Space for the 8-of-12 choice with 2.4 GHz off.
- 4 Optional follow-up:** press 1 (802.3af). Keith ruled it boots with fewer capabilities rather than not at all (2026-09-27); the tool shows one illustrative reduced set.

What it leaves out on purpose: one generic AP on one published guide; read your own AP's data sheet. The 802.3af behavior has no source and is labeled illustrative. A 2x2

client alone may notice little; the loss is in streams shared among many clients. Dual-port 802.3at feeds, IoT and scanning radios, and short cables are left out.

Pairs with: PoE Budget, PoE Reference. No lab handout yet.

Course Handouts

These are the WLAN Pros printed reference cards, each bundled in the app as a PDF. They are not simulators: there is nothing to play, step or set, and there is no Present button. Every card opens in the same viewer, with **Zoom in, Zoom out, Fit page to window**, page arrows, pinch or double-tap to zoom, and **Share or download** to hand a student the PDF. The cards display offline, so a room with poor Wi-Fi does not stop them. A screen reader announces the card name and the zoom gesture but cannot read what is printed on the card.

The Teacher's Guide use is simple: put a card on the projector while you talk through it, and the students have the same card in their own copy of the app. The channel cards open the "Channels and the rules" sequence in the Teacher's Guide. Where else to hand a card out is our suggestion from what is printed on it, and those lines are marked (*inferred*).

The four checklists (Top 20, Extended, Extended non-advertised, Connection) are dated V.5, May 24, and scored: enter 1 or 0 per item per band and the card works out a percentage for 2.4, 5 and 6 GHz.

WLAN Pros Bubble Diagram

bubble-diagram

Teaches: every piece a Wi-Fi connection depends on, drawn as one map: the client and end user, the RF medium processes (DFS, RRM, MCS, contention, overhead), association, authentication and encryption paths, the AP, and the wired side out to DHCP, DNS, the controller, captive portal and the authentication database.

When to hand it out in a course: on day one, as the map of the course, and again at the start of troubleshooting. (*inferred*)

How to use it in class: project it and ask the room where a reported problem could live, then trace the path from the client to the service. It is the picture version of the Troubleshooting Causes card, so show the two together. It is the only landscape card, and the viewer was set up to show it undistorted.

Pairs with: Troubleshooting Causes; Association, Frame by Frame; 802.1X and EAP Ladder.

Wireless LAN Troubleshooting Causes

troubleshooting-causes

Teaches: the numbered list of places a Wi-Fi problem can come from, from the end user's skills and perceptions, through the client device, the RF medium, contention, the MCS decision, DFS, association, 802.11k/v/r, authentication and encryption, to the upper layers, the controlled port and the fixed access configuration.

When to hand it out in a course: at the start of the troubleshooting module. (*inferred*)

How to use it in class: read a trouble ticket aloud and have the room place it on a numbered row before anyone opens a tool. Each row names what to measure (retry rates, average MCS, channel utilization and so on), so it also works as a list of what to capture.

Pairs with: Bubble Diagram, Channel Utilization Meter, Rate Adaptation, Why Two Devices Disagree.

Top 20 Wi-Fi Checklist

top-20-checklist

Teaches: the 20 settings and measurements to check first when evaluating a Wi-Fi installation against best practice, per band: channel plan and widths, co-channel interference, minimum basic rates, SSIDs across bands, primary and secondary RSSI, AP transmit power, WPA2/WPA3, 802.11k/r/v, channel utilization and clients per radio.

When to hand it out in a course: at the start of the design or assessment module, before any site survey exercise. (*inferred*)

How to use it in class: project it and go row by row, asking the room to predict the value for each band before you point at it. Rows marked ^ are ones the community still argues about (transmit power, WPA2/3 transition, 802.11r, utilization targets), and they make good discussion. The green boxes are meant to be changed for a site's own requirements.

Pairs with: Extended Checklist, Channel Planner, Rate vs Range, Uplink vs Downlink.

Extended Wi-Fi Checklist

extended-checklist

Teaches: items 21 to 50, the next layer after the Top 20: beacon and DTIM intervals, legacy security and 802.11b rates off, 160 and 320 MHz widths, channels 144 and 165, 6 GHz discovery and power classes, hidden SSIDs and SSID count, Wi-Fi 7 MLO, 802.11d/e/h/i/w/u, MU-MIMO, transmit beamforming and OFDMA.

When to hand it out in a course: after the Top 20, once the class can explain each Top 20 item. (*inferred*)

How to use it in class: pick the rows marked "Not Recommended" and ask the room why a feature on the spec sheet would be turned off. Transmit beamforming and MU-MIMO lead straight into the MIMO and Beamforming simulator.

Pairs with: Top 20 Checklist, MIMO and Beamforming, OFDMA Resource Units, Multi-Link Operation, Power Save.

Extended Checklist (Non-Advertised Items)

extended-checklist-nonadvertised

Teaches: items 51 to 66, the settings that are not in the beacons, so a scan cannot see them and you must ask the network administrator: automatic channels and widths, client isolation, airtime fairness, DFS events, mesh, band steering, load balancing, auto RF, PoE budget, non-802.11 interference and firmware.

When to hand it out in a course: with the Extended checklist, or at the point a class learns what a survey tool can and cannot see. (*inferred*)

How to use it in class: ask the room which of these a site survey scan would show. The card's point is that none of them is in the beacons, so you have to ask. Then take the rows marked ^ (mesh, band steering) and run the matching simulator.

Pairs with: Band Steering, Airtime Fairness, Repeaters and Mesh Backhaul, DFS and Radar.

Wi-Fi Connection Checklist

connection-checklist

Teaches: the 14 checks to run from a client, in order: see the SSIDs, Wi-Fi 7 MLO and the MLD MAC, associate, authenticate, get an address, gateway and DNS from DHCP, ping the gateway, DNS, a remote IP and a remote name, check the client's MCS and transmit rate, then a speed test.

When to hand it out in a course: with association and roaming, or as the first page of any hands-on lab. (*inferred*)

How to use it in class: walk the list and ask the room what failure at each step tells you and what it rules out. The order is the connection sequence, so the first failing step points at the layer. The card's footnote says speed tests belong in a non-production environment because of the load they put on the channel.

Pairs with: Association, Frame by Frame; the Connected, No Internet: Captive Portals lesson; The Number on the Box vs the Number in Your Hand for step 14.

2.4 GHz Channel Allocations

channel-allocations-24ghz

Teaches: the 2.4 GHz channel map: channels 1 to 14, with 1 to 13 spaced 5 MHz apart, the 22 MHz 802.11b and 20 MHz OFDM channel widths, why 1, 6 and 11 are the three non-overlapping 20 MHz channels, the non-US four-channel plan, and the 802.15.4/Zigbee channels on the same band.

When to hand it out in a course: at the start of "Channels and the rules", per the Teacher's Guide.

How to use it in class: project it and ask why the US uses 1, 6 and 11 when the band has 11 channels, then show the Zigbee row sitting between them. The center-frequency formula ($2407 + 5 \times \text{channel number}$) is on the card.

Pairs with: Channel Planner, Adjacent Channels and AP Stacking, Legacy Protection Cost, Fourier and FFT's Swept vs FFT race (the same band, with Bluetooth and a microwave oven in it).

5 GHz Channel Allocations

channel-allocations-5ghz

Teaches: the 5 GHz channel map by U-NII band: 28 20 MHz, 14 40 MHz, 7 80 MHz and 3 160 MHz channels, the DFS channels, the TDWR channels, channel 144 and U-NII-4, and the power and indoor/outdoor rules for the US, Canada and Australia.

When to hand it out in a course: in "Channels and the rules", after the 2.4 GHz card, per the Teacher's Guide.

How to use it in class: count the 80 MHz channels with the room, then take away the DFS ones and ask how many are left for a busy site. Use the regulatory rows to show the answer changes by country.

Pairs with: DFS and Radar, Channel Planner, Adjacent Channels and AP Stacking.

6 GHz Channel Allocations

channel-allocations-6ghz

Teaches: the US 6 GHz channel map, 1.2 GHz of spectrum across U-NII-5 to U-NII-8: 59 20 MHz, 29 40 MHz, 14 80 MHz, 7 160 MHz and 3 320 MHz channels, the Preferred Scanning Channels, and the three power classes most people know.

When to hand it out in a course: in "Channels and the rules", after the 5 GHz card, per the Teacher's Guide.

How to use it in class: ask why a client does not scan all 59 channels, then point at the Preferred Scanning Channels in bold. Follow with the GVP card if the class is US-based and current on the rules.

Pairs with: 6 GHz Channel Allocations with GVP, 6 GHz Power and PSD, Channel Planner.

Modulation and Coding Schemes (MCS Index)

mcs-index-card

Teaches: the HT, VHT and HE MCS index side by side: modulation, coding rate and the PHY rate for each channel width and guard interval.

When to hand it out in a course: right after the Modulation Simulator, when the class is ready to turn a constellation into a data rate. *(inferred)*

How to use it in class: pick one row, say MCS 7, and ask the room to predict what doubles the rate: the width, the streams, or a shorter guard interval. Then read across the card. It is a printed card and separate from the in-app MCS Index tool, which is the live lookup.

Pairs with: Modulation Simulator, Rate vs Range, The Number on the Box vs the Number in Your Hand, the MCS Index tool.

6 GHz Channel Allocations with GVP

channel-allocations-6ghz-gvp

Teaches: the same US 6 GHz chart with all four power classes, including Geofenced Variable Power, which entered the rules effective 27 April 2026.

When to hand it out in a course: straight after the plain 6 GHz card, for US classes. *(inferred)*

How to use it in class: read the GVP row against Standard Power directly above it. They are authorized in the same two sub-bands, 5.925 to 6.425 GHz and 6.525 to 6.875 GHz, so GVP inherits the same gap in the middle of the band and does not reach U-NII-6 or U-NII-8. Tell the room it is rules on paper: as of this release no geofencing system has been approved and no GVP access point or client has been certified.

Pairs with: 6 GHz Channel Allocations, 6 GHz Power and PSD.